

aTOMos v3.6.1.18

Seeded emit engine

Exact operator seeds

Integer texture fetches and persistent VRAM

Ordered native EMIT and exact phase arithmetic

ASA/NA, JK, seam parity and log-spherical calculus

Matched Orekit forecasts and SPICE ephemerides

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Foundation Preserved R17 engine and physical contracts

Implementation Native CUDA C++, C++ and exact Python reference

An exact source expression remains authoritative. A derived spatial cache may accelerate a decision only under an explicit error bound and a complete resolution path. Persistent state commits once for each accepted hinge event.

Performance claims identify the measured operation and hardware. The retained physical laws and clocks do not imply a validated satellite receiver, photonic device or complete S2 replacement. Prepared with AI assistance.

Exact words and complete native artifact streams

R18 extends the native operator engine with an optional complete ordered EMIT journal. Artifacts compiled by the original seeded workflow can now be materialized from native GPU execution, including every successful emission inside fused chunks. Canonical Cell48, State64, faults, receipts and ASA/NA/JK words retain their existing meanings.

Component	Mechanism	Observable contract
Native journal	Lane-local capture and synchronous compaction	Every fresh EMIT in each lane's epoch order
Seeded materializer	Canonical decoder over contiguous chunks	Original artifact bytes after fault-free HALT
Source ownership	Canonical bytes, generation and boundary state	Explicit source and continuation checks
Phase and seams	Bounded exact 32-bit modular identities	Complete quotient, parity and lineage
Retained engine	Fused/reference and texture/global paths	Earlier word, spatial, event and physics contracts

The stream extends the finite transition proof: equal transitions produce the same event predicate, record and next state. Disjoint lane intervals preserve order independently of GPU scheduling. Proven capacity is allocated before mutable execution; source identity and complete boundaries govern continuation. Prefix and fault outcomes remain explicit. A compact EMIT journal is distinct from a full instruction trace or final receipt.

Two original seeded fixtures reproduce 3,461 and 131,517 artifact bytes under native execution. Fresh original Python replay verifies every ordered event field and final State64 word. The compiler still evaluates formal expressions on the host before encoding their result as EMIT cells. Native materialization does not yet move that evaluator or learner search onto the GPU.

PHI/TIME, KLEIN and LSYS phase expressions also gain exact bounded forms, with the signed-wide reference retained. Independent checks cover complete quotients, remainders, seams, signed extrema and feedback. Matched phase measurements report both gains and regressions; their final-only device costs do not measure complete artifact throughput. Orbital comparisons separately match physical inputs, frames and horizons; exact words alone supply no satellite-position accuracy ranking.

Preserved lineage and continuing scope

The complete R17 and earlier chapters follow the new R18 material. Earlier reports retain their original release, input, hardware and observation scope. Statements describing a then-missing feature remain historical; the R18 chapters identify what has since been implemented and checked.

The imported PHI remains a finite phase counter, distinct from exact $a + b\phi$ arithmetic; SDF0 still assigns zero. Lossless one-bit-plane transport preserves complete operator words. Broader native expression lowering, non-affine event discovery and dynamic geometry/cache rebinding remain continuing work. Physical applications require their declared models, observations and error criteria. Source provenance, final-byte document review and package integrity provide separate evidence.

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1 R18: complete native seeded EMIT streams

R18 closes a concrete gap found on returning to the original seeded workflow: a materialized artifact needs every fresh EMIT in order, including those inside a fused chunk. The optional journal records that stream while retaining the complete canonical transition, final receipts and ASA/NA/JK feedback. Final-only paths remain separate. `formal/SEEDED_EMIT_JOURNAL.md` specifies the complete contract; the native schema is `ATOMOS-EMIT-JOURNAL-R1`.

1.1 Transition and complete ordered stream

For lane i , let $z_i(k)$ contain complete VM state, error, receipt and optional feedback state at next dispatch epoch k . Preserve

$$z_i(k+1) = C_k(z_i(k)), \quad C_k = \text{commit}_k \circ \text{VM}_k \circ \text{inject}_k. \quad (1)$$

For pure execution, injection and word commit are identities. Append an event exactly when the actual receipt says both executed and emitted. A sticky state flag supplies no fresh event. EMIT-HALT supplies one; identical payloads at distinct epochs remain distinct. A VM emission and successful feedback commit are different observables, with final feedback status recording a refused commit.

The native record is 40 bytes: a uint64 epoch and eight uint32 fields. It records lane, pre-instruction cell, flags, payload, post-transition lineage, cell, branch and status. Lineage, cell and status come from the actual post-transition State64. This compact EMIT schema is distinct from the vendor's full instruction trace.

With $E_i(k)$ denoting the record when present, define

$$J_i[k, k+n) = \parallel_{j=0}^{n-1} \begin{cases} [E_i(k+j)] & \text{fresh VM emission,} \\ [] & \text{otherwise,} \end{cases} \quad (2)$$

where $\parallel$ denotes ordered concatenation. At most one event per transition gives $|J_i| \leq n$ and proven total capacity Ln for L lanes. Lane i writes its disjoint interval $[in, (i+1)n)$ in increasing epoch order. No global atomic append or GPU scheduling order defines output. Host compaction publishes lane-major storage with offsets for each lane.

For contiguous unchanged execution,

$$J_i[k, k+a+b) = J_i[k, k+a) \parallel J_i[k+a, k+a+b). \quad (3)$$

The R17 induction extends directly: equal pre-states give the same complete transition, emission predicate, record, appended prefix and post-state. Disjoint output intervals preserve this independently per lane. Across several lanes, concatenate corresponding lane subsequences; appending whole lane-major chunks gives a different global lane order.

1.2 Admission, ownership and terminal status

The VM and feedback journal APIs admit 0 through 256 requested epochs, at least one lane, and representable epoch/allocation bounds. Zero work observes an unchanged boundary without a new receipt or event. `max_events=0` selects automatic Ln capacity; a nonzero value below that bound refuses before transition or injection. A larger value does not enlarge the actual allocation beyond Ln .

The event buffer initially uses $40Ln$ bytes on each of GPU and host, plus lane counters, overflow indicators and complete initial/final boundary arrays of size $O(L)$. Coupled execution captures both feedback states. These allocations finish before corresponding mutable execution. The

immutable canonical source is retained separately. Runtime failure cannot produce a successful journal, and no rollback is promised after it.

The synchronous drain checks capacity, lane, epoch bounds and ordering, then captures all final VM words, errors and receipts, with applicable feedback words. A nonzero VM error or feedback status gives `Faulted`; otherwise all lanes halted gives `Complete`; otherwise `Prefix`. A complete library journal may contain no emissions. Artifact materialization separately requires at least one EMIT. Every requested epoch runs even after early HALT/fault, preserving the canonical final held receipt. Dispatch, execution, VM emission and accepted feedback counts remain different.

Each chunk retains a shared immutable source object: process-unique owner ID and every canonical `.tmg` byte read back from actual GPU-expanded program words. It survives VM destruction. Move preserves the source; reset changes generation. The owner ID is not a cryptographic identity. Continuation requires the same source object, owner, generation, lane count, execution mode, binding and complete word profile. It also requires contiguous epochs and exact previous-final/next-initial VM states, errors and applicable word states. Ordinary execution creates an explicit gap. The feedback borrower additionally rejects external advance/reset, even before a zero-work call.

Exclusive VM/lane ownership is required throughout the synchronous call, with no concurrent same-lane work on another stream. Continuation checks do not authenticate edited event rows or reconstruct a live owner from serialized IDs. Pinned source and independent deterministic replay provide the persisted output evidence.

1.3 Literal byte materialization

For flags F and payload W , admit count $c = (F \gg 8) \& 7$ only for $1 \leq c \leq 4$. With endian bit $b = (F \gg 11) \& 1$, byte ordinal j , $0 \leq j < c$, is

$$d_j = \begin{cases} j & b = 0, \\ c - 1 - j & b = 1, \end{cases} \quad B_j = (W \gg 8d_j) \& 255. \quad (4)$$

Big endian reverses the selected low c bytes; it does not select the high c bytes. Concatenating decoded events in one lane's epoch order gives the source-compatible artifact.

`wqk_materialize` requires one canonical initial lane and source bits 0 and 1, denoting seeded profile and emitted bytes. The horizon defaults to the source header. Before mutable execution, it reserves all requested events and up to four artifact bytes per epoch, and verifies GPU-expanded source equality with the complete input. It joins contiguous chunks and writes an artifact only after fault-free HALT with at least one EMIT. Prefix/fault reports do not write an artifact. Source, artifact and report paths must not alias.

The JSON contains all nine native event fields plus sequence, count and byte order, with complete final State64 and run parameters. External replay checks all twelve fields, lane zero, complete state and output bytes, and requires literal true artifact-written/source-equality assertions. These assertions are not authenticated credentials; pinned bytes and exact replay supply the substantive output comparison.

CLI timings describe host setup; journal allocation, execution, transfers, compaction and decoding; artifact write; and total through artifact persistence. JSON encoding/report output is excluded. They establish no device-only or matched external materializer speedup.

The original compiler still validates dependencies, evaluates any formal tree on the host, and lowers the result into Cell48 EMIT cells. R18 executes and materializes those cells natively. GPU `formal.evaluate`, arbitrary JSON evaluation and native learner search remain separate work. This exact stream contract adds no physical law or continuous precision.

2 R18: exact phase and seam arithmetic in 32 bits

The original WQK GPU sources contain a useful way to avoid wide signed intermediates while preserving the finite machine. R18 implements the bounded identities below in the shared transition. The old wide expressions remain selectable through `ATOMOS_WIDE_PHASE_REFERENCE` for matched comparison. The canonical C interpreter is unchanged. Full bounds and source relationships are given in `formal/EXACT_PHASE32.md` and the retained original-return audit.

2.1 PHI and TIME retain the complete quotient

For raw unsigned words A, B , let $a = \text{s32}(A)$ and $b = \text{s32}(B)$. Take $M = 2^p$, with $p = 12$ for PHI and $p = 14$ for TIME. The canonical Euclidean sum has $a + b = Mq + r$ and $0 \leq r < M$. For each input define

$$q_a = (A \gg p) - (A \gg 31)2^{32-p}, \quad r_a = A \& (M - 1), \quad (5)$$

and likewise for B . Quotient terms are converted to signed32 before subtraction. The sign correction follows from $\text{s32}(A) = A - (A \gg 31)2^{32}$, hence $a = Mq_a + r_a$. Now compute

$$t = r_a + r_b, \quad q = q_a + q_b + (t \gg p), \quad r = t \& (M - 1). \quad (6)$$

Substitution proves the required quotient and remainder. Each input quotient lies in $[-2^{31-p}, 2^{31-p} - 1]$, and $0 \leq t \leq 2M - 2$. The complete output quotient fits signed32: its PHI range is $[-1048576, 1048575]$ and its TIME range is $[-262144, 262143]$. Thus no signed-overflow assumption or wide sum is required.

The complete q remains observable. PHI tests $q \neq 0$ for wrap status and uses its parity for the selected branch/orientation behavior. TIME mixes the raw word of q into lineage. Keeping only parity or remainder would change execution. The canonical old key, cell identity, lineage postlude and successor rule remain intact.

2.2 KLEIN and LSYS use exact modular phase updates

For raw rho word R , the finite KLEIN update uses

$$\rho' = R \& (2^{20} - 1), \quad \epsilon = (R \gg 20) \& 1. \quad (7)$$

Signed interpretation changes the quotient by 2^{12} when the sign bit is set. That change is even, so remainder and seam parity are preserved. For an odd seam, set $\theta_0 = \Theta \& (2^{18} - 1)$. The source half-turn or reflection respectively yields

$$\theta' = (\theta_0 + 2^{17}) \bmod 2^{18}, \quad \theta' = (2^{17} - \theta_0) \bmod 2^{18}, \quad \phi' = (-\Phi) \bmod 2^{12}. \quad (8)$$

Unsigned wrapping followed by masking is exact because these periods divide 2^{32} . Preserve the source orientation, optional sheet, status, branch and normalization. This keeps the finite parity; it adds no winding history.

For LSYS, let $o = \text{orientation} \& 1$ and $b = \text{branch} \& 1$. With raw turn argument A , the canonical chirality/branch product is positive exactly when $o \oplus b = 1$. Thus

$$D = \begin{cases} A & o \oplus b = 1, \\ -A \pmod{2^{32}} & o \oplus b = 0, \end{cases} \quad \Phi' = (\Phi + D) \& (2^{12} - 1). \quad (9)$$

This equals the signed-wide phase expression even for -2^{31} arguments and raw, non-normalized phases. Rate division remains R17's separately proved truncation-toward-zero magnitude shift. PHI/TIME floor division and LSYS rate truncation are distinct contracts.

Equal subexpressions preserve the canonical transition and its ordered fused composition. Boundary checks, compiled resources and matched timings provide separate evidence. Fewer source operations alone establish no speedup, cache-hit rate or physical accuracy.

3 R18: journal correctness and phase measurements

These results concern native finite-word execution and artifact streams. Orbital comparisons have a separate contract and evidence chapter. The retained R17, S2 and physical-profile results keep their original scopes; they are not new measurements of R18 journal output.

3.1 Independent native and Python checks

Both the bounded-phase build and retained signed-wide build pass all four standalone native suites. The canonical VM suite passes 28,794 assertions; feedback passes 1,718 checks. The expanded fused suite passes 8,073,110 checks, including 4,306,912 State64 words, 2,691,820 receipt words and 804,264 feedback words. It covers signed extrema, complete PHI/TIME quotients, raw seam inputs, every LSYS shift and chunks 1, 7, 32 and 256.

The new journal suite passes 10,075,735 checks using the unchanged C transition and an independent scalar per-bit ASA/NA/JK reference. It compares 4,842,880 State64 words, 2,015,150 receipt words, 1,547,800 feedback words, 1,030,950 event fields and 439,904 materialized bytes. Cases exercise texture/global reads and reference/fused dispatch, repeated payloads, early/final HALT and faults, zero work, refused capacity, continuation, source lifetime/move, reset/external-advance refusal and nonzero starting epochs.

All 59 Python tests pass, including ten seeded verifier/admission tests. Altered-record cases cover every event field including lane, all sixteen final state words, interior/truncated bytes, journal truncation/duplication/reordering, incomplete status, epochs, strict completion booleans, malformed integer/duplicate JSON fields and changed pinned source evidence. The positive verifier unit test constructs a CPU reference-shaped report; actual native runs are separate evidence below. Logs are `review/r18_initial_checks.txt`, `review/r18_wide_checks.txt` and `review/r18_python_checks.txt`.

The complete R18 spatial build also passes all ten native suites, including the inherited spatial/S2 and feedback checks. The standalone narrow/wide builds above isolate the four word-machine suites for arithmetic comparison.

3.2 Original seeded artifacts reproduced by the native GPU

The original compiler generated each selected program and complete sidecar twice, identically to the stored originals. No external source, compiler or formal evaluator was edited. The source manifest pins the selected dependency closure and original artifacts. The family-authority evaluation also reproduced its direct canonical JSON result through 352,593 host formal steps.

Original fixture	Epochs	EMITs	Event fields	Bytes
world03 release	869	866	10,392	3,461
learner06 family authority	32,883	32,880	394,560	131,517

Both runs use fused native texture execution, with respective chunk sizes 32 and 256. Both reach HALT without fault. Every event's twelve fields agree in order, all sixteen final State64 words match, and every artifact byte equals fresh original Python replay/materialization. Before mutable execution, the CLI checks its complete input against the actual GPU-expanded program bytes. The verifier also requires literal true artifact-written and source-equality report fields.

The world03 output SHA-256 is

3849188eddecbb9904a9059680bbf0752a0ee27805b551984cb8391e68e805a0

and the family-authority output SHA-256 is

```
4e59666a7ccdc2505d94fe760f5d317f5302f8766a6bd8f1b022912890e5844a
```

Exact program, source-manifest and report hashes accompany the complete comparisons in `review/r18_world03_verified.json` and `review/r18_family_authority_verified.json`. Digests identify bytes; they do not replace the individual comparisons or establish authorship.

These graphs contain host-compiled EMIT cells. The evidence establishes native execution and complete original output equality. The formal expression tree was evaluated on the host before those cells were created. These verification runs alone establish no throughput advantage over another complete artifact materializer.

A subsequent full-build CLI check runs both original fixtures through all four reference/fused and texture/global combinations: eight completed cases, each with every event field, final word and output byte checked. Seven additional publication checks preserve existing output on zero/short prefix runs and reject completed traces without EMIT or aliased source, artifact and report paths. `review/r18_materialize_cli_checks.json` binds the executable and these results.

3.3 Compiled resources and actual machine instructions

Static `cuobjdump` inspection covers the exact narrow/wide executables used in the phase experiment: 20 kernels each, plus ten kernels in each separately linked materializer. Every inspected kernel reports zero stack and local bytes, with no LDL/STL. Texture transition variants contain five actual TLD sites each. Register counts are 54 for both pure-journal fetch paths and 78 for all four coupled-journal variants. This supports register retention for the inspected per-lane loop state; program reads, event stores and boundary publication still access memory.

The bounded forms reduce static instruction sites in all fourteen transition-containing bodies by 16 through 32, including padding and potentially unexecuted branches. Registers change little: pure fused texture rises from 49 to 50, injected final-only feedback falls from 75 to 74, and journal counts are unchanged. All builds contain no I2F, MUFU or F2I; the floating-group mnemonic that remains is HFMA2, using zero-register/constant forms. Exact word semantics therefore do not mean exclusively integer-unit mnemonics. `review/R18_NATIVE_CODE.md`, its JSON, complete disassemblies and resource dumps bind these findings to actual executable/source bytes. Static instruction counts establish no runtime instruction count, occupancy, cache-hit rate or speedup.

3.4 Matched bounded-phase and signed-wide execution

The phase experiment compares two R18 builds differing in selected phase/seam expressions. It uses the RTX 5070 Ti Laptop GPU, CUDA 12.8.61, MSVC 19.44 and `sm_120`, with 65,536 spread-initialized lanes. The 1,024-cell PHI/TIME and KIN2/KLEIN/LSYS/HINGE programs each run 256 epochs. A 1,048,576-cell synthetic mixed program, 48 MiB plus its header, runs 128 epochs. These are generated input programs with no precomputed VM output. Coupled cases use the declared `copy32` profile.

Twelve workload/profile/dispatch cases each compare texture and global reads in both builds, with five fresh-process repeats: 240 serialized trials. Variant order rotates and reverses; there is no warmup. Fused chunks contain 32 epochs. Every comparison agrees in full-array FNV-1a-64 and exact first-lane fields. This noncryptographic digest is separate from the independent individual-field correctness tests above.

Program	Feedback	Dispatch	Texture	Global
PHI/TIME	none	reference	1.003	1.023
PHI/TIME	none	fused	0.979	1.021
PHI/TIME	copy32	reference	1.002	1.007
PHI/TIME	copy32	fused	0.880	1.084
Seam/turn	none	reference	0.991	1.004
Seam/turn	none	fused	0.976	0.994
Seam/turn	copy32	reference	1.010	0.996
Seam/turn	copy32	fused	1.099	1.058
Mixed 48 MiB	none	reference	0.997	0.999
Mixed 48 MiB	none	fused	0.996	1.008
Mixed 48 MiB	copy32	reference	0.945	1.046
Mixed 48 MiB	copy32	fused	1.009	0.994

Entries are signed-wide median device time divided by bounded-phase median device time; values above one favor bounded-phase execution. The range is $0.880\times$ to $1.099\times$, with both regressions and gains. The fused PHI/TIME feedback texture case is slower under bounded expressions, while the fused seam/turn feedback case is faster. Mixed fused cases lie close to one. Five repeats without confidence intervals do not establish a statistically stable advantage near unity.

Cold host-complete ratios range from $0.973\times$ to $1.044\times$. They include setup and readback outside the device batch interval. Trials, medians, binary hashes and program hashes are in `review/r18_phase_runs.json`; reproduce the experiment with `tools/measure_phase.py`. These are final-only timings, distinct from complete ordered-journal costs. The bounded expressions have exact finite arithmetic proofs; these measurements establish no universal phase speedup, texture advantage, VRAM saturation or satellite-position improvement.

4 R18: matched satellite forecasts with Orekit

The new comparison establishes close numerical agreement with an independent Orekit implementation of the same frozen orbital problem. It does not establish better satellite-position accuracy than Orekit. Across eight cases, the maximum aTOMos CPU–Orekit position difference is 0.026760 m and the maximum CPU–CUDA difference is 0.000220015 m. Both engines show almost the same discrepancies against later external positions: all eight first-day sampled maxima are below 10 m, while some seven-day maxima reach kilometres.

4.1 The physical problem and observations are matched

The retained ORBIT-DYNAMICS-R2 models are evaluated for G05 (MEO), C03 (GEO), C06 (IGSO) and CHANDRA (HEO), from the 2 January and 2 February 2025 GPST epochs. The original models, training provenance, reference products, native binaries and scorer sources were frozen under 71 input hashes before these R18 numerical comparisons. No target-position fit, clock adjustment, frame alignment or physical parameter selection is performed here. Both date sets were already inspected in R10; neither is a newly blind R18 confirmation.

Each implementation receives the same initial six-state vector, degree/order 12 EGM96 coefficients, Sun/Moon forcing series, solar-pressure and cylindrical shadow rule, empirical RTN acceleration, relativity setting and frame series. Orekit 13.1.8 with Hipparchus 4.0.3 supplies its numerical propagator, Holmes–Featherstone gravity algorithm and relativity implementation. Independent Java adapters preserve the other frozen force equations. Orekit’s normalized gravity coefficients are converted from the same frozen unnormalized table. Its central term is added once, with automatic central attraction disabled. The final adapter uses absolute Cartesian position/velocity states and explicitly checks short-time motion; no orbit determination is performed.

The upstream [numerical-propagator API](#) describes configurable integration and force models; the [Holmes–Featherstone API](#) describes normalized coefficients and its stable gravity recurrence. The actual 13.1.8 binary and source archive are separately pinned in `review/r18_orbit_java_dependencies.json`; the live documentation links are explanatory references, not the version pin.

For GNSS, the independent SP3 parser retains the original GBM precise positions and their IGS20/IGb20 realization labels. Both engines use the same frozen GCRS-to-ECEF transform for scoring. CHANDRA uses the original Horizons geometric geocentric ICRF table, with calendar TDB converted to GPST through SOFA/ERFA. Each returned epoch must equal the exact decoded request epoch; all 15,553 rows match with zero engine/request timestamp offset. This checks alignment of the calculation, not the accuracy of a live clock.

Seven cases contain 2,016 future samples. February C03 contains 1,441 and has a 172800 s reference gap. Missing rows are not filled. CHANDRA’s final samples lie about 51.184 s before the nominal seven-day GPST endpoint; the files retain exact horizon offsets. Every first-day statistic below uses 288 available samples. Full reports also contain cumulative 1, 6, 12, 48 and 72 hour statistics, first sampled threshold exceedances and reference gaps.

4.2 Physical-reference discrepancy remains horizon dependent

For common reference position r_i and engine position p_i^a in the declared comparison frame, the measured quantities are

$$e_i^a = \|p_i^a - r_i\|_2, \quad E_{\text{RMS}}^a = \sqrt{\frac{1}{N} \sum_{i=1}^N (e_i^a)^2}, \quad E_{\text{max}}^a = \max_i e_i^a. \quad (10)$$

Table 1 reports metres, rounded to three decimals. “Arc” means all available future samples up to seven days; it is not a continuous-time bound. CUDA is omitted from this compact physical-error table because its separate maximum state disagreement with CPU is quantified in Table 2; its complete scores remain in the JSON report.

Epoch	Object	0–24 h maximum		Arc maximum		Arc RMS	
		CPU	Orekit	CPU	Orekit	CPU	Orekit
Jan	G05	1.057	1.056	12.998	12.975	5.294	5.281
Jan	C03	5.304	5.304	78.717	78.716	37.434	37.434
Jan	C06	1.172	1.172	8.084	8.083	2.943	2.943
Jan	CHANDRA	8.093	8.093	25635.951	25635.961	5412.235	5412.237
Feb	G05	3.072	3.069	71.800	71.773	33.113	33.099
Feb	C03	4.654	4.654	100604.174	100604.175	39382.862	39382.863
Feb	C06	2.108	2.108	7.629	7.630	3.494	3.494
Feb	CHANDRA	0.290	0.290	5663.182	5663.184	2055.685	2055.685

Table 1: Matched forecast discrepancy against later GBM/Horizons positions, in metres. Both engines solve the same frozen physical model.

The first-day maxima span 0.289813–8.092836 m for native CPU. G05’s seven-day maxima are 12.998 and 71.800 m; C06’s are 8.084 and 7.629 m. The larger discrepancies must remain visible: January CHANDRA reaches 25.636 km, February CHANDRA 5.663 km and February C03 100.604 km. Closely matching these curves in Orekit shows that a different integrator alone does not remove the long-horizon discrepancy. This experiment does not uniquely separate force-model limitations, unmodelled events and reference-data limitations, and does not infer a cause from a plotted jump.

4.3 Numerical agreement is a separate measurement

Define the same-epoch state-position difference $d_i^{a,b} = \|x_i^a[0:3] - x_i^b[0:3]\|_2$ in the common inertial coordinates. Standard Orekit integration uses a 10^{-6} m positional absolute tolerance and a 60 s maximum step; refinement uses 10^{-7} m and 15 s. Velocity absolute tolerances are respectively 10^{-9} and 10^{-10} m/s, with relative tolerance 2×10^{-14} . These are integrator controls, not certified physical-position tolerances.

Epoch	Object	Maximum state-position difference (mm)			
		CPU–CUDA	CPU–Orekit	CUDA–Orekit	Orekit refinement
Jan	G05	0.043202	26.652946	26.696136	0.026221
Jan	C03	0.040229	1.035783	0.995845	0.011058
Jan	C06	0.014359	1.081491	1.095529	0.021503
Jan	CHANDRA	0.019398	9.973403	9.992799	0.064834
Feb	G05	0.027756	26.759896	26.786545	0.018755
Feb	C03	0.024485	1.027233	1.050839	0.006472
Feb	C06	0.005661	1.009391	1.013778	0.027620
Feb	CHANDRA	0.220015	6.077876	6.297885	0.022174

Table 2: Numerical disagreement over each complete sampled arc. Orekit columns use the standard run; refinement compares its standard and tighter controls.

The maximum Orekit refinement displacement is 0.064834 mm, occurring in January CHANDRA; the GNSS maximum is 0.027620 mm. An independent force audit evaluates 65 points on each

standard trajectory against the retained Python R2 equations: 520 checks, with maximum acceleration discrepancy $1.115761 \times 10^{-15} \text{ m/s}^2$. The completed audit binds the exact final adapter, dependency manifest, requests and all 16 standard/refined state files. The tight run is scored against the same external rows in a second full report.

These results make the numerical contribution small compared with the measured metre-to-kilometre reference discrepancies in this experiment. They do not make the arithmetic exact. Lossless one-bit word packing preserves the encoded model and operators; native physical propagation still uses finite-precision floating-point arithmetic. The discrete exactness proofs elsewhere in this document and the numerical orbital agreement above are different guarantees.

4.4 Reference checks, runtime scope and reproducibility

The independent GBM/WUM product intersection yields seven-day RMS discrepancies of 0.02436/0.02447 m for G05, 0.16894/0.15315 m for C06 and 2.20440/0.93894 m for C03 (January/February). The largest C03 product discrepancy is 3.57510 m in January. These are observed cross-product differences, not a certified uncertainty floor; providers can share observations and modelling assumptions. CHANDRA has no second trajectory product in this experiment. The common frozen EOP transform and the GCRS/ICRF coordinate approximation remain stated limitations, as does retrospective product availability. Correlated sampled positions supply no general confidence level.

Native one-batch durations include worker startup, model initialization, query execution and teardown. CPU takes about 0.43–0.57 s for the GNSS cases and 2.23 s for CHANDRA; CUDA takes 21.2–24.0 s and 84.6–87.5 s respectively. These single trials are diagnostic, and show no GPU advantage for this particular workload and lifecycle. Orekit’s reported internal timing has a different boundary, so no cross-library speed ratio is claimed. The finite-word spatial and journal benchmarks retain their separate workloads and cannot substitute for this orbit comparison.

`formal/ORBIT_COMPARISON_PROTOCOL.md` states the frozen experiment; `formal/ORBIT_COMPARISON_RESULTS.md` records the results and exact hashes. The scoring script is `tools/compare_orbit_references.py`. Its standard and tight summaries are `review/orbit_comparison/matched_forecast_summary.json` and `review/orbit_comparison/matched_forecast_tight_summary.json`. The numerical admission record is `review/r18_orekit_numerical_audit.json`. Plot generation verifies these bindings and recomputes the displayed errors from the frozen reference rows and scored states.

NASA describes [SPICE as an observation-geometry system](#). Its separate SPK interpolation experiment in this release measures representation of a supplied frozen forecast. Neither that experiment nor the matched Orekit calculation establishes universal superiority over a space agency’s navigation system. What this adds is a reproducible external numerical comparison and explicit accuracy-versus-horizon evidence for the existing seeds.

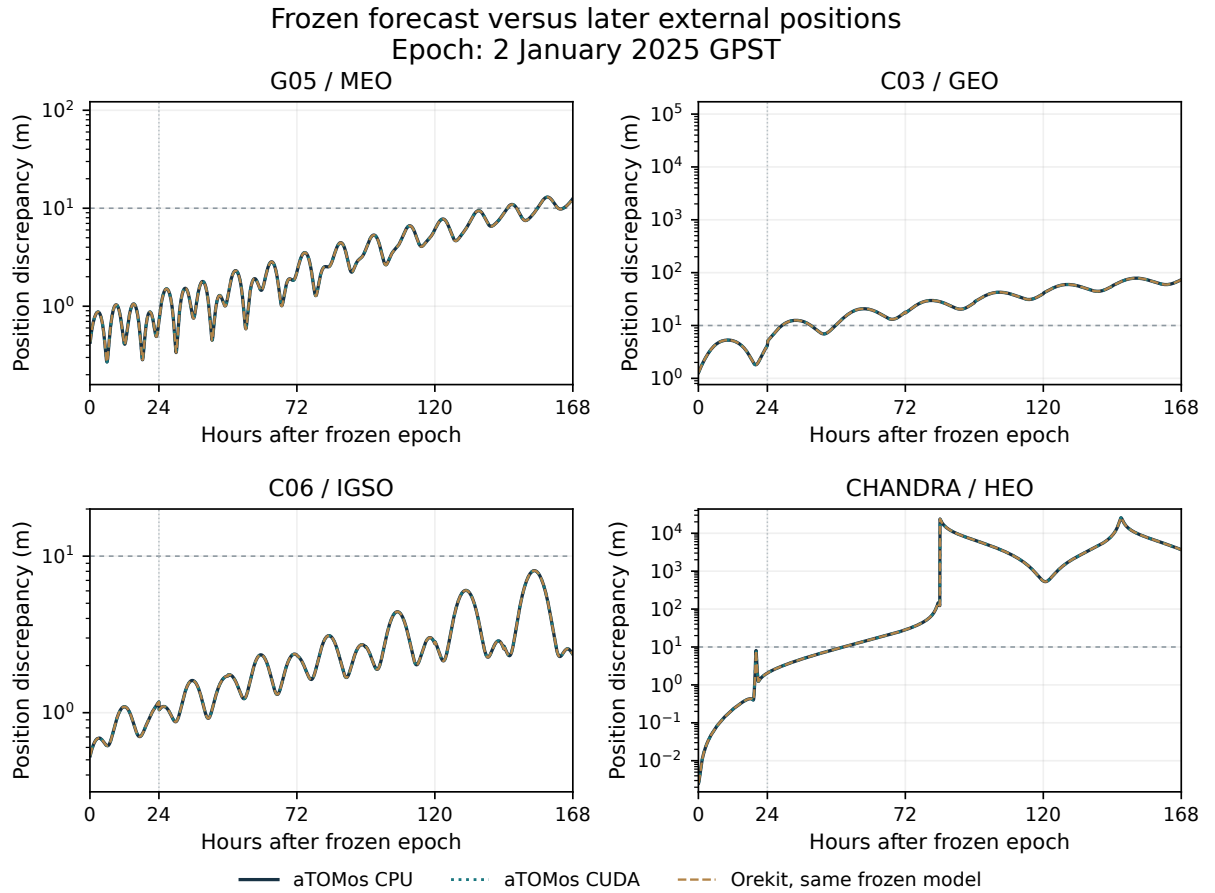


Figure 1: January forecast discrepancies. Curves from the three implementations nearly overlap. The horizontal 10 m line is descriptive; the vertical dotted line marks 24 hours. Per-object vertical scales match the February panels. Selected sampled error is shown, with no reference-derived alignment.

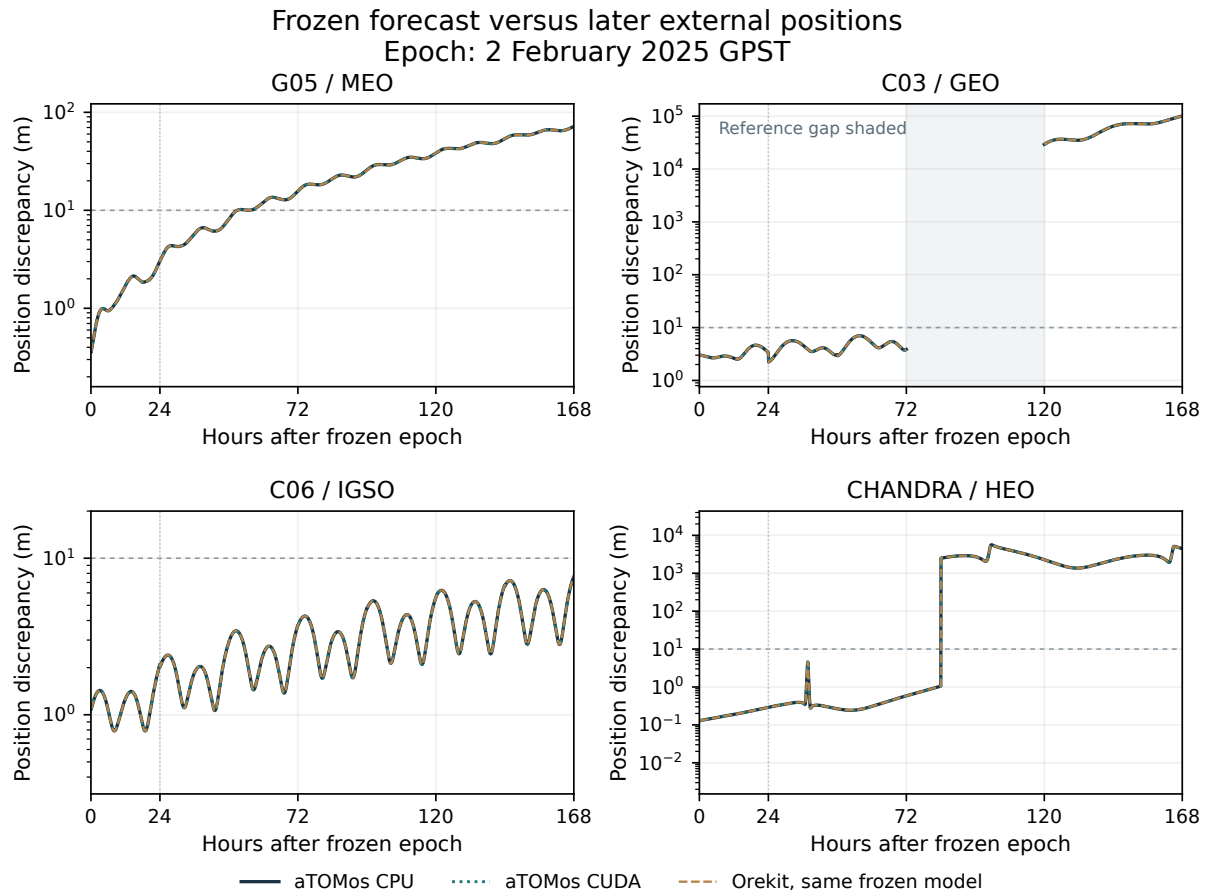


Figure 2: February forecast discrepancies. The missing two-day C03 reference arc is shaded and the curves are broken there. Sparse or absent observations do not establish performance between the retained samples.

5 R18: SPICE ephemeris representation and query

R18 constructs and queries actual CSPICE N0067 type 9 SPKs through Spiceypy 8.2.0. Every input state comes from the unchanged R10 native CPU ORBIT-DYNAMICS-R2 forecast. No future observation is supplied to SPICE and no model is refitted. This measures how well an SPK stores and reconstructs the same forecast; it is not independent satellite-position validation or forward physical propagation by SPICE. The underlying forecast still needs the separate external-reference comparison.

5.1 Same states, explicit time and frame conventions

The initial declared grid covers G05 (MEO), C03 (GEO), C06 (IGSO) and Chandra (HEO), from both January 2 and February 2, 2025, through seven days. These dates were used in earlier releases. Before state evaluation, the protocol fixes input intervals of 300, 900 and 1800 seconds crossed with polynomial degrees 7, 9 and 15: 72 cases. Every interval midpoint is compared with a native state, every stored node is checked, and queries beyond either coverage end must fail. The complete model, binary, interface, dependency and output hashes are recorded in `review/r18_spice_protocol.json` and the comparison report.

For GPS epoch coordinate g_0 and model offset t , use

$$T = (2444244.5 - 2451545.0)86400 + g_0 + 51.184 + t, \quad E = \text{UNITIM}(T, \text{TDT}, \text{TDB}). \quad (11)$$

Here T and E are TT and TDB seconds past J2000. The pinned NAIF LSK gives $T = E - K \sin \epsilon$, $\epsilon = M + e_B \sin M$ and $M = M_0 + M_1 E$, hence

$$\frac{dT}{dE} = 1 - K \cos \epsilon M_1 (1 + e_B \cos M). \quad (12)$$

SPK velocities are multiplied by dT/dE on writing and converted back on reading. Metres become kilometres in the SPK. States are geometric relative to Earth with aberration correction NONE. Model GCRS axes are transported under an explicit identity-orientation convention to the SPICE J2000 label; this does not validate a relativistic GCRS/ICRF coordinate transformation.

The tested TT round trips agree exactly at their floating-point inputs. CSPICE's short LSK correction differs from geocentric ERFA by up to 15.320 microseconds in January and 22.406 microseconds in February; both SPK nodes and queries use the same CSPICE convention. The stored model Julian TT epoch differs by -8.8215 microseconds from the GPS-derived value. Absolute ET doubles here have approximately 0.119 microsecond spacing. These numerical interfaces remain finite precision despite lossless seed transport.

5.2 Observed interpolation errors and a useful storage profile

At the initial 300-second, degree-9 setting, the six GNSS cases have midpoint maximum position differences of 0.455–1.082 mm. Chandra instead reaches 3.211 m in January and 1.667 m in February. Its coarse 900-second, degree-9 case reaches 8.716 km; the 1800-second, degree-15 February case reaches 45,580.644 km. All failures remain in the report. A higher polynomial degree does not repair every undersampled trajectory or improve every boundary.

After those results, a separately declared adaptive follow-up tests Chandra intervals of 30, 60 and 120 seconds with degrees 9 and 15. It is not a blind confirmation. Its denser native grid preserves every earlier reference state exactly. Degree-9 results are:

Interval (s)	SPK bytes / seven days	January max (mm)	February max (mm)
30	1,134,592	0.497	0.484
60	569,344	0.497	0.704
120	286,720	1.012	0.752

Degree 15 sometimes worsens edge errors, reaching 14.626 mm in the February 30-second case. The measured starting profile is degree 9 with 300-second GNSS nodes and 60-second Chandra nodes: at most 1.083 mm added position error at tested midpoints, with 114 KiB per GNSS object and 556 KiB for Chandra. The 120-second Chandra option uses 280 KiB and also stays below 1.013 mm at the tested midpoints. These are sampled reconstruction results for these trajectories, not continuous or future physical bounds.

The original experiment checks 72,648 nodes and 72,576 midpoints; the follow-up adds 141,132 nodes and 141,120 midpoints. A fresh process reopens all 84 SPKs and reproduces every reported maximum exactly. Every coverage rejection and node-preservation check passes. Full protocols, arrays, SPKs and verification reports are retained under `review/r18_spice*`.

5.3 Cost, limits and physical meaning

Five repeated warm batches of 2,016 shuffled queries at the original 300-second, degree-9 profile take median 0.457–0.463 s through the native GNSS worker API, about 1.793 s for Chandra, and 0.00875–0.00935 s through the SpiceyPy query API. SPICE is 49.4–51.9 times faster on these GNSS API cases and 197.0–204.9 times faster on Chandra. It reuses a precomputed 116,736-byte SPK, while the native path still executes RK4 steps from checkpoints and crosses IPC/JSON. Precomputation is excluded and reported separately. Host load is uncontrolled. These are storage/compute and interface measurements, not equal arithmetic work, GPU throughput or a general product ranking.

For native forecast x_N , SPK reconstruction x_S and external physical reference x_* at a common time and frame,

$$\|x_S - x_*\| \leq \|x_S - x_N\| + \|x_N - x_*\|. \quad (13)$$

This experiment measures the first term at specified samples. It enables aTOMos forecasts to be distributed through established SPICE geometry interfaces while preserving their measured accuracy to the chosen interpolation budget. It does not improve the second term or establish a physical accuracy advantage over NASA software. Detailed equations, chronology and reproduction commands are in `formal/SPICE_EPHEMERIS_COMPARISON.md`. The primary interface definitions are NASA NAIF's [SPK type 9](#) and [uniform-time conversion](#) documentation; the numerical results above are local measurements.

6 R17: fused word execution with exact boundaries

R17 moves several consecutive per-lane epochs into one native kernel chunk. It retains the R16 canonical Cell48 instruction, its mandatory postlude, the literal two ASA/NA stages, and JK feedback. The reference launch modes remain available. This changes when the mutable state is published, while preserving the operator sequence inside each chunk. The full contract is `formal/FUSED_WORD_EXECUTION.md`; the source authority remains the imported, pinned C/Python TOMAGI implementation and the R16 feedback binding.

6.1 The complete per-lane transition

For lane i , write $z_i = (s_i, f_i, r_i, w_i, k)$ for its complete State64, sticky VM fault, last receipt, optional feedback state, and next dispatch epoch. Let P denote the immutable program, seed, bank descriptors, word profile, and injection binding. A coupled epoch is

$$C_k = \text{commit}_k \circ \text{VM}_k \circ \text{inject}_k. \quad (14)$$

The extended transition also advances k by one. For pure VM execution, injection and word commit are identities; the canonical transition and receipt behavior remain.

Injection reads the old committed q , applying the enabled low/high-word mapping to raw rho and v_ρ only under the existing eligibility rule. The VM reads that injected state, computes its pre-key, executes the canonical body and postlude, and issues the receipt at epoch k . Commit then reads this actual receipt and the same old feedback word, applies the declared word bodies and two masks, and performs JK once for a fresh accepted EMIT. The three operations cannot be commuted.

All raw words, alias-sensitive writes, floor-versus-truncating division, word wrapping, and old-key/old-cell lineage inputs retain their source meanings. A refused VM instruction holds the state presented to the VM; it does not undo an earlier authorized injection. A feedback fault suppresses later work only where the reference binding specifies that behavior. No recovery or extra emission is introduced by fusion.

6.2 Inductive equivalence of a fused chunk

A fused thread loads one complete lane state, repeats the same ordered transition n times locally, and publishes the final state. Denote this operation by F_n . For $n > 0$, its required equality is

$$F_n(z_i; P, k) = (C_{k+n-1} \circ \cdots \circ C_{k+1} \circ C_k)(z_i; P). \quad (15)$$

Equality covers every State64 word, VM fault, complete final receipt, all feedback fields, and next epoch. A permitted zero-length call is identity: it neither executes an instruction nor changes a receipt.

Proposition 6.1 (Exact fused execution). *Equation (15) holds if the two paths use the same integer transitions and immutable inputs, each thread exclusively owns its lane, and no intermediate observer or mutable cross-lane dependency is required inside the chunk. The epoch range and resources must be admitted, and launches sharing mutable state must be ordered.*

Proof. Both paths start from the same complete state. Suppose they agree after j iterations. Their next injection sees the same old word and eligibility; the VM therefore sees the same injected state and immutable program and computes the same instruction, postlude, fault result, and receipt at $k + j$. Commit sees the same old feedback state and receipt and produces the same word and metadata. Thus they agree after $j + 1$. Induction gives the final equality. Every lane reads only its own mutable state and common immutable data, so the proof applies independently to all lanes without a cross-lane barrier. $\square$

The ownership premise requires one ordered CUDA stream or explicit cross-stream dependencies. Sequential host calls alone do not order independent streams. A future opcode that reads another lane's mutable state, a program edit, or an external sensor sample inside a chunk would change these hypotheses and require another boundary or synchronization contract. The existing borrow lifetime, state-generation, and epoch checks remain in force; direct DeviceView mutation must not bypass them.

6.3 Halt, fault, and final-receipt fidelity

Dispatch count is distinct from successful instruction count and from accepted hinge count. A chunk starting at k with $n > 0$ requested epochs ends with

$$k_{\text{next}} = k + n, \quad \text{epoch}(r_{\text{final}}) = k + n - 1. \quad (16)$$

These equations still hold if the lane halted or faulted earlier. The fused loop evaluates the later reference no-execution epochs too.

If EMIT-HALT occurs in the final requested iteration, its receipt is freshly emitted and that payload commits once. If it occurs earlier, the final receipt is a later halted no-execution receipt; the feedback state still retains the earlier emission count and committed word. A HALT executed in the final iteration retains its successful instruction receipt, normalization, and lineage, with no successor change.

A fault first raised in the final iteration retains that refusal's available instruction metadata. A fault already sticky before the last iteration produces the reference faulted no-execution receipt, including the appropriate default fields and final epoch. Raw initially halted state is not normalized to simplify this case. Existing word-state faults retain the exact reference ordering and effects.

An optimization that merely breaks on HALT or fault can leave a stale receipt. Such a shortcut would need an additional proof that it constructs the exact final held state and receipt; the ordinary local loop needs no shortcut. A final receipt describes one dispatch. Cumulative accepted EMIT count is held separately by the feedback profile, and a final pure-VM receipt alone does not count earlier emissions. Recommitting an already consumed final receipt remains a hold.

6.4 Chunk composition and observation

For admitted lengths a, b , the same transition law gives

$$F_{a+b}(z; P, k) = F_b(F_a(z; P, k); P, k + a). \quad (17)$$

Hence any ordered partition of a requested run preserves its final state and total epoch count. Each completed chunk publishes an observation boundary. An application requiring a callback, state edit, sensor input, or receipt consumer after every logical epoch must choose chunk length one or preserve the needed finer boundary. Final-state equivalence is not identity of all intermediate global-memory traces.

The native `VM.enqueue_batch` accepts chunks from zero through 256 epochs; zero holds and a larger chunk rejects before mutation. The feedback `run_steps_fused` admits a chunk size from one through 256, with default 32, and partitions the requested unsigned 32-bit total count. A zero total holds state, receipt, and epoch while resetting its timing record; the chunk-size argument is still validated. The complete epoch range must fit before dispatch. These finite chunk limits are execution choices, not arithmetic approximations or permission to skip internal emissions. Each eligible internal EMIT reaches commit.

6.5 Exact LSYS rate division without signed overflow

LSYS divides each signed rate x by 2^s , truncated toward zero, where $s = \text{clamp}(\arg_1, 0, 30)$. For its raw word w , let $b = w \gg 31$. With all word negations evaluated modulo 2^{32} , the shared transition uses

$$m = \begin{cases} w & b = 0, \\ -w \pmod{2^{32}} & b = 1, \end{cases} \quad d = m \gg s, \quad w' = \begin{cases} d & b = 0, \\ -d \pmod{2^{32}} & b = 1. \end{cases} \quad (18)$$

The unsigned magnitude m equals $|x| \leq 2^{31}$, even for $x = -2^{31}$. Thus $d = \lfloor |x|/2^s \rfloor$, and sign restoration returns exactly the raw word of $\text{trunc}(x/2^s)$. No overflowing signed negation is

needed; shift zero preserves the minimum signed word. An arithmetic right shift of a negative signed value would generally implement a different rounding rule.

This identity lowers the four LSYS rate divisions in the common reference/fused helper. It does not alter the phase update or replace PHI/TIME floor division. Equal subexpressions preserve the transition proof above; compiled instructions and timing remain separate evidence.

6.6 Texture reads, local storage, and evidence scope

The texture path continues to fetch immutable integer Cell48 records through raw element texture reads, including canonical key searches. The global path reads the same records. Neither introduces floating interpolation, filtered values, rasterization, or altered operator bits. Mutable lane state is held in local program variables during a chunk and published at its end.

For $N > 0$ coupled epochs, the reference uses $2N$ kernel launches without injection and $3N$ with injection; fixed chunk size c reduces these to $\lceil N/c \rceil$ fused launches. Setup and readback are excluded from these counts. The internal logical epochs and accepted emissions remain the same; a launch-count reduction is not a measured speed ratio.

Local variables do not establish physical register residency. Registers, spills to local memory, occupancy, cache hits, and scheduling depend on the compiled kernel and device. Their measurements are separate from this equivalence proof. A texture object alone promises neither cache hits nor speedup.

Reference and fused comparisons must use matched seeds, initial states, profiles, epoch counts, and output boundaries. Setup, device batch, host dispatch/completion, final readback, and digest/output costs remain distinct. Comparing a trace-producing reference run with an unobserved fused batch would include an extra workload difference.

R17 specializes execution of the finite word machine and its declared feedback. It does not lower all XOP expressions, move the exact rational affine event solver onto the GPU, supply cross-lane world synchronization, or establish physical accuracy. The inherited phase, winding, arithmetic, event, and physical-profile distinctions remain unchanged.

7 R17: native evidence and return to the original

The final R17 implementation was built with CUDA 12.8.61 and MSVC 19.44 for `sm_120`. Measurements use the RTX 5070 Ti Laptop GPU, with 46 SMs, 36 MiB L2, and 12,820,480,000 reported VRAM bytes. The host is a Core Ultra 7 255HX. GPU jobs were serialized. All R17 trials use the same final executable; its SHA-256 is recorded in the measurement and compiled-code reports.

7.1 Independent correctness and compiled execution

All nine native test suites pass, including the inherited spatial and feedback suites. The new fused suite passes 1,513,078 checks against the unchanged canonical C machine and an independent per-bit ASA/NA/JK model. Its field comparisons include 722,912 State64 words, 451,820 receipt words, and 292,264 feedback words; other checks cover errors, epochs and API contracts. The inherited native VM suite adds 28,794 assertions, and its feedback suite adds 1,718 checks. All 49 inherited Python tests pass.

The fused cases cover all 16 opcodes, alias-sensitive raw fields, multiple lanes, both fetch paths and feedback bindings, four word profiles, key hits and misses, forced two-cell banks, halted and faulted lanes, stale EMIT flags, nonzero starting epochs, an ordered nondefault stream, and chunks of 1, 7, 32 and 256 across 263 requested epochs. LSYS cases cover every shift from 0 through 30, clamped arguments, negative odd values and signed extrema, including -2^{31} . This is finite-machine evidence; the exact transition proof supplies the general composition argument.

The compiled audit inspects actual SASS and resource declarations, binding their bytes to the executable and native source. Integer texture variants contain actual TLD sites. All six fused variants have zero declared stack and local-memory bytes, with no LDL/STL instructions in the inspected kernels. The final register counts and comparison with the preserved initial build are recorded in `review/R17_NATIVE_CODE.md`. These facts concern this compiler target; they do not establish occupancy, cache hits or performance on another GPU.

The initial LSYS division was integer-valued but compiled through a reciprocal-and-correction sequence. The final unsigned magnitude shift and sign restoration removes that need while retaining signed truncation toward zero. Some floating-unit instructions can still materialize constant bit patterns in compiler output. Exact integer state semantics must not be confused with a claim that every generated opcode uses an integer unit.

7.2 Matched device and complete-host measurements

The tiny workload is the original three-cell source fixture, with 96 requested epochs and identical initial lane words. The larger workload contains 1,048,576 synthetic Cell48 instructions: 48 MiB of operator cells, plus the 128-byte header. Its 15 non-HALT opcodes, arguments and successor links are generated from an explicit recipe; lane initialization spreads raw rho, lineage and starting cell. It runs 128 epochs on 65,536 lanes. The larger program is generated input, rather than precomputed VM output.

Each case compares R17 reference/fused dispatch and texture/global reads. The tiny cases additionally run the unchanged R16 executable. Three fresh process trials per variant give 132 total runs, with rotated ordering and no warmup. Every case has matching full-array FNV-1a-64 and exact first-lane fields, with the expected accepted EMIT totals in the tiny coupled cases. FNV is noncryptographic; the independent tests separately compare complete individual fields. Chunk size is 32 throughout the measured fused runs.

Program	Lanes	Mode	Ref. tex. Device median (ms)	Fused tex. Device median (ms)	Tex. gain Reference/fused	Global gain
3 cells	1	pure	0.803	0.082	9.76×	7.78×
3 cells	1	copy32	2.337	0.295	7.92×	7.01×
3 cells	4,096	pure	0.749	0.064	11.66×	14.92×
3 cells	4,096	copy32	2.387	0.212	11.25×	11.35×
3 cells	262,144	pure	5.992	0.336	17.82×	20.28×
3 cells	262,144	copy32	16.051	0.830	19.33×	20.05×
48 MiB	65,536	pure	3.003	0.897	3.35×	4.21×
48 MiB	65,536	copy32	5.767	1.135	5.08×	6.21×

The table reports medians of device batch time in milliseconds. Across these 16 matched R17 comparisons, device ratios range from 3.347× to 20.277×. The unchanged R16 tiny-fixture comparisons give 6.460× to 19.710×. Setup, final readback and host work are outside device batch time. These are measurements of the stated word machine workloads, with three repeats, rather than a universal speed bound.

Cold host-complete ratios range from 0.965× to 1.098×: 13 of 16 cases have a lower fused median, and three have a higher median. Initialization, allocation, upload, final readback and hashing remain substantial costs. For example, a large synthetic table has a different setup cost from the three-cell fixture. Device speed ratios must not be presented as complete application speedups.

The texture path is not uniformly faster than global reads; the raw trials retain both results. A 48 MiB program exceeds the reported L2 capacity, but that alone does not establish that every cell is visited, that a cache is saturated, or that the device reaches a memory-bandwidth limit. No new S2 operation is measured here. Earlier S2 results retain their original inputs and correctness contracts.

7.3 What returning to the original source changes

The fresh audit returns to the user's TOM World Query Kernel 0.6.0 directory, preserving its files. Two concrete mechanisms deserve further integration. First, the original GPU sources express several phase and seam operations with exact 32-bit modular arithmetic. Their input ranges, floor-division behavior and overflow treatment must be retained when replacing wider intermediate arithmetic. Second, the original seeded artifact workflow consumes a complete ordered stream of executed EMIT records. A final receipt is insufficient to reconstruct that stream.

The seeded compiler validates dependencies and evaluates its formal tree on the host, then encodes the result and lowers literal bytes through `emit.graph`. R17's native transition port does not yet make that formal evaluator a GPU program. A useful next integration is an opt-in, lossless EMIT sink with explicit capacity and ordering, verified against the original seeded compiler and materializer. It must retain every fresh EMIT inside a fused chunk, including repeated payloads and EMIT-HALT. Its additional output work requires separate matched measurements.

The original packed key and Morton notation are also distinct. RADIX reads the canonical packed key, while lineage incorporates canonical cell identity. A derived cache may rearrange physical storage only while preserving those logical keys, indices and successor semantics. Reordering the canonical program by opcode or Morton code would change execution.

`review/R17_ORIGINAL_RETURN.md` and `review/R17_SEEDED_RUNTIME_AUDIT.md` identify the source locations and bounded next steps. The reports distinguish reused mechanisms, verified R17 behavior and capabilities still absent. The source recipe, trials, compiled audit, checks and mathematical proof remain independently inspectable in the release.

8 R16: a source-pinned native word machine

R16 imports the finite TOMAGI 1.0 execution semantics from Tom Klootwijk's World Query Kernel 0.6 package and gives them a native CUDA realization. Its Cell48 operators execute against resident State64 words. An explicit fresh-emission binding connects that machine to the inherited two ASA/NA mask stages and JK state; a separate exact rational adapter discovers affine plane crossings. These are two bounded engine profiles within the retained aTOMos formalization.

The source authorities are `src/c/tomagi.h`, `src/c/tomagi.c`, the source Python TOMAGI implementation, and `spec/TOMAGI_1_0_FORMAL_DEFINITION.md` in the supplied WQK package. The release provenance manifest pins file contents and preserves the source attribution. No license grant or independent authorship finding is inferred from a notice. The complete integration contract is `formal/WQK_INTEGRATION.md`.

The source opcode PHI advances a finite phase. It is distinct from the exact $a + b\phi$ algebra. SDF0 assigns zero; it does not calculate Euclidean distance. KLEIN retains a reduced chart and wrap parity; it does not retain the full winding lift. These precise meanings preserve compatibility without changing the inherited physical or geometric contracts.

8.1 Canonical words, address, and admission

For an integer z , define unsigned reduction $u_{32}(z) = z \bmod 2^{32}$ and signed interpretation s_{32} of that word. Thus

$$\text{wrap}_{32}(z) = s_{32}(u_{32}(z)), \quad \text{mod}_m(z) = z - m \lfloor z/m \rfloor, \quad m > 0. \quad (19)$$

Modulo addition of unsigned raw words implements the source wrapping addition without signed overflow. Where the source specifies a wider intermediate, reduction occurs after that intermediate is evaluated. Signed division in LSYS truncates toward zero; phase and tick wrap counts use floor division. They are different operations for negative inputs.

The canonical little-endian stream contains a 128-byte header and N 48-byte cells, exactly $128 + 48N$ bytes. It declares the eight-byte magic `TOMAGI1\0`, version `0x00010000`, cell size 48, state size 64, and six zero reserved words. Admission requires $N > 0$, an entry below N , opcodes in $[0, 15]$, successors below N , and strictly increasing unique unsigned keys (K_{hi}, K_{lo}) .

State64 comprises sixteen raw 32-bit words. The signed coordinate and rate fields are

$$(\rho, \theta, t, \phi, v_\rho, v_\theta, v_t, v_\phi).$$

They are followed by unsigned orientation, sheet, branch, cell, lineage, and output; residual is signed and status unsigned. Cell48 has twelve words: two key words, opcode, flags, four signed arguments, two successors, payload, and auxiliary word.

The constructor's `header_state()` retains the entire initial state. Its `entry_state()` changes only the cell index to the declared entry; that is the constructor's initial lane. Explicit `set_states()` preserves supplied raw words and resets sidecar faults, receipts, and dispatch epoch. It does not normalize an initial branch or phase before the first instruction.

With periods

$$(M_\rho, M_\theta, M_t, M_\phi) = (2^{20}, 2^{18}, 2^{14}, 2^{12}), \quad (20)$$

the pre-instruction key is the exact bit-field concatenation

$$K = \text{mod}_{M_\rho}(\rho)2^{44} + \text{mod}_{M_\theta}(\theta)2^{26} + \text{mod}_{M_t}(t)2^{12} + \text{mod}_{M_\phi}(\phi). \quad (21)$$

It is a finite address, not a metric. Physical coordinates, units, and chart identifications need an explicit application binding.

8.2 Instruction bodies and the mandatory postlude

Each row below denotes the instruction body before the shared postlude. The source write order is part of its semantics: SET and JIT1 can address any state field, including branch, status, cell, and lineage. A JIT1 target that aliases branch can modify its earlier parity assignment.

Opcode	Source body retained in R16
0 NOP	No body change.
1 SET	Write argument bits to field flags & 15.
2 JIT1	Source hash/parity from seed and pre-state; assign branch, then add the selected signed perturbation with word wrapping.
3 KIN2	Update each rate, then its position by the updated rate, modulo 2^{32} .
4 PHI	Wide signed phase addition; Euclidean reduction by 2^{12} , wrap status, selected branch and orientation action.
5 TIME	Wide signed tick addition; reduction by 2^{14} , wrap-parity branch and conditional lineage mix.
6 SDF0	Residual zero, ZERO status set, branch one.
7 CONE	Radial/angular interval maximum; wrap residual first, then test signed residual ≤ 0 .
8 SPHERE	Radial and optional cyclic-phase interval maximum; wrap residual first, then test signed residual ≤ 0 .
9 KLEIN	Reduce signed rho; odd wrap parity applies the selected theta map, phase negation, orientation and optional sheet flip.
10 RADIX	Select the declared bit of the pre-instruction key; shifts outside $[0, 63]$ refuse.
11 HINGE	If branch low bit is one, add coordinate arguments with wrapping and apply selected flips.
12 LSYS	Signed orientation/branch phase turn and rate division by $2^{\text{clamp}(arg_1, 0, 30)}$, truncating toward zero.
13 PROJECT	Assign payload to output.
14 EMIT	Assign payload, set EMIT, optionally HALT.
15 HALT	Set HALT.

The CONE/SPHERE residual wraps before its inside test. R16 preserves that finite behavior. An application wanting the unwrapped interval geometry must additionally prove a no-overflow domain. Neither opcode is a general Euclidean signed-distance field.

For PHI, for example, the exact wider intermediate is

$$r = s_{32}(\phi) + arg_0, \quad w = \lfloor r/2^{12} \rfloor, \quad \phi_{\text{body}} = r - w2^{12}. \quad (22)$$

The branch is the selected wrap-parity or half-phase bit. This is not a multiplication by the golden ratio. Likewise, KLEIN uses the parity of $\lfloor s_{32}(\rho)/2^{20} \rfloor$ and does not preserve that integer after reduction. A 32-bit lineage hash cannot recover an unbounded winding count.

Every successful body, including HALT and EMIT-HALT, then normalizes θ, t, ϕ , masks orientation and branch to one bit, and updates lineage. It does not globally normalize rho or sheet. Writing i^- for the executed cell and K^- for the pre-instruction key,

$$L^+ = \text{mix}_{32}(L_{\text{body}} \oplus \text{payload} \oplus \text{aux} \oplus K_{hi}^- \oplus \text{rotl}_{32}(K_{lo}^-, 7) \oplus b^+ \oplus i^-). \quad (23)$$

The source unsigned hash retains its exact constants and operations. If the resulting state is not halted, the normalized branch chooses a successor. With REKEY enabled, an exact lookup of the new key takes precedence on a hit; a miss sets REKEY_MISS and uses that successor. HALT takes no successor. An already halted lane executes no instruction.

The native fault sidecar reports bad cell, opcode, or RADIX shift. A refused transition publishes an error receipt and leaves State64 unchanged. This result is separate from a successful source transition. The reviewed source OpenCL HINGE uses a whole-word nonzero branch test, where C/Python use branch&1. Raw initial branch 2 exposes the difference. R16 follows the C/Python low-bit rule; agreement on already-normalized states would not detect this discrepancy.

8.3 Literal one-bit planes and immutable texture records

For 32 source words w_j , with zero padding only for missing terminal lanes, define the transport and its inverse:

$$P_b = \sum_{j=0}^{31} ((w_j \gg b) \& 1) 2^j, \quad w_j = \sum_{b=0}^{31} ((P_b \gg j) \& 1) 2^b. \quad (24)$$

The two transformations are inverse because both address the same bit (j, b) . Signs, flags, operands, and instructions are preserved. This packing changes storage order, not operator meaning or arithmetic precision.

The native upload reconstructs the header and cells on the GPU from raw integer plane textures. Decoded Cell48 records remain immutable, banked within device limits, and are read by three `tex1Dfetch<uint4>` operations per cell. The matching global-memory variant reads the same records. Element reads use integer indices and no filtering or normalized-value conversion. The canonical source, decoded program, and mutable State64/receipt arrays are separate objects.

Program-bank capacities obey device resource limits and the signed texture-index range. Integer texture access establishes use of that hardware path, not a cache-hit rate, permanent cache residency, or a speed guarantee. Memory access and execution measurements belong to the recorded evidence. This profile does not extend the lowering domain of every inherited exact expression or transcendental operator.

8.4 Fresh EMIT and actual state feedback

The versioned binding ATOMOS-TOMAGI-EMIT-FEEDBACK-R1 wraps the pure VM. Before an enabled coupled step, it copies the low and high 32-bit halves of the old hinge word q_k to raw State64 rho and v_ρ , respectively; halted or faulted lanes are skipped. This is an explicit application state mapping, not an implicit change to the pure TOMAGI transition. The VM can still run without the binding.

After one canonical instruction, a fresh receipt is eligible exactly when that instruction executed successfully and was EMIT:

$$e_{k,i} = [executed_{k,i} = 1] \wedge [opcode_{k,i} = 14] \wedge [error_{k,i} = 0]. \quad (25)$$

Each (epoch, lane) is consumed at most once. The drive word is the zero-extended 32-bit payload. PROJECT, an instruction after an earlier EMIT, an already halted lane, a refusal, or a duplicate receipt supplies no fresh event. EMIT-HALT supplies exactly one. Equal payloads in different successful EMIT epochs remain distinct events.

Let P be the present mask, $M = 2^{64} - 1$, and let the declared Boolean word body give $x = X(q_k, d_k)$.

The literal two mask stages are

$$\begin{aligned} a_i(x) &= x \& A_i \& P, & n_i(x) &= a_i(x) \& N_i, \\ T_i(x) &= \begin{cases} n_i(x), & n_i(x) \& B_i = 0, \\ 0, & \text{otherwise,} \end{cases} & y_k &= T_1(T_0(x)). \end{aligned} \quad (26)$$

Here NA is the retained mask stage, not logical complement. The declared J, K word bodies read the same old q_k and the derived y_k , then commit

$$q_{k+1} = [(J_k \& (M \oplus q_k)) \mid ((M \oplus K_k) \& q_k)] \& P. \quad (27)$$

An accepted emission toggles the separate event parity once. This fixed-chart binding leaves chart orientation zero; VM orientation remains its own state field and is not a geometric seam declaration.

Ordered native dispatches make the committed q_{k+1} available to the next enabled pre-step binding, without a whole-state host readback. The coupler's lifetime, state-generation, and expected-epoch checks detect replacement or advancement through the managed VM API. Direct DeviceView writes or low-level launches remain caller-owned operations and must not bypass this ownership while coupled execution is active. Dispatches sharing mutable lanes require one ordered stream or explicit cross-stream dependencies; host call order alone does not order distinct CUDA streams. These conditions preserve the declared execution sequence; they do not make the finite recurrence a physical measurement.

8.5 What is deliberately outside this import

The WQK finite learner's exact candidate survival, explicit ambiguity, and separate validation gates are useful design material. Its delivered candidate families remain evaluated source material in R16. They are not imported as additional CPU toy families or claimed as native GPU model search. The source compiler's `formal.evaluate` can compute an answer in Python and lower its serialized bytes to EMIT cells. Running those cells executes the materialization, not the preceding Python search or proof.

A future exact calibration profile would need to retain unresolved eligible candidates, not discard overflow or unsupported evaluation as a mismatch. Unique selection requires exactly one verified fitting candidate and none unresolved. Noise, observability, physical units, and independent observations remain part of that application's model. The retained physics chapters supply their existing physical equations; this source import supplies a concrete finite execution and event interface.

9 Exact affine event discovery and atomic hinges

The R16 profile `ATOMOS-EXACT-AFFINE-EVENTS-R1` specializes the source world-query continuation idea to rational affine trajectories and the existing editable aTOMos plane guards. The implementation is `python/atomos_events.py`, using `Engine.step_certified_crossing` in `python/atomos_hinge.py`. It is an exact host event adapter, distinct from the native finite-word VM in Section 8. It does not claim GPU analytic root discovery or dynamic GPU spatial refitting.

The reviewed source basis is the WQK package's `spec/TOM_WORLD_QUERY_KERNEL_0_4_REBUILD.md`, continuation schema, and `src/python/tom_world04r/` implementation. R16 retains the useful event/continuation structure while specifying strict rational admission, complete semantic replay, simultaneous conflict handling, and terminal-horizon behavior.

9.1 One plane definition and one immutable epoch

An epoch E contains rational start t_0 , finite rational horizon $H > t_0$, state x_0 , rates v , a finite set of unconsumed relations, and fixed nominal frame and time convention. Each scalar field declares seven integer SI dimension exponents; a rate has that field's dimension divided by time. Between events,

$$x(t) = x_0 + v(t - t_0), \quad t \in [t_0, H]. \quad (28)$$

This equation is the current epoch's conditional local prediction; an earlier accepted event replaces its remaining continuation. It does not assert that an arbitrary accelerating physical body is exactly affine.

A relation selects three length fields and binds them to the existing literal plane expression in a hinge seed. With rational normal n and offset d , the guard and its pullback are

$$\begin{aligned} g(x) &= n \cdot x - d, \\ g(x(t)) &= a(t - t_0) + c, \quad a = n \cdot v, \quad c = n \cdot x_0 - d. \end{aligned} \quad (29)$$

The adapter derives n, d from the actual plane graph; the relation does not carry an independent second guard equation. The field names, seed identity, and nominal frame are part of the binding. A nonunit normal is valid for its zero set and crossing signs, but g is then a defining function rather than a Euclidean signed distance. Unsupported nonlinear, spherical, or otherwise unrecognized guard graphs reject this profile rather than falling back to approximate root sampling.

Exact input means integers or reduced rational values. Serialized num/den limbs must be actual integers with positive denominator. Floating values, Boolean integer aliases, and lossy integer coercion are excluded. Rational reduction changes representation, not the represented value.

9.2 Complete earliest roots and exact exclusions

Each remaining relation r declares a closed active interval, inclusive scalar support intervals, and scalar equalities. Its root search window is

$$W_r = (t_0, H] \cap [\alpha_r, \beta_r]. \quad (30)$$

Thus a root at the current start is excluded, including a zero created by an atomic reset at that timestamp. There is no implicit zero-time event cascade.

When $a_r \neq 0$, the sole root is

$$\tau_r = t_0 - \frac{c_r}{a_r}. \quad (31)$$

It is admitted only if $\tau_r \in W_r$ and every support/equality gate holds exactly at $x(\tau_r)$. When $a_r = 0, c_r \neq 0$, there is no root. When $a_r = c_r = 0$, the guard is identically zero and has no isolated crossing; the result remains unresolved unless a time or gate exclusion already proves the relation irrelevant over the whole forward window.

For an affine scalar field x_j , its exact range on a closed enclosing interval $[l, u]$ is

$$I_j = [\min(x_j(l), x_j(u)), \max(x_j(l), x_j(u))]. \quad (32)$$

If I_j is disjoint from a required support interval, or does not contain a required equality value, that gate excludes the entire window. Including the open window's lower endpoint in this enclosure is conservative. Failure of this exclusion test does not assert that all gates can hold simultaneously; it preserves a possible relation. For a nonconstant guard, the subsequent exact gate test at its unique root resolves it.

Let $C(E)$ index the relations with admitted roots, after checking every unconsumed relation. If nonempty, select

$$\tau_* = \min_{r \in C(E)} \tau_r, \quad \mathcal{E}_* = \{r \in C(E) : \tau_r = \tau_*\}. \quad (33)$$

Priority and relation identity order the certificate within these exact ties. They neither move the root nor authorize partial chronological execution inside a simultaneous set. A certificate records all candidate roots as well as the complete selected earliest set. An unresolved eligible relation prevents a false completeness claim.

Proposition 9.1 (Earliest-event completeness in the affine profile). *If every eligible relation is either exactly resolved or excluded by the stated interval proof, Equation (33) contains every admissible event at the least admissible time in $(t_0, H]$.*

Proof. A nonconstant affine guard has exactly the root in Equation (31). A constant nonzero guard has none. An excluded relation cannot satisfy a required gate in its window. Every remaining isolated root is checked for window membership and all gates. Enumeration therefore neither omits an admissible root nor adds an inadmissible one. Exact rational minimum and equality then select precisely all least-time roots. The assertion covers the declared finite relation set and affine algebra only. $\square$

9.3 A crossing certificate without an epsilon step

At an admitted nonzero-slope root,

$$g(x(\tau_* + s)) = as, \quad \sigma_- = -\text{sgn}(a), \quad \sigma_+ = \text{sgn}(a). \quad (34)$$

These signs follow algebraically for $s < 0$ and $s > 0$; the runtime does not choose a numerical epsilon. The hinge adapter re-derives slope, root, point, and source seed binding before using the certified sides. Its existing chart orientation, if present in the guard convention, corrects both signs consistently; this fixed-frame crossing itself declares no reversing seam.

The post-side inside bit drives the selected lane through the actual two stored mask stages and the existing word J, K bodies. The update reads an old hinge snapshot and commits once, with event parity toggled once. Other lanes retain their declared bodies; the optional seam lane does not receive a toggle from this fixed-frame event.

If several simultaneous relations name the same hinge, they generate one pulse only when their complete origin, velocity, root, and plane-seed bindings agree. Different bindings to that hinge are a conflict. Merely sharing a timestamp is insufficient to identify two physical crossings.

At $\tau_* = H$, σ_+ is the sign of the algebraic affine extension. It does not claim an observed or evolved future interval. The horizon event yields a `terminal_state` point snapshot; it does not construct a zero-width successor affine segment. With no admitted events, the adapter advances the affine state to H and finalizes without a hinge pulse.

9.4 Atomic simultaneous state and rate updates

Let $x^- = x(\tau_*)$ and v^- denote the common pre-event snapshot. For a field with simultaneous updates, the supported rules are

$$z^+ = \begin{cases} c, & \text{all updates are set to the same value } c, \\ z^- + \sum_j \delta_j, & \text{all updates are add,} \\ z^- \oplus \delta_1 \oplus \dots \oplus \delta_m, & \text{all updates are admissible xor.} \end{cases} \quad (35)$$

XOR is limited to nonnegative dimensionless integer state and is not defined for rates. Mixed modes or unequal simultaneous set values reject. Unmentioned fields retain their pre-event values. State and rate maps are handled separately, so a new rate does not retroactively change the interval just realized.

World edits and all hinge edits are computed on clones. Only after every check succeeds are the successor epoch, fired relation IDs, hinge states, transition record, and accepted certificate published together. No failed candidate partially mutates the world. The successor at $\tau_* < H$ starts a new affine interval using x^+, v^+ ; the successor at H is terminal.

9.5 Semantic replay, finite progress, and source corrections

A hash certifies bytes, not that an asserted event follows from an epoch. The adapter regenerates the expected certificate $C(E)$ from the current immutable epoch and admits a submitted certificate $\hat{C}$ only when

$$\text{canonical}(\hat{C}) = \text{canonical}(C(E)). \quad (36)$$

This checks the root, source epoch, all candidates, complete simultaneous membership, and all gate-derived claims. Altering a root or omitting an earlier/tied relation and calculating a fresh hash does not satisfy this equality. The source profile supplies the update bodies, so a certificate cannot substitute different state edits.

An identical already accepted certificate returns a duplicate result without a second transition. The same identity with changed bytes rejects. Packed session loading reconstructs the initial profile, replays each accepted certificate semantically, and compares the derived final epoch and transitions; unpacking bytes alone is not accepted replay evidence.

Each successful event time is strictly later than the current start, and at least one previously unconsumed relation is marked fired. For m initial once-only relations there are at most m event sets, followed if necessary by a single event-free horizon finalization. A configured event-set budget may stop earlier with an explicit resource result; unresolved relations and conflicting edits stop without commit. This argument does not extend to arbitrary rearming, newly created relations, or zero-time cascades.

Three source implementation issues are corrected at this boundary. The reviewed path accepts $\tau = H$ but then tries to create a forbidden zero-length OpenSegment; R16 uses a terminal snapshot. The reviewed rational dictionary parser can truncate floating limbs when its schema has not been run; R16 requires actual integer limbs. The reviewed event application can accept a rehashed claim without replaying its full semantics; R16 recomputes the complete expected event set. These concrete corrections strengthen the finite event contract. They do not add a claim of arbitrary world prediction, a GPU analytic solver, or a new physical conservation law.

10 R16 execution evidence

The R16 build ran on the NVIDIA GeForce RTX 5070 Ti Laptop GPU, compute capability 12.0, using CUDA 12.8 and Visual Studio 2022 Build Tools. This section records the new word and event profiles. The later S2 and large-memory chapters preserve their original R15 measurements.

10.1 Independent semantics and feedback checks

All eight native CTest suites passed. The TOMAGI suite completed 28,794 assertions against the unmodified source C implementation. It covers all sixteen opcodes, all SET/JIT1 target fields, raw initial branch 2, signed extremes, wrap and division behavior, successor selection, REKEY, bank boundaries, exact upload reconstruction and fresh/sticky emission receipts. Both integer-texture and ordinary-load paths are checked.

The feedback suite passed 1,718 checks. Its three-cell literal source compiles to 272 canonical bytes. The unmodified source Python transition function generates a separate 24-step reference, including every State64 and feedback word. Both native fetch paths match every recorded word

at every step; 257 lanes also match after a complete device batch. Additional cases cover whole-word absorption, full 64-bit inputs, repeated identical fresh payloads, EMIT-HALT, sticky status, refused VM steps, duplicate receipts and managed state-generation changes.

The fixture originally numbered receipts from one while the VM begins at zero. The first feedback run exposed that discrepancy. The generator now distinguishes one-based display steps from zero-based receipt epochs; subsequent checks pass. This correction changed the fixture's expected receipt identity, not the VM transition. The retained report records the current program and expected-byte hashes.

The Python suites passed all 49 tests: 22 inherited hinge tests and 27 new event tests, including 100 independently constructed rational-root cases. A separate public-API evidence script passed 36 checks and preserves its own literal profiles, exact certificates and transitions. It confirms the turnaround roots $1/4$, $1/2$ and $3/4$, horizon-event handling, duplicate holds, and byte-exact semantic replay. Rehashed false roots, omitted earlier events, incomplete simultaneous sets and packed sessions missing a certificate are rejected.

10.2 Matched native word runs

The command-line runner executes one ordered device batch and then reads all final arrays. The paired report uses the same three-cell RADIX/EMIT program, 96 steps, three fresh-process repetitions and alternating fetch order. Each lane starts identically. "Copy32" enables the explicit emitted-word feedback profile; "pure" executes the imported VM without that binding.

Profile	Lanes	Device batch median (ms)		Host-complete median (ms)	
		Texture	Global	Texture	Global
Pure	1	0.738	0.786	83.330	83.076
Pure	4,096	0.812	0.731	82.269	79.922
Pure	262,144	5.926	5.764	127.801	126.556
Copy32	1	2.181	2.110	83.479	80.220
Copy32	4,096	2.132	2.210	79.486	83.853
Copy32	262,144	15.481	15.333	168.847	170.766

All six cases agree across fetch paths and repetitions on the full-array FNV-1a-64 digest and the exact first-lane fields. The digest is noncryptographic; these large comparisons supplement the independent word-by-word correctness suites. The largest coupled case accepts 12,582,912 fresh emissions across 25,165,824 VM instructions. A separate one-lane runner result matches the independent 24-step Python fixture.

Device time uses CUDA events over the ordered batch. Host-complete time includes cold setup, allocations, upload, dispatch, final readback and digest; it excludes JSON output. The source report preserves every trial and the separate setup, batch-wall and readback times. No per-step host readback is included in the batch. The tiny immutable program is shared by many lanes; this does not measure a large program working set or a saturated memory bus. Global loads have the lower device median in four of six cases. These are finite instruction workloads, with no corresponding S2 query output.

10.3 Reproducible release inputs

The new executable is `wqk_run`; its inspectable source is `tools/wqk_run.cpp`. `tools/measure_wqk.py` creates the paired report. The standalone CMake option `ATOMOS_SPATIAL=OFF` omits inherited S2-dependent targets while retaining the VM, feedback and their checks. The full build retains all previous native profiles.

11 An exact seed and event-driven hinge engine

R15 selects an engine core with exact algebraic operands, finite expression graphs, binary guard results and delayed state feedback. Geometry, branch history and chart orientation are separately typed parts of its state. The R14 electromagnetic, mechanical, thermal, clock and observation equations remain the physical models reached through this interface. A geometric branch does not itself supply force, energy, elapsed proper time or a detector reading.

11.1 The golden-ratio pair is an exact arithmetic domain

Let $\phi = (1 + \sqrt{5})/2$ and represent $a + b\phi$ by $\text{Phi}(a, b)$ with $a, b \in \mathbb{Q}$. Its representation is unique because ϕ is irrational. The defining identity $\phi^2 = \phi + 1$ gives

$$\begin{aligned} (a, b) + (c, d) &= (a + c, b + d), \\ (a, b)(c, d) &= (ac + bd, ad + bc + bd), \\ \phi(a, b) &= (b, a + b), & \phi^{-1}(a, b) &= (b - a, a). \end{aligned} \quad (37)$$

The last two maps are exact inverses. For a nonzero pair its field inverse is

$$(a + b\phi)^{-1} = \frac{(a + b) - b\phi}{a^2 + ab - b^2}. \quad (38)$$

The denominator is the product with the conjugate $a + b(1 - \phi)$ and cannot vanish for a nonzero rational pair. These are algebraic identities; a rational approximation such as 1.618... is not substituted for ϕ .

For exact sign selection put $A = a + b/2$, $B = b/2$, so the value is $A + B\sqrt{5}$. If $B = 0$, use the sign of A . If $A = 0$, use the sign of B . If A, B have the same sign, that is the answer. Otherwise

$$\text{sgn}(A + B\sqrt{5}) = \text{sgn}(A) \text{sgn}(A^2 - 5B^2). \quad (39)$$

The last comparison is rational and is justified by comparing two positive magnitudes before squaring. Thus this domain admits exact comparisons without evaluating a rounded square root. Transcendental expressions outside the selected domain retain the inherited unresolved-result convention.

11.2 One-bit planes retain arithmetic carries and the whole expression

Each rational coefficient has an integer numerator and a positive denominator; canonical reduction uses their greatest common divisor. Integer magnitudes, signs, lengths, denominators, operator tags and references all belong to the source. For a group of w serialized operands, their bit plane k can be packed as

$$B_k = \sum_{j=0}^{w-1} b_{j,k} 2^j, \quad b_{j,k} = (B_k \gg j) \& 1. \quad (40)$$

Declared lengths and padding make this a bijection on the finite payload. One-bit encoding changes the layout, not the number of mathematical degrees of freedom. A variable-length exact coefficient is not reduced to one scalar bit or one fixed-width word.

For nonnegative binary integers the full adder is explicitly

$$\begin{aligned} c_0 &= 0, & s_i &= a_i \oplus b_i \oplus c_i, \\ c_{i+1} &= (a_i \wedge b_i) \vee [c_i \wedge (a_i \oplus b_i)]. \end{aligned} \quad (41)$$

The final carry is retained. Subtraction uses an explicit signed-integer operation or sufficient sign extension and two's-complement addition, with the complete signed result retained. Fixed logical words still use their declared modulo- 2^w arithmetic, a different type. The pair update $(a, b) \mapsto (b, a + b)$ therefore needs addition with carry; a shift or XOR alone does not implement multiplication by ϕ . Numerator/denominator arithmetic likewise retains all carries and any required additional limbs.

The seed's finite directed acyclic graph stores literal and input nodes, selected arithmetic, affine points and guard bodies. A literal is an exact operand; an input node binds one immutable sample. Operand order and grouping are part of the graph, including sequential arithmetic folds. Lazy branches evaluate only their demanded operands. Shared subexpressions may be cached under the same operand identities without changing the denotation.

The source's literal "zero-order" convention is left-to-right sequencing with uniform operator precedence. For the explicit sequence $(v_0, (\circ_1, v_1), \dots, (\circ_m, v_m))$, its body is

$$a_0 = v_0, \quad a_i = a_{i-1} \circ_i v_i \quad (1 \leq i \leq m), \quad \text{Fold} = a_m. \quad (42)$$

Every intermediate value is type checked. Explicitly grouped subexpressions are operand nodes with their own bodies. Thus an ungrouped addition followed by multiplication is folded in that order; a compiler must not impose the usual multiplication-before-addition parse on this source convention. The stored AST records the selected grouping unambiguously.

"Zero-order" here says nothing about derivative order, absence of dynamics or $O(1)$ computational complexity. Work depends on visited graph nodes, operand bit lengths, comparisons and generated descendants. An exact expression may still be expensive to evaluate. Independently selected physical evolution retains its actual body; the fold does not replace the inherited ROOT, ODE or DIFF domains when those are requested.

11.3 Guard values, boundary ownership and an accepted event

A geometric guard is a signed defining function g whose selected inside is $g < 0$. A plane may use $g(x) = n \cdot x - c$, with $n \neq 0$; a sphere may use

$$g_{\text{sphere}}(x) = \|x - a\|^2 - R^2, \quad R > 0. \quad (43)$$

The latter is a squared polynomial guard, not a distance in metres. It has the sphere's exact sign while avoiding a square root in that predicate. The sign theorem for rational ϕ pairs applies when all active operands belong to that domain.

Let $\kappa \in \{0, 1\}$ record the current local coorientation and set $g_c = (-1)^\kappa g$. The guard bit is $b_g = [g_c < 0]$. Equality is a separate BOUNDARY result by default, with no transition commit. An application may explicitly assign boundary ownership to inside or outside. An unavailable demanded sample is MISSING_INPUT; an unsupported demanded operator is UNKNOWN. Neither status is a false guard.

The event record binds an immutable identity and content to the selected sample, seam and event-order inputs. Physical acquisition additionally supplies its declared timestamp and availability binding; a logical sequence counter alone is not elapsed physical time. A new accepted event supplies a hinge pulse $h \in \{0, 1\}$: it can be a side change after a defined nonboundary baseline, an explicitly declared crossing, or a declared nontrivial seam transition. An initial sample can establish the baseline without a pulse. A repeated identical identity is not a new pulse; reusing an identity with changed content is invalid. Continuous trajectory crossings not represented by this acquisition/event policy are not proved detected merely because sampled endpoints have the same sign.

11.4 Every hinge crossing and a reversing seam are different bits

Let p be hinge parity, κ coorientation parity, and $\epsilon \in \{0, 1\}$ identify whether the admitted seam reverses the selected local sign. The next parities are

$$p^+ = p \oplus h, \quad \kappa^+ = \kappa \oplus (h \wedge \epsilon). \quad (44)$$

Thus every accepted hinge pulse changes p , whereas an orientation-preserving hinge leaves κ unchanged. The determinant -1 of the coefficient-pair map in equation (37) concerns an algebraic basis; it is not by itself a physical chart-orientation theorem.

The inherited two independently stored ASA/NA mask sets act in their declared order and produce the word y from the common old snapshot. Define $b_{\text{lat}} = \text{bit}(y, d)$ at a reserved drive lane d . It is this post-mask bit, rather than an unconditionally substituted raw guard, that the selected latch records. The selected default drive adapter sets

$$J_d = h \wedge b_{\text{lat}}, \quad K_d = h \wedge \neg b_{\text{lat}}. \quad (45)$$

On a distinct optional seam lane s set $J_s = K_s = h \wedge \epsilon$. Every other lane retains its base J, K body. Assemble the word masks and use the same synchronous JK identity on all lanes:

$$q^+ = (J^* \wedge \neg_w q) \vee (\neg_w K^* \wedge q). \quad (46)$$

Equation (45) holds when $h = 0$ and sets the drive lane to b_{lat} when $h = 1$; the seam lane toggles exactly on a reversing pulse. Custom word equations must retain this body to claim this particular drive adapter; an arbitrary user-supplied J, K expression need not implement it. Disjoint allocation prevents either adapter from overwriting the other's semantics. This is local lane preservation for one step: later base expressions may read those changed bits, so it is not a global noninterference claim.

Guard, mask, parity, chart-lift, consumed-event and word outputs are computed from one old state and accepted record, then committed together. A seam's new raw guard must be paired with the matching new coorientation before comparing physical sides; applying old κ to a new reversed guard would invent a crossing. No partial state is committed for an unresolved demanded result. Physical time and an external plant can continue while a digital event waits.

11.5 Seam compatibility, winding and a graph beyond the branch tree

For overlapping charts U_i, U_j , declare an invertible transition T_{ij} and a sign $\sigma_{ij} = (-1)^{\epsilon_{ij}}$. The selected exact local guard compatibility is

$$g_j(T_{ij}x) = \sigma_{ij}g_i(x). \quad (47)$$

For general defining functions a positive factor could preserve sign, but it must then be an explicit different contract; exact SDF transport has its stronger metric conditions in Section 12. An input record asserting this scalar equality does not by itself prove that both arguments denote the same physical point. That premise is supplied by the actual chart map and coordinate binding.

On a triple overlap the maps and signs obey

$$T_{ik} = T_{jk} \circ T_{ij}, \quad \sigma_{ik} = \sigma_{jk}\sigma_{ij}. \quad (48)$$

When using lifted periodic coordinates these equalities are interpreted with the explicitly retained deck/winding action, not by dropping a full turn. For example, $\vartheta = \bar{\vartheta} + 2\pi w$ with $w \in \mathbb{Z}$ and $\bar{\vartheta} \in [0, 2\pi)$ retains the complete angle. Normalizing its address updates w ; the residue alone is insufficient to invert the lift. The inherited half-turn and reflective-Klein chart actions remain separately named maps.

For a closed seam walk C , the product $\prod_{(i,j) \in C} \sigma_{ij}$ is its sign holonomy. If it is -1 , a globally single-valued choice of local signs cannot trivialize that loop. This is a statement about the specified chart/sign bundle, not a proof that an arbitrary embedded solid has a globally defined signed distance or a particular physical Klein-bottle geometry. A branch tree alone has no cycle; repeated bit flips along a tree cannot establish such holonomy. The engine therefore distinguishes its expansion tree from an additional seam graph containing declared return and cross edges. Every return is an explicit state transition at a later accepted event. It does not impose a self-consistent eigenvector or an instantaneous circular AST.

11.6 CUDA realization: immutable source and derived VRAM epochs

The native realization has two kinds of device data. The authoritative source is the immutable packed operator/operand payload and its version identity. Expansion products, bounds, classifications and work queues are derived VRAM data. Their key includes source identity, state/event epoch, parameters and the represented interval. Rebuilding a cache is permitted only if the same declared semantic result or a certified enclosure is preserved.

For the raw-word path, a linear texture object exposes unsigned integer elements and uses `tex1Dfetch` with integer indices and `cudaReadModeElementType`. No filtered interpolation, normalization, color conversion or texture-coordinate wrapping is a semantic operation on the source bits. Bounds, alignment and resource size are checked explicitly. The relevant descriptor semantics are documented in [HC1: NVIDIA CUDA Runtime API, Texture Object Management](#).

A producer may expand the read-only seed into a separate output allocation. A completed kernel and explicit stream/event dependency precede a consumer reading that output as a texture in the next epoch. No kernel reads through the texture path from storage that it concurrently modifies through global writes. The reason is concrete: NVIDIA specifies that the texture cache is not coherent with same-launch global writes; see [HC2: CUDA C++ Best Practices Guide, Texture Memory](#). Double buffering or another explicit ownership schedule can satisfy this contract. A device pointer or a launch-order assumption across unsynchronized streams is not a completed dependency.

The device lowering declares its supported subset. A kernel performing native bounded-width integer word operations is not thereby a complete arbitrary-precision rational- ϕ evaluator. A kernel consuming projected IEEE coordinates is not thereby the exact AST evaluator. Each supported predicate either executes its exact body, or returns a proved enclosure/decision with a defined exact fallback for unresolved cases. Unsupported bodies preserve the source and route to a supported evaluator; they are not silently rounded into different semantics. Correctness checks establish the tested subset, and benchmarks separately report its time, transfers and resource use.

Hardware chooses texture-cache residency, replacement and scheduling. The formal contract guarantees neither a cache hit nor a speed advantage over ordinary global loads. Seed expansion in VRAM is a realizable dataflow and implementation choice whose capacity and throughput are measured for the actual GPU. It does not mean an infinite tree resides in finite memory or that one-bit storage alone computes its operators. Self-reference remains the explicit delayed event/state recurrence above.

12 Exact spatial templates without a prescribed raster

This new spherical profile supplies coordinate geometry and direct primitive evaluation to the hinge engine. It is distinct from the inherited horizontal chart $\rho = \log(H/r_0)$, $\theta = \text{atan2}(N, E)$ and from the quantized WANT.W64 address. An expression-template table stores complete functions and parameters. It is not a table of sampled distances whose interpolant is silently declared exact.

12.1 Spherical logarithmic coordinates retain the Euclidean metric

For a fixed positive reference length r_0 , let $\rho \in \mathbb{R}$, polar angle $0 < \theta < \pi$, and azimuth φ modulo 2π . In fixed Cartesian axes,

$$r = r_0 e^\rho, \quad x = o + r e_r, \quad e_r = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta). \quad (49)$$

The orthonormal tangents are $e_\theta = (\cos \theta \cos \varphi, \cos \theta \sin \varphi, -\sin \theta)$ and $e_\varphi = (-\sin \varphi, \cos \varphi, 0)$. Differentiation gives the metric and volume element

$$ds^2 = r^2(d\rho^2 + d\theta^2 + \sin^2 \theta d\varphi^2), \quad dV = r^3 \sin \theta d\rho d\theta d\varphi. \quad (50)$$

Thus equal coordinate increments do not have equal physical length. The standard orthogonal-coordinate gradient/divergence construction is given in [SC1: MIT Mathematical Methods, lecture 14](#); substitution $r = r_0 e^\rho$ gives the following explicit bodies for a smooth scalar f :

$$\begin{aligned} \nabla f &= \frac{1}{r} \left(e_r f_\rho + e_\theta f_\theta + e_\varphi \frac{f_\varphi}{\sin \theta} \right), \\ \nabla^2 f &= \frac{1}{r^2} \left(f_{\rho\rho} + f_\rho + f_{\theta\theta} + \cot \theta f_\theta + \csc^2 \theta f_{\varphi\varphi} \right). \end{aligned} \quad (51)$$

These factors are required if a selected gravity, diffusion or field equation is expressed in this chart. Removing them changes the physical equation. The origin is outside the logarithmic chart; the polar-axis and angular-seam coordinate ambiguities use an explicit alternate chart or absent-chart status. They are not physical holes. Full azimuth winding is retained separately from a periodic address.

For twice differentiable coordinate functions of time and a translating origin $o(t)$, with the Cartesian axes held fixed, direct kinematics are

$$\begin{aligned} \dot{x} - \dot{o} &= r(\dot{\rho} e_r + \dot{\theta} e_\theta + \sin \theta \dot{\varphi} e_\varphi), \\ \ddot{x} - \ddot{o} &= r(a_\rho e_r + a_\theta e_\theta + a_\varphi e_\varphi), \\ a_\rho &= \ddot{\rho} + \dot{\rho}^2 - \dot{\theta}^2 - \sin^2 \theta \dot{\varphi}^2, \\ a_\theta &= \ddot{\theta} + 2\dot{\rho}\dot{\theta} - \sin \theta \cos \theta \dot{\varphi}^2, \\ a_\varphi &= \sin \theta \ddot{\varphi} + 2\dot{\rho} \sin \theta \dot{\varphi} + 2 \cos \theta \dot{\theta} \dot{\varphi}. \end{aligned} \quad (52)$$

These are derivatives of the same seeded trajectory, rather than finite differences of rendered points. Rotating axes additionally require R14's frame-velocity and acceleration terms. A specified trajectory supplies kinematics; its physical realization still needs the corresponding force and energy model. Exact expression differentiation does not identify an unknown trajectory from an uncalibrated observation.

12.2 Distance, defining function and occupancy remain different outputs

For a declared solid A with boundary ∂A , its signed Euclidean distance is the distance to ∂A , negative inside, positive outside and zero on the boundary. Where differentiable away from exceptional closest- point sets, its gradient has unit norm. In the chart above this becomes

$$f_\rho^2 + f_\theta^2 + \csc^2 \theta f_\varphi^2 = r^2. \quad (53)$$

The metric-weighted equation is a necessary local property; boundary data, sign and the selected global distance definition are still required. The unweighted expression $f_\rho^2 + f_\theta^2 + f_\varphi^2 = 1$ is not the Euclidean SDF condition in logarithmic spherical coordinates.

A sphere has the exact direct SDF $f(x) = \|x - a\| - R$. Its squared guard in equation (43) has the same sign but different units and gradient. A normalized plane has $f(x) = n \cdot x - c$, $\|n\| = 1$. The

retained segment and finite-cone templates likewise provide explicit piecewise distance bodies under their own domains. An absolute-value pyramid inequality is a valid defining predicate when its region is intended; it is not automatically a Euclidean distance. A scalar sphere formula cannot be replaced by an unspecified angular LUT and an assertion of identical shape.

Occupancy is $[f < 0]$ with its declared boundary policy. Boolean combinations of defining functions retain their predicate logic. For example, $\min(f_A, f_B) < 0$ describes a union, but inside overlapping solids that minimum need not be the distance to the union's exterior boundary. Therefore the engine exposes whether a returned scalar is a true distance, a distance bound, a defining guard, or only a Boolean. A proof requiring a Lipschitz constant uses the actual selected function, not its informal SDF label.

12.3 Rigid and uniform-scale transport is exactly solvable

Let f_0 be a signed distance in material coordinates. For a translation $a(t)$, rotation $R(t) \in SO(3)$ and uniform scale $\lambda(t) > 0$, set

$$x = a + \lambda R X, \quad X = R^T(x - a)/\lambda, \quad f_t(x) = \lambda f_0(X). \quad (54)$$

For every pair X, Y , the transform multiplies their Euclidean separation by λ . Taking the infimum over the original boundary proves that (54) is exactly the transformed signed distance. On its differentiable domain, $\nabla_x f_t = R \nabla_X f_0$, preserving unit normal length.

Write $\dot{R} = [\omega]_\times R$ in world components. The velocity of a material point, expressed at its current position, is

$$v(x, t) = \dot{a} + \omega \times (x - a) + (\dot{\lambda}/\lambda)(x - a). \quad (55)$$

Differentiating $f_t(a + \lambda R X) = \lambda f_0(X)$ at fixed X gives

$$\partial_t f_t + v \cdot \nabla_x f_t = (\dot{\lambda}/\lambda) f_t. \quad (56)$$

Rigid transport has $\lambda = 1$ and zero material derivative. On the moving boundary $f_t = 0$, the same relation gives its normal velocity. Corners and medial sets keep their nondifferentiability status; a unit normal is not invented where the selected derivative has no value.

For an arbitrary diffeomorphism Φ , the pullback guard $g_t = f_0 \circ \Phi^{-1}$ preserves the transformed boundary and inside sign, but

$$\nabla_x g_t = (D\Phi)^{-T} \nabla_X f_0 \quad (57)$$

generally has nonunit length. Dividing by this local length does not in general recover the global Euclidean distance. If the full transformation is globally bi-Lipschitz with constants $0 < \ell \leq L$, the actual transformed distance satisfies the useful bound

$$\ell |f_0(X)| \leq |f_t(\Phi(X))| \leq L |f_0(X)|. \quad (58)$$

This follows by taking the boundary infimum of the two distance inequalities. Local Jacobian values alone do not establish those global constants on an arbitrary domain. A deformation or its inverse supplied only through an ODE retains that solution/inversion obligation.

12.4 Direct template evaluation has a precise computational scope

A template-table entry contains an identifier, complete expression body, parameters, domains, frame/time binding and declared output type. At supplied x, t the selected sphere, plane, affine-coordinate or admitted algebraic guard can be evaluated directly. Their derivatives can be supplied as exact bodies. No spatial raster or marching search is required for that selected evaluation. This is the useful meaning of a direct or “query-free” primitive profile; it does not remove the supplied point/time or the work of evaluating its expression.

Finding a first ray hit instead asks for a selected root of $f(o + \tau d) = 0$. Finding a general nearest feature asks for a minimum with its domain and selection conditions. A closed primitive may supply a finite formula; an arbitrary composite, deformation or trajectory may require root selection, a search, or an unresolved result. Exact encoding does not turn these different tasks into one constant-cost lookup. A finite dictionary of templates is not a dictionary containing every possible continuous shape.

12.5 PSI excludes a certified descendant set, not future events

Let N identify a branch and its declared descendants $\mathcal{D}_N$ over the time interval I . Here descendant points include every point of a represented primitive's extent, not only one origin per instance. Its bound must assert

$$\forall D \in \mathcal{D}_N, \forall t \in I : \|P_D(t) - P_N(t)\| \leq R_N(t). \quad (59)$$

Primitive extent, transformed children, motion and any represented uncertainty must all be included. A center-only bound is insufficient for a finite solid. For a 1-Lipschitz query guard f_t with excluded side $f_t > 0$, the condition

$$f_t(P_N(t)) > R_N(t) \quad (60)$$

implies $f_t(P_D(t)) > 0$ for every enclosed point: subtract the distance bound from the center value. For a whole-interval exclusion the condition must hold throughout I , or follow from a common interval lower bound. Equality is not excluded. For Lipschitz constant L_f , replace the right side by $L_f R_N$. If evaluation returns $\underline{f}_N \leq f_t(P_N)$, a certified decision uses $\underline{f}_N > L_f R_N$, including all additional margins in compatible units.

One concrete slab version is

$$|n \cdot P_N - c| > w/2 + R_N + d_N + \delta_N, \quad \|n\| = 1, \quad (61)$$

where R_N encloses projected descendants, d_N bounds their projected deformation and δ_N bounds remaining projected uncertainty. It is the retained corrected PSI proof: every descendant, not merely an undeformed parent hull, is on the excluded side. Monotone-looking branch indices or a positive depth counter cannot replace the quantified enclosure.

A finite AABB hierarchy realizes another enclosure. Each parent contains all child bounds; leaf bounds include their actual points/primitive extents. After motion, changed leaves are refitted and their ancestors recomputed, or a previously certified all-time bound is used. Geometry and bounds carry the same epoch before pruning; mixed old/new bounds do not establish enclosure. A self-referential return that changes the root or transforms descendants starts a new bound evaluation unless the old certificate explicitly covered that return. An infinite descendant family requires a proved finite bound or cannot be pruned by that finite-radius certificate.

Pruning here removes only a proved irrelevant geometric evaluation for a specified query and epoch. It does not skip accepted hinge events, JK updates, clock progression, winding or future seam returns. Diffraction, thermal coupling, force or sensor contributions require their R14 error/energy obligations before they are omitted from a physical model. Keeping those obligations preserves the distinction between a fast spatial engine and a proved physical simulation.

13 Implemented native profiles and measured scope

The new implementation has three separately declared numeric domains. The Python reference uses arbitrary-precision rational coefficients in $\mathbb{Q}(\phi)$. The native exact program uses bounded integer coefficients in $\mathbb{Z}[\phi]$. The spatial index preserves supplied binary64 words, uses conservative geometric filters and exact normalized-direction membership. The hybrid ID-list path uses official S2 robust predicates for final decisions; the resident count path certifies definite decisions

on the GPU and exposes unresolved queries for selected complete S2 host resolution. None of these statements equates a finite input word with an unknown physical measurement.

13.1 Exact operator words on the GPU

The native profile ATOMOS-EXACT-GPU-ZPHI31-R1 admits

$$z = a + b\phi, \quad a, b \in \mathbb{Z}, \quad |a|, |b| \leq B = 2^{30} - 1.$$

It carries at most 128 expression nodes and 128 input slots, a frame identity, and seven signed SI dimension exponents. The supported bodies are literal, input, addition, subtraction, multiplication, multiplication by ϕ , and the plane defining function $n \cdot p - c$. The root dependency closure is evaluated in dependency order. This is a bounded native subset, not a complete XOP parser or arbitrary rational/transcendental evaluator.

For multiplication, the largest temporary coefficient sum is at most $3B^2$, which is below $2^{63} - 1$. Every retained coefficient is checked against B . Exact sign is obtained from $2z = A + b\sqrt{5}$, $A = 2a + b$. When A and b have opposite signs, unsigned comparison of A^2 with $5b^2$ determines the sign; $|A| \leq 3B$ makes both squares fit in 64 unsigned bits. Thus no floating projection enters these arithmetic decisions. A result outside the declared coefficient domain returns unresolved instead of wrapping or rounding.

Canonical 32-bit operator words are transposed in complete 32×32 bit-plane blocks. An integer texture supplies the packed words to a native inverse-transpose kernel. After expansion completes, a separate immutable texture exposes decoded operator words to the evaluator. A readback through that texture is compared with the original canonical payload. Packing preserves every bit; a transpose by itself is not a compression-ratio guarantee.

The native integration example uses $g(x) = \phi x$ as a defining function in one declared frame. Sampled endpoint nonzero sign changes produce explicitly numbered events; this policy does not claim to recover every continuous crossing between observations. There is no chart reversal in this fixed-frame example. Its explicitly selected positive-side drive is $d = \mathbf{1}\{g > 0\}$, distinct from an inside bit defined with $g < 0$. The adapter uses $x = d$, ordered mask sets and $J = y$, $K = \neg y$. Boundary and unresolved outputs hold state. Repeating an event does not toggle either parity again. This connects evaluated operator words to persistent native hinge state across an explicit host synchronization boundary. This example retains that host adapter; the separate resident count-to-word and bounded integer recurrence profiles below execute delayed feedback on the device. None evaluates the complete arbitrary-expression XOP catalog on the GPU.

13.2 Conservative spherical texture filtering

Each GPU hierarchy node contains outward-rounded bounds and an escape link; traversal is stackless. Raw integer texture loads reconstruct point words. Binary32 projections are used only for the broad-phase squared chord-distance test. Let $u = 2^{-24}$ and $R^2 \in [0, 4]$. For accepted unit-direction seeds, outward bounds, round-to-nearest and the declared non-fused arithmetic, the documented leaf and node overestimate bounds are respectively below $185u$ and $161u$. Both fit within a conservative $256u$ envelope.

The implemented candidate threshold rounds the radius upward and adds $2^{-14} = 1024u$. Even the addition's rounding leaves more than $1019u$ above the authoritative radius. A true member therefore survives this filter under the stated premises. The CPU then applies the shared exact normalized- direction predicate. A candidate-buffer overflow invokes a complete host query; it never truncates the answer. The detailed inequalities and resource limits are in `formal/NATIVE_GPU.md`.

The preserved original point bytes and outward node cache are distinct. No rounded filtered point

becomes an updated source root. The present spatial object is immutable; a source change requires a new build. The timed native radius path includes query uploads, launches, synchronization, candidate readback, exact refinement and sorted host-visible output. The hybrid pipeline uses up to 20 persistent CPU participants for refinement. An overflow query is rerun completely on the host index. That work and its thread participation are recorded with the pipeline timing.

13.3 Resident query epochs and device feedback

The count profile uploads immutable source words, outward hierarchy bounds and a query table once per object. Repeated kernels return the two integers (I, U): definite inside and unresolved counts. The binary64 distance body has a proved absolute error below $256 \cdot 2^{-52}$; the wider 2^{-40} threshold margin supplies definite inside/outside certificates. Complete traversal gives $I \leq \text{Count} \leq I + U$. Only unresolved queries need the selected complete host resolver. A count-only result does not materialize identities and must be compared with the same cardinality output.

The selected feedback binding reads the old word's low bit to choose one of two immutable queries. A following kernel consumes a resolved summary and executes the configured two ASA/NA stages and synchronous JK update. Its declared initialization pulse and subsequent drive-change pulses have been checked over ten device epochs; unresolved inputs hold state, replay holds, and stale epochs reject. The chart is fixed. This path performs the query selection and state commit without per-epoch host data readback.

A separate working-set executable uses exact bounded integer ϕ operators, old-word operands and ordered masks with persistent global ping-pong state. It reconstructs immutable records from bit planes, compares integer texture fetches with ordinary global loads, and visits every record in streaming and coprime affine-permuted epochs. The quick check passed four working-set sizes. Completed sweep sizes, full-run validation and timings belong to the following resident chapter's generated results; no larger completed size follows from allocation alone. This is an operator/memory workload, not an S2 query test.

13.4 Hardware texture execution

The measured machine uses an RTX 5070 Ti Laptop GPU, 12,227 MiB reported VRAM, CUDA 12.8, driver 591.59 and an Intel Core Ultra 7 255HX with 20 logical processors. The Release build targets compute capability 12.0. Host predicates use strict floating-point compilation; CUDA filters disable fused multiply-add.

The initial validated native build (CUDA source at commit 9cfe094) contains TLD.LZ in the texture specialization and LDG.E in the matching global-memory specialization. Nsight Compute recorded 169,426 texture-load sectors for a correctness-workload launch. These are direct compiled and executed texture-path observations. They do not measure a cache hit ratio or establish persistent cache residency. The profiler's duration is kept separate from unprofiled benchmark timing.

13.5 Correctness and S2 comparison

The exact Python suite contains 22 passing checks. The native spatial suite reports 37,837 passing checks against independent exhaustive robust-predicate queries, including exact nearest ordering and radius membership. The native texture suite compares 512 radius queries across texture, ordinary GPU reads and exact host results, and exercises overflow, immutable event payload, duplicate/stale events, Boolean seam flags and rounding-mode rejection. The native exact-program, operator-to-hinge, resident count and selected feedback checks are recorded in the release's validation report. All six native test targets passed in the extended build. The exact Python suite is

reported separately from those native targets.

Official S2 is pinned to commit `079611b654ad89afd9c3c3a1796d64bdd6a6b340`. Both implementations use the same fixed point words, identities and inclusive radius inputs. S2's conservative distance query is refined with its exact predicate. A persistent worker pool and reusable S2 query objects remove timed thread creation. All CPU configurations (1, 4 and 20 configured threads), every sample and the matching global-memory GPU path are retained in the JSON report.

All 36 measured workload answers matched. The CPU BVH was faster than the best tested S2 CPU configuration in 36/36 cases, with observed ratios from 1.43 to 7.86. The complete texture pipeline was faster than that S2 configuration in 22/36 cases. Ordinary GPU reads were faster than texture reads in 21/36 cases. These are measured median results for the declared radius workload and hardware, not a universal ranking.

Five timed trials follow a warmup for each method. The table reports complete host-visible milliseconds. S2 and BVH columns use the best of the recorded 1/4/20-thread configurations; every configuration remains in the JSON report. An asterisk marks a capped query count, whose requested count is retained in the machine-readable report. A ratio above one favors the texture pipeline.

Data	Points	Queries	S2	BVH	Texture	Ratio
uniform	1024	1	0.0010	0.0005	0.0928	0.01
uniform	1024	256	0.1920	0.0759	0.1940	0.99
uniform	1024	4096	1.3882	0.4231	0.3779	3.67
uniform	65536	1	0.0010	0.0007	0.1226	0.01
uniform	65536	256	0.2305	0.0970	0.2973	0.78
uniform	65536	4096	1.3702	0.6068	0.5498	2.49
uniform	262144	1	0.0016	0.0010	0.1399	0.01
uniform	262144	256	0.2361	0.0915	0.2961	0.80
uniform	262144	4096	1.6846	0.5962	0.5869	2.87
clustered	1024	1	0.0057	0.0018	0.1439	0.04
clustered	1024	256	0.2161	0.0881	0.2686	0.80
clustered	1024	4096	2.4198	0.8359	1.2282	1.97
clustered	65536	1	0.4404	0.1562	0.3044	1.45
clustered	65536	256	3.5139	1.5114	1.8005	1.95
clustered	65536	4096	43.7240	18.7162	17.7211	2.47
clustered	262144	1	0.1955	0.0665	0.2659	0.74
clustered	262144	256	4.8300	1.9548	2.6720	1.81
clustered	262144	4096	58.2124	25.6991	24.5095	2.38
great circle	1024	1	0.0035	0.0010	0.1694	0.02
great circle	1024	256	0.1899	0.0690	0.3241	0.59
great circle	1024	4096	1.9783	0.5231	1.4359	1.38
great circle	65536	1	0.0247	0.0062	0.1562	0.16
great circle	65536	256	1.0354	0.3826	0.8049	1.29
great circle	65536	4096	7.3944	2.6444	3.0526	2.42
great circle	262144	1	0.0683	0.0185	0.2144	0.32
great circle	262144	256	1.4311	0.5496	0.9787	1.46
great circle	262144	4096	15.6874	4.9170	5.4164	2.90
duplicates	1024	1	0.0151	0.0044	0.1572	0.10
duplicates	1024	256	0.5013	0.1777	0.4475	1.12
duplicates	1024	1953*	1.6740	0.5596	1.1451	1.46
duplicates	65536	1	2.2128	0.3022	0.5615	3.94

Data	Points	Queries	S2	BVH	Texture	Ratio
duplicates	65536	30*	3.4573	0.8774	1.4016	2.47
duplicates	65536	30*	3.3480	0.8067	1.5072	2.22
duplicates	262144	1	13.1645	1.6747	2.3979	5.49
duplicates	262144	7*	13.8840	2.2573	2.7745	5.00
duplicates	262144	7*	14.0007	2.0574	2.7055	5.17

The first complete report is retained as `review/s2_full.json`. The current report includes the subsequent bounded-overflow and parallel exact-refinement optimization; no answer semantics were changed.

Construction and the first device initialization are reported separately. Per-query allocations may be retained for reuse. The reported static resident byte count excludes later query and hinge scratch allocations; it is not peak VRAM. Workloads with duplicate points explicitly cap total potential output at one million identities; requested and actual query counts are both recorded. No timing from a capped workload is presented as if all requested queries ran.

The ID-returning comparison above covers spherical point radius queries. Dynamic refits, query-coherence studies for that spatial workload, nearest and region throughput, full log-spherical expression lowering and the complete S2 feature set remain outside this measurement. The separate resident count and word-permutation experiments use their own contracts and do not fill those missing comparisons. Broader engine superiority therefore remains unestablished. The experiment measures hierarchy traversal, batching, memory access and refinement. It does not isolate a performance benefit from ϕ arithmetic, Klein topology or log encoding; those retained formal mechanisms require their own matched application measurements.

14 Resident count predicates and delayed device feedback

The resident profile keeps point seeds, a derived hierarchy and a query batch in device memory across selected launches. Its output is a count and a resolution status. This is a different interface from returning every point identity; comparisons must use the same requested output. A separate working-set experiment evaluates delayed exact word feedback. Its timing does not by itself measure a spherical-query improvement.

14.1 The exact distance target and a bounded binary64 body

For the original finite binary64 triples p, q accepted by the pinned S2 unit-length predicate, define

$$P = p/\|p\|, \quad Q = q/\|q\|, \quad D = \|P - Q\|^2, \quad 0 \leq D \leq 4. \quad (62)$$

The supplied binary64 radius $R^2 \in [0, 4]$ denotes its exact dyadic value; membership is $D \leq R^2$. The stored triples are not silently assumed to have exact norm one. This normalized-direction meaning is the one declared by the official [S2 distance predicate](#).

Let $\varepsilon = 2^{-52}$ and assume strict binary64 round-to-nearest, gradual underflow and the displayed non-fused operation sequence. S2's computed norm tolerance is 5ε . Its three-square evaluation error gives $|||p||^2 - 1| < 9\varepsilon$, hence $|||p|| - 1| < 5\varepsilon$ and the conservative coordinate bound $|p_i - P_i| < 8\varepsilon$. Apply the same bound to q .

The direct device body is

$$d_i = \text{RN}(p_i - q_i), \quad s_i = \text{RN}(d_i^2), \quad \widehat{D} = \text{RN}(\text{RN}(s_1 + s_2) + s_3). \quad (63)$$

Writing $\Delta_i = P_i - Q_i$, raw coordinate error plus subtraction rounding gives $|d_i - \Delta_i| < 18\varepsilon$. Since $|\Delta_i| \leq 2$, the square error before rounding is below $18\varepsilon(4 + 18\varepsilon) < 73\varepsilon$; product rounding adds less than 3ε . Each of the two additions adds less than 8ε . Therefore

$$|\widehat{D} - D| < 3(76\varepsilon) + 2(8\varepsilon) = 244\varepsilon < 256\varepsilon. \quad (64)$$

The detailed proof includes the elementary absolute underflow allowance 2^{-1075} and the norm-admission bound in `formal/RESIDENT_PREDICATE_PROOF.md`. No overflow is possible in this bounded coordinate body. This is an error certificate in dimensionless squared-chord units, not a physical position tolerance.

14.2 A wide interval produces definite or unresolved results

Use the exactly representable margin $m = 2^{-40} = 4096\varepsilon$ and

$$t_- = \text{RN}(R^2 - m), \quad t_+ = \text{RN}(R^2 + m). \quad (65)$$

Each threshold's rounding error is below 3ε , including zero, subnormal radii and $R^2 = 4$. Thus

$$\begin{aligned} \widehat{D} \leq t_- &\implies D < R^2 \quad \text{definitely inside,} \\ \widehat{D} > t_+ &\implies D > R^2 \quad \text{definitely outside,} \\ \text{otherwise} &\quad \text{unresolved by this interval.} \end{aligned} \quad (66)$$

For example, the first implication follows from $D < \widehat{D} + 256\varepsilon < R^2 - (4096 - 3 - 256)\varepsilon$ after the threshold comparison. The second follows from the corresponding lower bound. An exact boundary cannot be assigned to outside by either proved rule.

The resident implementation uses componentwise equality of the original triples as an exact zero-distance certificate, including signed zeros. A rounded distance equal to zero is not: distinct subnormal separations can square to zero. Distinct proportional triples may also denote the same direction and require an appropriate exact certificate or unresolved result. The endpoint $R^2 = 4$ separately accepts every admitted direction from equation (62).

14.3 Complete counts and a visible resolution path

The conservative binary32 hierarchy filter can reject only proved outside subtrees. Surviving raw binary64 point words reach the predicate above. If I points are definitely inside and U remain unresolved, then

$$I \leq \text{Count}(q, R^2) \leq I + U. \quad (67)$$

Only $U = 0$ establishes a complete resident count. Otherwise the query status remains unresolved until a selected exact host query supplies the whole answer. Partial counts are not final answers. The count profile does not truncate an ID list or inherit a candidate-capacity prefix as its result.

Seeds, query words and bounds share the selected epoch. Derived outputs use separate storage, and their consumers wait for completed producer launches. An immutable point hierarchy can be reused; changed geometry needs a new valid bound and binding. Texture instructions do not guarantee a cache hit or that every resident VRAM word is resident in the texture cache.

14.4 The implemented count-to-word feedback binding

The `ResidentFeedback` profile assigns immutable query alternatives Q_i^0, Q_i^1 to lane i . A count kernel reads the old 64-bit state word q_i^n and evaluates

$$(I_i^n, U_i^n) = \text{ResidentCount}(Q_i^{q_i^n \& 1}). \quad (68)$$

A following commit kernel consumes that summary. If $U_i^n > 0$, its word, parity, hinge count and last committed epoch hold; the status is unresolved. This feedback profile does not silently run a host resolver. For a resolved summary, the drive is $d_i^n = [I_i^n > 0]$. A new resolved epoch uses the explicit pulse-on-first rule

$$h_i^n = [\text{uninitialized}] \vee [d_i^n \neq d_i^{\text{last}}]. \quad (69)$$

Replaying the last committed epoch holds; an older epoch is rejected. Only resolved, ordered epochs update the remembered drive and epoch. Thus the first sample is a declared initialization event, not an inferred geometric crossing.

Let T_x, T_J, T_K be the supplied bitwise four-entry Boolean tables, let M be the valid-word mask, and write the two ordered ASA/NA stages as

$$\mathcal{A}_j(v) = \begin{cases} 0, & (v \& A_j \& N_j) \& B_j \neq 0, \\ v \& A_j \& N_j, & \text{otherwise.} \end{cases} \quad (70)$$

For an accepted pulse, all three tables refer to the same old word:

$$y = \mathcal{A}_1(\mathcal{A}_0(T_x(q, d) \& M)), \quad (71)$$

$$J = T_J(q, y) \& M, \quad K = T_K(q, y) \& M, \quad (72)$$

$$q^+ = ((J \& \neg q) \vee (\neg K \& q)) \& M, \quad p^+ = p \oplus 1. \quad (73)$$

Otherwise the logical state holds. Exhausted hinge-count storage rejects the commit. This profile has a fixed chart, so its orientation stays zero.

The source and query table outlive the feedback object. Each external phase acquires a fresh invalidated device view, and calls sharing that source are serialized. The count kernel reads state; the distinct commit kernel writes it. Their common stream orders these operations before the next count kernel, so a second state buffer is unnecessary for this two-kernel binding. No per-epoch host data readback is required. This is delayed feedback through a geometric predicate, with unresolved results still visible.

14.5 A separate exact integer working-set profile

The working-set executable binds a smaller independent operator profile. Each immutable 16-byte record holds integer $a, b \in [-1023, 1023]$, an opcode $o \in \{0, 1, 2, 3\}$ and a 32-bit salt w . Its old state is a 32-bit word q and a parity bit. Define

$$s = (q \& 31) - 16, \quad c = (q \& 3) - 1, \quad d = ((q \gg 2) \& 3) - 1, \quad (74)$$

where subtraction occurs in a signed integer domain. With $\phi = (1 + \sqrt{5})/2$, the selected exact coefficient pair is

$$(a', b') = \begin{cases} (a + s, b), & o = 0, \\ (b, a + b), & o = 1, \\ (ac + bd, ad + bc + bd), & o = 2, \\ (a - s, b), & o = 3. \end{cases} \quad (75)$$

The third row is literal multiplication by $c + d\phi$, using $\phi^2 = \phi + 1$. No floating-point approximation to ϕ is evaluated. The bounds $|a'| \leq 4092$, $|b'| \leq 6138$ and $|2a' + b'| \leq 14322$ give

$$(2a' + b')^2 \leq 205119684, \quad 5b'^2 \leq 188375220 < 2^{31} - 1. \quad (76)$$

Both squares and every preceding coefficient temporary fit signed 32-bit arithmetic. The optimized device body uses that width without narrowing loss; the independent CPU recurrence retains 64-bit intermediates. Same-sign terms are handled directly; the square comparison is

used only for opposing signs in $2(a' + b'\phi) = 2a' + b' + b'\sqrt{5}$. This proof is specific to the small procedural coefficient domain above. It does not narrow the separate native expression profile whose coefficients may approach $2^{30} - 1$.

For $\sigma = \text{sign}(a' + b'\phi) = 0$ the state holds. Otherwise the drive word is w for $\sigma < 0$ and $\neg w$ for $\sigma > 0$; $x = q \oplus \text{drive}$ passes through the two fixed mask stages in equation (70). The selected $J = y, K = \neg y$ specialization gives $q^+ = y$ and toggles parity. Every launch epoch supplies one fresh internal event opportunity per record; repeated nonzero signs do not imply repeated physical side crossings. Old q changes the next operands and word body, making this an explicit delayed self-reference.

14.6 Bank coverage and coherence

The immutable seed records are expanded from integer bit planes through raw integer texture reads. In this executable, only immutable records use the texture/global comparison; the two state buffers use global memory in both variants. Every record reads its epoch's old buffer and writes the distinct next buffer. All chunks finish before buffer roles change. The packed scratch producer and its texture consumer are also separate completed launches. These boundaries follow the ordering and texture-coherence requirements in the official [CUDA stream](#) and [texture-coherence](#) documentation; they do not assume coherent simultaneous texture reads and global writes to the same address.

For N records, streaming visits ordinal k directly. The other pattern uses

$$i(k) = (\alpha k + \beta) \bmod N, \quad \gcd(\alpha, N) = 1, \quad 0 \leq k < N. \quad (77)$$

Checked integer products avoid overflow, and coprimality makes this a bijection. Thus each full epoch visits every record exactly once, including every bank, rather than probing a small subset of a large allocation. The declared working set contains $16N$ seed bytes and $16N$ bytes in the two persistent state buffers. Its size is admitted against available VRAM with a reserve; only completed runs establish an exercised size. This accounting does not mean that all VRAM allocations fit in hardware caches.

14.7 Exact address and digest implementation

The optimized address body preserves equation (77). Streaming uses $i = k$ directly. For a power-of-two record count, the remainder is the low-bit mask. For every other admitted $N \geq 2048$, put $W = 2^{64}$ and, for the checked numerator $0 \leq x = \alpha k + \beta < W$, use

$$m = \left\lfloor \frac{W}{N} \right\rfloor, \quad q_0 = \left\lfloor \frac{xm}{W} \right\rfloor, \quad r = x - q_0N, \quad i = \begin{cases} r - N, & r \geq N, \\ r, & r < N. \end{cases} \quad (78)$$

The device obtains q_0 from the high half of an exact unsigned 64-bit product. Since $0 \leq x/N - xm/W \leq x/W < 1$, q_0 is either $\lfloor x/N \rfloor$ or one less. Hence $0 \leq r < 2N$ and one subtraction produces the exact remainder. The implementation computes m on the host from $W - 1$, avoiding a stored integer equal to W ; the existing checked affine product and addition remain required. This replaces division, not the permutation or its coverage contract.

Each full bank has a verified capacity $C = 2^b$ within the device texture element limit and the selected cap. Its index and offset are exactly $i \gg b$ and $i \& (C - 1)$. The final bank retains its actual valid count. Both immutable-record texture and global-load variants use the same address mapping, operands, recurrence and old/new state allocation rules.

The 128-thread digest now reduces four warp summaries, then combines those summaries in the first warp after one block barrier. All lanes participate; invalid ordinals contribute zero XOR, zero sum and zero count. XOR and unsigned addition modulo 2^{64} are associative, and the block count is

at most 128, so regrouping changes no digest bits. The implementation contract and independent-oracle limits are recorded in [formal/WORD_CACHE_BENCHMARK.md](#). These exact transformations provide no speed claim without the corresponding compiled checks and matched measurements.

14.8 Validation and measurement scope

Paired texture/global trials reset both state buffers to the same procedural snapshot outside timing. Per-epoch validation combines full-record XOR, sum and count digests with sampled independent CPU state replay. For $N \leq 65536$, an additional full CPU digest is computed. Digests can collide; they are evidence distinct from exhaustive bitwise state comparison. The timed kernel includes arithmetic, masks, JK and digest work. The reported $32N$ logical bytes per epoch count seed read, old-state read and next-state write; their effective rate is not a hardware DRAM-bandwidth measurement.

The count predicate and this word recurrence have different outputs and costs. Their measurements keep resident kernel time, transfers, host resolution, matched global loads and validation overhead separate. Neither a large working set nor their shared texture path establishes a speed advantage for the other's task, a cache-hit guarantee or physical navigation accuracy.

15 Measured resident counts and word working sets

All 27 resident count cases were measured; every complete answer matched.

The texture path with complete host-visible counts beat the best measured S2 configuration in 20 of 27 cases and the best measured CPU implementation in 13 of 27 cases.

These results concern inclusive spherical *counts* over the supplied exact IEEE seeds. They are not ID-return, general-S2, physical-position-accuracy or photonics measurements.

The word-cache results below are the preserved baseline in `review/word_cache_full.json`; later optimizations require separate measurement records.

15.1 Every resident count case

All entries are median milliseconds; CPU cells show the chosen worker count in parentheses. T/G mean integer texture/global loads. Device time is per resident launch, with five launches per sample. Complete time includes launch, count readback and any host exact fallback. The last column is unresolved points / texture fallback queries / global fallback queries.

Family	N	Q	S2 best	BVH best	T dev.	T comp.	G dev.	G comp.	U/Ft/Fg
uniform	16384	256	0.1165 (20)	0.0476 (20)	0.2154	0.2685	0.2031	0.2571	0/0/0
uniform	16384	4096	1.038 (20)	0.3499 (20)	0.2341	0.3236	0.2237	0.3168	0/0/0
uniform	16384	65536	11.42 (20)	3.458 (20)	1.304	1.776	1.276	1.826	0/0/0
uniform	262144	256	0.1888 (20)	0.0799 (20)	0.2536	0.3096	0.2447	0.2975	0/0/0
uniform	262144	4096	1.52 (20)	0.4807 (20)	0.2771	0.3811	0.2642	0.3538	0/0/0
uniform	262144	65536	19.96 (20)	6.063 (20)	1.587	2.15	1.552	2.007	0/0/0
uniform	1048576	256	0.2667 (20)	0.1105 (20)	0.3203	0.3787	0.3051	0.3612	0/0/0
uniform	1048576	4096	2.365 (20)	0.628 (20)	0.3155	0.4242	0.3005	0.3899	0/0/0
uniform	1048576	65536	28.74 (20)	6.631 (20)	2.793	3.913	2.754	3.92	0/0/0
clustered	16384	256	1.423 (20)	0.234 (20)	1.529	1.629	1.428	1.531	0/0/0
clustered	16384	4096	13.8 (20)	2.232 (20)	1.573	1.847	1.472	1.73	0/0/0
clustered	16384	65536	213.4 (20)	26.94 (20)	12.57	13.1	12.36	12.85	0/0/0
clustered	262144	256	3.168 (20)	0.4402 (20)	8.505	8.633	8.06	8.193	0/0/0
clustered	262144	4096	35.02 (20)	4.232 (20)	8.864	8.964	8.366	8.503	0/0/0
clustered	262144	65536	548.8 (20)	66.61 (20)	64.18	64.95	63.23	63.95	2/2/2
clustered	1048576	256	4.473 (20)	0.5397 (20)	18.65	18.79	17.76	17.9	0/0/0

Family	N	Q	S2 best	BVH best	T dev.	T comp.	G dev.	G comp.	U/Ft/Fg
clustered	1048576	4096	52.52 (20)	6.573 (20)	18.65	18.79	17.77	17.9	0/0/0
clustered	1048576	65536	744 (20)	76.54 (20)	132.3	133.3	130.4	131.5	7/7/7
great_circle	16384	256	0.563 (20)	0.0843 (4)	0.2874	0.3448	0.272	0.3235	0/0/0
great_circle	16384	4096	3.613 (20)	0.502 (20)	0.3082	0.4366	0.2926	0.4139	0/0/0
great_circle	16384	65536	42.77 (20)	5.61 (20)	2.04	2.505	2.002	2.438	0/0/0
great_circle	262144	256	1.495 (20)	0.1508 (20)	1.138	1.25	1.082	1.197	0/0/0
great_circle	262144	4096	10.49 (20)	1.129 (20)	1.142	1.41	1.09	1.341	0/0/0
great_circle	262144	65536	175.9 (20)	20.45 (20)	7.901	8.489	7.776	8.395	0/0/0
great_circle	1048576	256	2.122 (20)	0.2988 (20)	2.943	3.098	2.79	2.975	0/0/0
great_circle	1048576	4096	25.14 (20)	3.806 (20)	3.935	4.06	3.779	3.9	0/0/0
great_circle	1048576	65536	380.9 (20)	44.99 (20)	21.58	22.19	21.23	21.78	0/0/0

There were 9 unresolved point decisions across 2 cases; 9 queries per access mode required complete host fallback. Texture beat global loads in 0 of 27 device-time comparisons and 2 of 27 complete-time comparisons. The GPU-over-CPU gain therefore does not establish a texture-cache advantage.

The first GPU construction cost 2547.1377 ms, including 2546.8671 ms reported source preparation/upload. This cold setup is excluded from repeated-query timings; later setup costs remain in the complete report.

Device-only counts are complete only where the unresolved count is zero. Every 1/4/20-worker configuration and raw trial is retained in `review/resident_summary.json`; best means the lowest measured median, not an unreported preferred configuration.

S2 uses its public closest-point API with reused candidate vectors, followed by exact refinement/counting. The BVH count avoids ID allocation. Setup is excluded. GPU device-mode order alternates; CPU/configuration and host-complete mode blocks have fixed order. Five warm trials provide no confidence interval or universal speed guarantee.

15.2 Word/state coverage through the memory hierarchy

NVIDIA GeForce RTX 5070 Ti Laptop GPU reports 36 MiB of L2 (37748736 bytes). The largest working set was 9.744 GiB, 277.2 times L2 and 81.61% of reported device memory. The configured VRAM reserve remained available.

This is a separate bounded integer phi-expression and ASA/NA+JK word/state workload, *not an S2 comparison*. Working bytes include immutable records plus both persistent state buffers. Every record is visited in every epoch; zero-sign holds may leave a value unchanged.

Working MiB	L2 multiple	Records/epoch	Allocated MiB	Sampled	CPU oracle
0.0625	0.001736	2048	0.09888	124	Full replay
0.25	0.006944	8192	0.3815	130	Full replay
1	0.02778	32768	1.511	131	Full replay
4	0.1111	131072	6.028	131	Sample replay
16	0.4444	524288	24.05	131	Sample replay
18	0.5	589824	26.05	131	Sample replay
36	1	1179648	44.05	131	Sample replay
64	1.778	2097152	72.05	131	Sample replay
72	2	2359296	80.05	131	Sample replay
128	3.556	4194304	136.1	131	Sample replay
256	7.111	8388608	264.1	131	Sample replay
512	14.22	16777216	520.1	131	Sample replay
1024	28.44	33554432	1032	132	Sample replay
2048	56.89	67108864	2056	136	Sample replay
4096	113.8	134217728	4104	144	Sample replay
8192	227.6	268435456	8200	160	Sample replay

Working MiB	L2 multiple	Records/epoch	Allocated MiB	Sampled	CPU oracle
9978	277.2	326959104	9986	169	Sample replay

All epochs compare complete aggregate digests across access modes, traversal patterns and trials. Full CPU replay is restricted to the small sets flagged above; larger sets use sampled independent word replay. Digest agreement can collide and is not a byte-for-byte readback proof of every large buffer. Exact seed/state byte counts, free-memory observations and source receipts are retained in the summary JSON.

15.3 Every word-cache timing trial

Each cell lists trials in increasing trial order. Times are milliseconds for 3 measured epochs per trial. GPU sums exclude reset/warmup, digest readback and sampled verification. Host wall includes measured epochs, digest aggregation/readback and per-epoch sampled verification; reset/warmup are outside the timer. Raw per-epoch values remain in the JSON.

Streaming

MiB	L2 ratio	Texture GPU	Global GPU	Texture wall	Global wall
0.0625	0.001736	0.0943	0.09088	0.4775	0.4435
		0.06342	0.08512	0.3175	0.3453
		0.1115	0.08682	0.4189	0.4072
		0.103	0.08291	0.411	0.4639
0.25	0.006944	0.1041	0.09619	0.4149	0.4567
		0.08954	0.06877	0.4951	0.3928
		0.1001	0.08506	0.4529	0.4391
		0.08781	0.1161	0.4355	0.3581
1	0.02778	0.06947	0.0887	0.3398	0.4339
		0.1098	0.09859	0.4743	0.4767
		0.1087	0.09677	0.4717	0.4536
		0.0975	0.1707	0.3979	0.5152
4	0.1111	0.2865	0.2965	0.8373	0.8749
		0.2964	0.2728	0.8681	0.8475
		0.2933	0.2816	0.8343	0.821
		0.4168	0.4044	1.133	1.075
16	0.4444	0.3638	0.3873	0.9255	1.069
		0.3957	0.3713	1.086	1.033
		0.6731	0.6843	1.612	1.856
		0.6868	0.758	1.793	2.011
36	1	0.7426	0.692	1.987	1.857
		1.334	1.256	3.259	3.063
		1.242	1.149	2.858	2.814
		1.243	1.136	2.868	2.79
64	1.778	1.368	1.286	3.295	3.029
		1.428	1.362	3.367	3.215
		1.4	1.35	3.236	3.174
		2.508	2.394	5.562	5.457
72	2	2.599	2.435	5.689	5.465
		2.534	2.349	5.66	5.613
		4.955	4.758	10.89	10.89
		5.044	4.807	10.91	10.73
128	3.556	5.356	5.128	11.54	11.7
		10.62	10.02	23.77	22.34
		9.883	9.388	21.51	21.1
		9.796	10.79	21.02	22.74
256	7.111	19.93	18.97	43.12	42.31
		19.58	20.1	42.99	45.29
		20.1	20.27	43.45	45.11
		40.27	37.88	88.64	84.86
512	14.22	39.72	39.69	86.52	88.85
		40.54	37.34	88.7	83.51
		78.44	75.78	170.9	167.6
		79.82	75.06	173.5	166.9
1024	28.44	80.53	74.62	173.7	167.5
		176.1	168.7	360.7	353.9
		176.6	168.5	361.6	354.6
		176.4	170.9	361	356.2

MiB	L2 ratio	Texture GPU	Global GPU	Texture wall	Global wall
		215.2	205.6	441.7	432.3
		217.7	205.6	443.3	432.8
9978	277.2	217.2	203.9	445.3	428.2

Seeded coprime affine permutation

MiB	L2 ratio	Texture GPU	Global GPU	Texture wall	Global wall
		0.08128	0.08086	0.3697	0.4077
		0.08371	0.09946	0.3995	0.4421
0.0625	0.001736	0.07434	0.06886	0.2667	0.3064
		0.08173	0.1023	0.4128	0.3091
		0.05789	0.08483	0.3696	0.439
0.25	0.006944	0.06675	0.07347	0.2881	0.2909
		0.08902	0.08717	0.3428	0.4495
		0.08368	0.08528	0.3707	0.3998
1	0.02778	0.08973	0.1067	0.4084	0.4131
		0.1156	0.1452	0.4454	0.4714
		0.132	0.1134	0.4357	0.4211
4	0.1111	0.1162	0.09738	0.4114	0.3176
		0.3593	0.3567	0.8998	0.908
		0.3602	0.3686	0.9185	0.92
16	0.4444	0.3588	0.3426	0.9292	0.9077
		0.4314	0.4838	1.153	1.149
		0.4907	0.4725	1.192	1.122
18	0.5	0.4392	0.426	1.15	1.072
		1.045	1.054	2.173	2.079
		1.063	1.089	2.233	2.311
36	1	1.027	1.092	2.189	2.311
		5.074	4.98	6.78	6.718
		5.005	4.96	6.69	6.592
64	1.778	5.017	4.928	6.639	6.621
		5.314	5.328	7.23	7.21
		5.344	5.317	7.39	7.174
72	2	5.374	5.306	7.244	7.139
		11.15	11.04	14.32	14.22
		11.09	10.93	14.28	14.09
128	3.556	11.13	11	14.52	14.09
		23.14	22.84	29.41	28.76
		23.16	22.78	29.29	28.82
256	7.111	23.06	23.13	28.97	29.33
		47.34	46.38	59.75	58.67
		46.82	46.47	58.81	58.28
512	14.22	46.84	46.74	58.99	59.4
		96.93	96.6	121.7	121.2
		97.79	96.18	122.6	120.4
1024	28.44	97.4	97.04	121.5	120.5
		195.9	192.7	244	240.4
		194.2	193.5	241.9	241.5
2048	56.89	196.1	193	243.1	240.3
		387.6	386.7	481.8	481
		388.3	386.5	481.8	480.7
4096	113.8	387.3	387.4	481.5	484.5
		798	774.2	985.8	962.9
		776.6	799	966.8	988
8192	227.6	797.9	774.3	988.9	965.8
		980.1	942.7	1211	1174
		965.2	959.1	1198	1193
9978	277.2	984	943.8	1219	1177

Texture beat global loads in 7 of 34 GPU-median comparisons and 9 of 34 wall-median comparisons. Cache hit rates were not measured and no persisting-L2 window was set. Logical GB/s is workload throughput, not measured physical DRAM bandwidth or a universal texture advantage.

Reproducible source receipts. Measured benchmark source commit: 4d8992292a7fa0fc43fc050ec090512552f7d067. Subsequent source edits are not implied to have been measured by this record.

All trials, setup timings and metadata are preserved in `review/resident_summary.json`. Its inputs are `review/resident_full.json` and `review/word_cache_full.json`.

Resident SHA-256: `c8c25de5a13b9f559c65081ebd597547c1a5d41753edc09deff7db858fd4a94d`.

Word-cache SHA-256: `adb44ca24fbab2e4534bae716ca94213bd28ba4cf1908ae4d62fc0b0fbf93946`.

16 Measured exact word-kernel optimization

Across 17 equal working sizes, the optimized implementation beat baseline in 59 of 68 GPU-median comparisons and 54 of 68 host-wall-median comparisons. GPU baseline/optimized ratios range from 0.8048 to 1.254.

This is the bounded integer phi-expression, ASA/NA+JK and digest workload, *not an S2 comparison*. Both load paths receive direct streaming addresses, exact bank shift/mask indexing, integer reciprocal remainder with one correction, proved-safe 32-bit coefficient arithmetic and warp digest reductions. Seeds, recurrence, masks, reset/warm policy and independent 64-bit CPU oracle are unchanged.

Pairs require equal working bytes, seed, epoch/trial/warm counts, chunk size and bank layout. Every corresponding trial's address parameters and every epoch's XOR, modular-sum and visit-count digest match, including the final epoch. Digest collisions remain possible; large sets retain sampled CPU replay. All raw trials and unmatched sizes are retained in `review/word_optimization_summary.json`.

16.1 Memory extent and matching

Baseline maximum: 10462691328 working bytes (9.744 GiB), 277.2 times the reported 36 MiB L2, 81.61% of reported device memory. It covers 326959104 records over 20 banks per epoch; allocation requests total 10471136648 bytes and 1084227584 bytes were reported free after allocation.

Optimized maximum: 10462691328 working bytes (9.744 GiB), 277.2 times the reported 36 MiB L2, 81.61% of reported device memory. It covers 326959104 records over 20 banks per epoch; allocation requests total 10471136648 bytes and 1084227584 bytes were reported free after allocation.

Working bytes include 16N immutable bytes and both 8N-byte state buffers. Every record is visited each epoch. The maxima are independent observations; they are paired only when their actual sizes match. This extent is not active-warp occupancy, measured cache residency or zero-headroom allocation. Shared-device free-memory changes are not exclusive process peak measurements.

16.2 All full-sweep comparisons

17 paired sizes and 612 equal trial/epoch digest pairs; 3 measured epochs, 3 trials per path and 1 warm epoch(s). Chunks contain 8388608 logical bytes. GPU/wall wins: 59/68 and 54/68.

Table entries are median milliseconds for all measured epochs. B/O means baseline divided by optimized; ratios above one favor optimized. GPU time excludes reset/warm, digest readback and sampled checks. Wall time includes measured epochs and those checks, with reset/warm outside.

streaming / integer texture

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.0943	0.08122	1.161	0.4189	0.3674	1.14

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.25	0.006944	0.103	0.1016	1.013	0.4149	0.2989	1.388
1	0.02778	0.08781	0.08858	0.9913	0.4355	0.4075	1.069
4	0.1111	0.1087	0.09421	1.154	0.4717	0.4312	1.094
16	0.4444	0.2933	0.2812	1.043	0.8373	0.8822	0.9491
18	0.5	0.3957	0.405	0.9771	1.086	1.214	0.8949
36	1	0.6868	0.693	0.9911	1.793	1.785	1.004
64	1.778	1.243	1.198	1.037	2.868	2.844	1.009
72	2	1.4	1.34	1.044	3.295	3.099	1.063
128	3.556	2.534	2.397	1.057	5.66	5.477	1.033
256	7.111	5.044	4.809	1.049	10.91	10.66	1.023
512	14.22	9.883	9.494	1.041	21.51	21.18	1.016
1024	28.44	19.93	19.11	1.043	43.12	42.81	1.007
2048	56.89	40.27	38.57	1.044	88.64	84.7	1.046
4096	113.8	79.82	77.15	1.035	173.5	169.5	1.023
8192	227.6	176.4	173.1	1.019	361	356.9	1.012
9978	277.2	217.2	210.6	1.031	443.3	436.9	1.015

streaming / global

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.08682	0.08317	1.044	0.4072	0.3513	1.159
0.25	0.006944	0.08291	0.08154	1.017	0.4567	0.3635	1.256
1	0.02778	0.0887	0.08218	1.079	0.4339	0.4136	1.049
4	0.1111	0.09859	0.08989	1.097	0.4767	0.4784	0.9964
16	0.4444	0.2816	0.2481	1.135	0.8475	0.8686	0.9757
18	0.5	0.3873	0.3692	1.049	1.069	1.215	0.8797
36	1	0.692	0.641	1.08	1.857	1.855	1.001
64	1.778	1.149	1.118	1.027	2.814	2.719	1.035
72	2	1.35	1.226	1.101	3.174	3.049	1.041
128	3.556	2.394	2.217	1.08	5.465	5.159	1.059
256	7.111	4.807	4.44	1.083	10.89	10.05	1.083
512	14.22	10.02	9.086	1.103	22.34	20.88	1.07
1024	28.44	20.1	17.98	1.118	45.11	41.48	1.087
2048	56.89	37.88	36.71	1.032	84.86	83.11	1.021
4096	113.8	75.06	72.75	1.032	167.5	166.6	1.005
8192	227.6	168.7	165.5	1.019	354.6	349.2	1.016
9978	277.2	205.6	199.4	1.031	432.3	426.8	1.013

seeded coprime affine permutation / integer texture

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.08128	0.08938	0.9094	0.3697	0.3586	1.031
0.25	0.006944	0.06675	0.08294	0.8048	0.3696	0.472	0.7831
1	0.02778	0.08902	0.08749	1.018	0.3707	0.4353	0.8516
4	0.1111	0.1162	0.1046	1.111	0.4357	0.3839	1.135
16	0.4444	0.3593	0.3366	1.067	0.9185	0.9739	0.9431
18	0.5	0.4392	0.461	0.9529	1.153	1.311	0.88
36	1	1.045	0.9039	1.156	2.189	1.928	1.136
64	1.778	5.017	4.851	1.034	6.69	6.479	1.033
72	2	5.344	5.161	1.035	7.244	6.934	1.045
128	3.556	11.13	10.94	1.017	14.32	14.18	1.01
256	7.111	23.14	22.61	1.023	29.29	28.67	1.022
512	14.22	46.84	46.31	1.011	58.99	58.79	1.003
1024	28.44	97.4	95.83	1.016	121.7	120.2	1.013
2048	56.89	195.9	193.2	1.014	243.1	240.9	1.009

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
4096	113.8	387.6	388.9	0.9969	481.8	484.8	0.9937
8192	227.6	797.9	809	0.9862	985.8	997.2	0.9885
9978	277.2	980.1	978.7	1.001	1211	1212	0.9988

seeded coprime affine permutation / global

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.08086	0.07987	1.012	0.4077	0.3357	1.214
0.25	0.006944	0.08483	0.06765	1.254	0.3091	0.3977	0.7772
1	0.02778	0.08717	0.08454	1.031	0.4131	0.382	1.081
4	0.1111	0.1134	0.1126	1.007	0.4211	0.3951	1.066
16	0.4444	0.3567	0.3196	1.116	0.908	0.8294	1.095
18	0.5	0.4725	0.4818	0.9808	1.122	1.274	0.8807
36	1	1.089	0.9243	1.178	2.311	2.039	1.133
64	1.778	4.96	4.907	1.011	6.621	6.544	1.012
72	2	5.317	5.224	1.018	7.174	6.992	1.026
128	3.556	11	10.83	1.015	14.09	13.93	1.012
256	7.111	22.84	22.61	1.01	28.82	28.55	1.009
512	14.22	46.47	45.92	1.012	58.67	58.1	1.01
1024	28.44	96.6	95.4	1.013	120.5	120.5	1
2048	56.89	193	191.8	1.006	240.4	239.3	1.004
4096	113.8	386.7	383	1.01	481	477.3	1.008
8192	227.6	774.3	770.4	1.005	965.8	963.1	1.003
9978	277.2	943.8	934	1.01	1177	1167	1.008

16.3 Non-power-of-two check: 3 MiB

1 paired sizes and 80 equal trial/epoch digest pairs; 10 measured epochs, 2 trials per path and 1 warm epoch(s). Chunks contain 8388608 logical bytes. GPU/wall wins: 2/4 and 3/4.

Table entries are median milliseconds for all measured epochs. B/O means baseline divided by optimized; ratios above one favor optimized. GPU time excludes reset/warm, digest readback and sampled checks. Wall time includes measured epochs and those checks, with reset/warm outside.

streaming / integer texture

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
3	0.08333	0.3223	0.3277	0.9835	1.403	1.326	1.058

streaming / global

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
3	0.08333	0.2945	0.346	0.8512	1.381	1.426	0.9683

seeded coprime affine permutation / integer texture

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
3	0.08333	0.3877	0.3135	1.237	1.469	1.323	1.111

seeded coprime affine permutation / global

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
3	0.08333	0.3872	0.3396	1.14	1.476	1.441	1.024

Unpaired baseline bytes: [65536, 262144, 1048576]; unpaired optimized bytes: [].

16.4 Four-size quick check

4 paired sizes and 32 equal trial/epoch digest pairs; 2 measured epochs, 1 trials per path and 1 warm epoch(s). Chunks contain 8388608 logical bytes. GPU/wall wins: 6/16 and 8/16.

Table entries are median milliseconds for all measured epochs. B/O means baseline divided by optimized; ratios above one favor optimized. GPU time excludes reset/warm, digest readback and sampled checks. Wall time includes measured epochs and those checks, with reset/warm outside.

streaming / integer texture

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.0711	0.06093	1.167	0.4281	0.2789	1.535
0.25	0.006944	0.105	0.04128	2.544	0.3693	0.2294	1.61
1	0.02778	0.06138	0.0585	1.049	0.2487	0.2976	0.8357
4	0.1111	0.1419	0.0671	2.114	0.3451	0.3029	1.139

streaming / global

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.04486	0.04518	0.9929	0.2722	0.2404	1.132
0.25	0.006944	0.05638	0.04451	1.267	0.244	0.2731	0.8934
1	0.02778	0.03901	0.06202	0.629	0.1613	0.2714	0.5943
4	0.1111	0.04467	0.05101	0.8758	0.2835	0.2835	1

seeded coprime affine permutation / integer texture

MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
0.0625	0.001736	0.04803	0.06118	0.785	0.2913	0.1514	1.924
0.25	0.006944	0.03629	0.0673	0.5392	0.2135	0.294	0.7262
1	0.02778	0.06259	0.06701	0.9341	0.2558	0.2683	0.9534
4	0.1111	0.05661	0.06006	0.9425	0.2948	0.2187	1.348

seeded coprime affine permutation / global

	MiB	L2 ratio	B GPU	O GPU	B/O GPU	B wall	O wall	B/O wall
	0.0625	0.001736	0.0639	0.05987	1.067	0.2568	0.2753	0.9328
	0.25	0.006944	0.05638	0.06454	0.8736	0.2906	0.2573	1.129
	1	0.02778	0.0432	0.04634	0.9323	0.2013	0.3116	0.646
	4	0.1111	0.07523	0.07846	0.9588	0.2988	0.269	1.111

16.5 Scope and reproducibility

Baseline and optimized runs were separate; scheduling, device clocks and desktop activity may differ. The few trials give no confidence interval. Quick checks have one trial per path; the 3 MiB reciprocal check has ten epochs and two trials. Cache-hit and traffic counters belong to separate profiling evidence. Logical bytes/time does not measure physical DRAM bandwidth or a universal texture advantage.

Baseline source commit: 4d8992292a7fa0fc43fc050ec090512552f7d067. Optimized source commit: 59a7e7d37be24d381cc4be4c7b2a793778995ecf. Optimized CUDA source SHA-256: 742c25e582bd2d5c4d0868051b61a03d6d224d477029caaff949ba552b2890d2.

All six raw report receipts, every trial and per-epoch digest are preserved in `review/word_optimization_summary.json`; readable details are in `review/WORD_OPTIMIZATION.md`.

17 Native instructions and hardware-counter receipts

The preserved optimized `sm_120` disassembly contains `TLD.LZ` in the texture epoch variants and ordinary loads in the global variants. Compiler resources report 28 registers per texture thread and 26 per global thread, with a 40-byte stack. Local stack loads/stores remain visible; no zero-stack claim is made.

Nsight Compute 2025.1.0 sampled the first 8 MiB logical chunk of a 512 MiB working set in each mode, with 2,048 blocks of 128 threads. Kernel replay was enabled; caches were not flushed and clocks were not locked. Each mode used a separate process.

Mode	DRAM %	SM %	L1/TEX hit %	L2 hit %	Warps %
stream texture	58.55	37.67	43.15	43.56	84.89
stream global	66.48	48.80	43.24	43.19	83.05
affine texture	32.20	3.27	3.39	23.63	90.25
affine global	32.49	3.02	8.17	20.52	90.60

DRAM and SM are percentages of reported peak sustained elapsed throughput. Warps denotes peak sustained active warps. The L1/TEX ratio aggregates the shared unit; it does not isolate immutable seed traffic. These are selected-chunk observations, not full-working-set averages or cache-residency guarantees.

The earlier streaming-texture baseline sample reported 37.06% DRAM throughput, compared with 58.55% in the optimized sample. This does not establish bandwidth saturation. Profiler replay changes execution; the separate unprofiled sweeps supply performance timings.

Commands, source hashes, selected launches and raw receipts are retained in `review/word_profile_summary.json`. Replay behavior follows the [official Nsight Compute profiling guide](#).

18 Exact operator seeds: a mathematical contract

This chapter restates the XOP-R1 operator-seed contract introduced in release 3.6.1.11 and adopted by the unified field profile. Definitions specify a representation and its meaning; propositions establish conditional mathematical consequences. An implementation must still discharge the stated parsing, arithmetic and proof-checking obligations. By this retained specification alone, no new runtime, machine-checked proof, test, simulation or measured improvement is asserted. R15 executable subsets and their evidence are specified separately in the preceding chapters. The implemented R10 orbital seed and its binary64 arithmetic keep their existing format and interpretation.

18.1 Bits, values and exact expressions are distinct types

Write $\text{Bits}(w) = \{0, 1\}^w$ for a finite raw bit string. Its unsigned interpretation is an explicitly requested map

$$\text{uint}(b) = \sum_{i=0}^{w-1} b_i 2^i.$$

A binary64 payload belongs to $\text{Bits}(64)$ until an explicit conversion is applied. It does not become an arbitrary-precision real by being packed in one-bit planes. A finite binary64 value can be converted to the exact dyadic rational denoted by that value; this preserves its rounded input value, not an unknown physical quantity. Signed zeros are distinct raw payloads but the same rational zero. Infinity and NaN have no value in the exact real scalar types.

The scalar hierarchy is

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{A} \hookrightarrow \text{RealTerm}, \quad \mathbb{A} = \overline{\mathbb{Q}} \cap \mathbb{R}.$$

Here RealTerm is finite syntax for a possibly partial real expression, not a claim that every real number has a finite description. Coercions preserve denotation and are explicit in the typed graph. Booleans, finite words, tuples, vectors, matrices, functions and streams have separate type constructors. Dimensioned scalars also carry a seven-component rational SI exponent vector; addition requires equal dimensions, multiplication adds dimensions, and transcendental functions require dimensionless arguments. Angle conventions and coordinate frames are explicit semantic annotations, not inferred from an untyped number. A framed vector's type carries its fully qualified nominal frame identity; a framed matrix carries source and target identities. Each identity contains namespace, major/minor version and label. The full annotation is serialized and participates in sharing. A frame change requires an explicit typed template and its coordinate-realization obligations, not a relabelled vector. Bare mathematical vectors carry no implied physical frame.

An integer is $(s, a_0, \dots, a_{k-1})$ with $s \in \{0, 1\}$ and

$$z = (-1)^s \sum_{i=0}^{k-1} a_i \beta^i, \quad \beta = 2^{64}, \quad 0 \leq a_i < \beta.$$

Zero has $s = 0, k = 0$; otherwise $a_{k-1} \neq 0$. There is no format-level bound on k . A resource-limited implementation must return an explicit resource result, never silently discard high limbs. A rational is (p, q) with $q > 0$, $\gcd(|p|, q) = 1$; its zero is $(0, 1)$.

An algebraic real has a primitive irreducible polynomial $P \in \mathbb{Z}[X]$ with positive leading coefficient and an index selecting one of its distinct real roots in increasing order. This fixes its value and canonical mathematical identity. A rational isolating interval is a separate witness, not part of that identity. Checking irreducibility, isolating roots and obtaining a canonical minimal polynomial are explicit exact-algebra obligations. A general operator term has canonical *syntax*; different syntax may denote the same real.

18.2 Exact arithmetic on finite machine words

The literal full adder takes three bits $a, b, c \in \{0, 1\}$ and returns

$$s = a \oplus b \oplus c, \quad c' = (a \wedge b) \vee (c \wedge (a \oplus b)), \quad a + b + c = s + 2c'.$$

For subtraction with incoming borrow d , its bit and outgoing borrow are

$$r = a \oplus b \oplus d, \quad d' = (\neg a \wedge b) \vee (d \wedge \neg(a \oplus b)), \quad a - b - d = r - 2d'.$$

Feed the outgoing carry or borrow of position i to position $i + 1$, starting with zero. Signed values are handled by separate sign and magnitude, not by discarding an infinite sign extension. A product begins with partial bits $p_{i,j} = a_i \wedge b_j$ in position $i + j$; finitely many full-adder chains sum these shifted rows, retaining the final carry. Exact magnitude comparison scans from the most significant padded bit, with equality flag $e = 1$ and greater flag $g = 0$, updating simultaneously

$$g' = g \vee (e \wedge a \wedge \neg b), \quad e' = e \wedge \neg(a \oplus b).$$

This comparison supplies the branch in division; no approximate sign oracle is needed for integers. The bit equations also establish the limb implementation: $a + b = (a \oplus b) + 2(a \wedge b)$ and the two carry terms are disjoint, giving the full-adder identity by substitution. Summing the bit identities with powers of two telescopes all internal carries and borrows.

Limb storage does not force modular arithmetic. For nonnegative addition, pad the shorter operand with zero limbs, set $c_0 = 0$, and define

$$u_i = a_i + b_i + c_i, \quad r_i = u_i - \beta \lfloor u_i / \beta \rfloor, \quad c_{i+1} = \lfloor u_i / \beta \rfloor.$$

Append the final nonzero carry. A native implementation may realize u_i with two-word operations or bitwise carry logic; it cannot assume that u_i fits one limb. Subtraction with $A \geq B$ uses $c_0 = 0$ and

$$u_i = a_i - b_i - c_i, \quad c_{i+1} = \lfloor u_i < 0 \rfloor, \quad r_i = u_i + \beta c_{i+1}.$$

Signed addition selects addition or magnitude subtraction according to signs and exact magnitude order, then normalizes zero.

Multiplication is finite convolution followed by carrying, starting with $c_0 = 0$:

$$v_j = c_j + \sum_{i+\ell=j} a_i b_\ell, \quad r_j = v_j \bmod \beta, \quad c_{j+1} = \lfloor v_j / \beta \rfloor.$$

Continue carrying until it vanishes. The accumulator v_j is itself an exact multiword quantity; using one overflowing hardware word would violate the rule. For division by $D > 0$, process dividend bits from most to least significant:

$$Q_0 = R_0 = 0, \quad u_i = 2R_i + b_i, \quad e_i = \lfloor u_i \geq D \rfloor, \quad R_{i+1} = u_i - e_i D, \quad Q_{i+1} = 2Q_i + e_i.$$

The dividend prefix processed after step i is $DQ_i + R_i$, with $0 \leq R_i < D$. Signed Euclidean division is the unique $z = dq + r$ with $d \neq 0$ and $0 \leq r < |d|$; sign conversion must preserve this convention, including negative dividends. Integer powers with nonnegative exponent use finite repeated multiplication; negative powers produce a rational only when the base is nonzero.

Proposition 18.1 (No mathematical fixed-width overflow). *For finite integer operands, the preceding finite procedures return their exact sum, difference, product, or Euclidean quotient and remainder, provided every intermediate carry is retained and sufficient storage is available.*

Proof. For addition and subtraction, multiply each defining equation by β^i and sum: the carry terms telescope, leaving the represented operand relation. Convolution represents the product by distributivity, and carrying preserves each coefficient's value. For division, the prefix invariant follows by substitution, while $0 \leq R_i < D$ implies $0 \leq 2R_i + b_i < 2D$ and hence preserves the remainder bound. These arguments are conditional on retaining every term; they establish no bound on time or allocated memory. $\square$

18.3 Positive roots and elementary functions

$\text{SQRT}(x)$ denotes the unique $y \geq 0$ satisfying $y^2 = x$, with domain $x \geq 0$. $\text{ROOT}(n, x)$ requires integer $n \geq 1$: for even n , it is the nonnegative root and requires $x \geq 0$; for odd n , it is the unique real root. An arbitrary polynomial root uses the algebraic root index above, or an explicit existence-and-uniqueness predicate for a real-term root. No unnamed branch is selected. Thus $\sqrt{x^2} = |x|$ for real x , and cancellation to x requires $x \geq 0$. Complex branches are outside this real profile.

The real elementary declarations have ordinary mathematical denotations:

$$\begin{aligned} \exp x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, & \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, & \log x &= \int_1^x \frac{du}{u} \quad (x > 0), \\ \text{atan } x &= \int_0^x \frac{du}{1+u^2}, & \pi &= 4 \text{atan}(1). \end{aligned}$$

These definitions permit finite operator terms for infinite-process values. They are not a request to store an infinite series or an assertion that a finite limb evaluator can return all digits. A certified finite approximation needs separate truncation and rounding bounds. A raw approximate evaluation cannot serve as an exact equality proof.

$\text{ATAN2}(y, x)$ is undefined at $(0, 0)$ and uses range $(-\pi, \pi]$:

$$\text{atan2}(y, x) = \begin{cases} \text{atan}(y/x), & x > 0, \\ \text{atan}(y/x) + \pi, & x < 0, y \geq 0, \\ \text{atan}(y/x) - \pi, & x < 0, y < 0, \\ \pi/2, & x = 0, y > 0, \\ -\pi/2, & x = 0, y < 0. \end{cases}$$

The literal source OTAN2 remains a distinct named template with its own equation and domain; it is never an alias silently replaced by this operator. $\tan x$ requires $\cos x \neq 0$; inverse sine and cosine use ranges $[-\pi/2, \pi/2]$ and $[0, \pi]$ and domain $[-1, 1]$. Floor is the unique integer n with $n \leq x < n+1$; wrapping is $x - P \lfloor x/P \rfloor$ for $P > 0$, with the integer winding retained separately.

18.4 Decisions, partiality and honest result states

Exact rational order is decided by integer cross multiplication with positive denominators. Algebraic order and equality can be decided by exact polynomial arithmetic, common-root analysis and real-root isolation. For unequal algebraic roots, sufficiently refined disjoint rational isolating intervals determine the order; common roots are handled algebraically instead of waiting forever for interval separation. Supplying those algorithms is an implementation obligation.

There is no general equality or sign oracle for unrestricted transcendental terms. COMPARE returns one of LT, EQ, GT, or UNKNOWN, each decided result carrying a valid justification. FLOOR , branch selection and nonzero checks may likewise remain unresolved. Outcomes distinguish a value or a retained term, an unmet domain obligation, an unknown decision, and exhausted resources. These are not numeric zero. IF is lazy: only the selected branch must be defined once its predicate is proved; an undecided predicate preserves the conditional term.

An exact finite decimal input denotes a rational with its declared decimal scale. A measurement with error is a value plus an explicit uncertainty model; giving its central value an exact rational spelling removes no measurement error. This distinction applies equally to station coordinates, timing and force coefficients in an orbital seed.

18.5 Scoped operator graphs and named dependencies

An operator seed contains finite typed records. A node is

(opcode, result type, ordered operand references, opcode payload).

The fixed XOP-R1 catalog in `formal/OPERATOR_CATALOG.md` supplies each opcode's arity, type relation, denotation, domain and payload fields. An unknown opcode is rejected; a user-defined formula is a named template built from catalog operators, with an explicit interface and body. There is no implicit "evaluate arbitrary text" opcode or undeclared exact solver.

Each scope has an ordered node table and ordered exported references. Local node operands refer only to earlier nodes. Lambda, finite sum/product, integral, derivative and solution declarations carry nested scopes. Bound variables use de Bruijn indices: zero denotes the nearest binder, and larger indices count outward through lexical binders. The type environment fixes every index. Substitution shifts indices when passing a binder and is capture-avoiding. Nested scopes cannot reference an outer local node by an accidental numeric ID; captured values enter through explicit binders. Names are display metadata; renaming them does not change variable binding.

References have distinct tags for a local node, bound variable, old state slot, current sample value, dependency export, raw sample timestamp, or converted GPST timestamp. A template application additionally identifies its template and supplies all parameters in declared order. Dependency records carry the complete canonical seed/template bytes in the self-contained form. A digest is a lookup aid; acceptance requires the expected complete bytes and interface. A dependency-only manifest is explicitly an open package until all required bytes are supplied. Dependency and template-call graphs are finite and acyclic in this profile. Finite summation and delayed state do not imply unrestricted recursive template calls.

Sample declarations identify their type, absolute time convention, provenance and binding mode. A time convention has a versioned semantic profile identity; the registered GPST convention supplies its stated scale and epoch, while a custom convention includes a complete dependency/template mapping and its domain. A name alone does not define a clock, resolve a leap second or prove monotonicity. An inline sample is an indexed literal accompanied by its absolute timestamp; an external sample is a declared input requirement. `SAMPLE` selects the value bound to the current logical step, with its timestamp retained. Missing data is unresolved input, not a guessed next sample. Raw-time reference tag 5 returns the sample's exact rational in its declared source-clock coordinate. Converted-time tag 6 explicitly applies the versioned mapping and returns GPST SI seconds; the value and both timestamp references read one immutable sample snapshot. Integration arguments are explicit expressions such as $t = \tau_{\text{GPST}} - \tau_{\text{epoch}}$ and $h = \tau_{\text{GPST}} - \tau_{\text{previous}}$, with matching time units and epoch convention. A custom raw clock tick is not implicitly a second, and a missing mapping cannot be replaced by an assumed offset. A modular phase, spatial key, hash or template identifier does not supply the state, timestamp or sample bytes it refers to.

18.6 Canonical variable-length wire contract

The canonical semantic stream begins with the eight ASCII octets `XOPSEED1`, followed by profile numbers (1, 0) and the ordered tables specified in the catalog. Every natural number uses minimal unsigned base-128 encoding: digit payloads occupy seven low bits, the high bit marks continuation, the least significant digit comes first, and the final high bit is zero. Zero is one zero octet; a multi-octet number cannot have a zero final payload. Thus counts and references are not limited to a fixed machine word.

A byte string has a natural-number length followed by exactly that many octets; a sequence has an element count followed by exactly that many schema-defined elements. An integer has a sign octet, a limb count and that many little-endian 64-bit limbs. Rational and algebraic literals use

the canonical value forms already defined. Strings used as identifiers are restricted ASCII byte strings; free-form provenance is opaque bytes and does not participate in name lookup. Types, references and opcode payloads are encoded by their declared tagged schemas. Nonminimal counts, redundant leading limbs, trailing bytes, nonzero padding, invalid references and payload fields outside the selected schema have no canonical decoding.

Canonical here means one encoding for the same ordered, scoped syntactic record. It does not mean that all mathematically equal expressions, all sharing patterns, or all valid topological node orders receive one universal byte string. Structural sharing is permitted only for identical typed syntax under the same bindings and immutable dependency identities. The strategy history records any rewrite that changes the syntax while preserving its meaning.

Proposition 18.2 (Unique literal and record decoding). *Given the catalog, a finite well-formed canonical stream has one decoded record and its canonical re-encoding returns the original stream.*

Proof. The continuation terminator uniquely delimits each minimal natural-number encoding. Lengths and fixed-width limbs uniquely delimit strings and literals. The record schema fixes field order and the selected tag fixes each variant. Induction over finite sequences and nested scopes therefore determines every field and its boundary. Rejecting alternative zero forms, redundant lengths and trailing bytes makes re-encoding identical. This is a structural proof of the specified grammar, not evidence about a parser implementation. $\square$

To pack literal one-bit planes, let C be the canonical octet stream, of length L . Pad its last block with zero octets to a multiple of 512 octets. Interpret each block as 64 little-endian words $w_0, \dots, w_{63}$ and set

$$a_{r,c} = ((w_r \gg c) \& 1), \quad p_c = \sum_{r=0}^{63} a_{r,c} 2^r.$$

The envelope is eight ASCII octets XOPPLAN1, then the minimal encoding of L , then the plane words in increasing block and plane order. Its size fixes the required block count; after inverse transpose, padding must be zero and the first L octets must decode as XOPSEED1. Empty input has no seed grammar, although the underlying byte permutation can act on an empty byte string.

Proposition 18.3 (Bit-plane involution and information accounting). *The padded block transform T satisfies $T(T(W)) = W$. Including L makes packing injective on canonical seeds. The transform changes organization, not information content or the number of bits required by its uncompressed padded blocks.*

Proof. The output bit at (c, r) is the input bit at (r, c) . Transposing again restores every coordinate. The explicit length removes padding ambiguity. Consequently distinct canonical inputs cannot share an envelope. A bijective permutation cannot itself identify multiple arbitrary inputs with a smaller code space. $\square$

This envelope is newly named. It neither changes nor retroactively reinterprets R10's uint64/binary64 codec. A conversion from R10 must declare whether it preserves raw bits, converts finite values to dyadic rationals, or replaces an approximation by a separately justified symbolic model.

18.7 Causal recurrence and the literal two-mask kernel

The state table declares each slot's type, initial expression and next-state expression. Reading slot j at logical step t is precisely

$$s_j(0) = \text{init}_j, \quad s_j(t+1) = E_j(s(t), d(t)); \quad s_j = \text{DELAY}(\text{init}_j, E_j).$$

All right-hand sides read the same old state and the same input snapshot. The new tuple commits simultaneously after all required domains and decisions are resolved. A self-loop in an instantaneous operator DAG is inadmissible: every recurrence cycle must cross a declared `DELAY`. An algebraic fixed-point equation is a separate solution declaration needing existence, uniqueness and branch obligations; it is not an execution order.

For w -bit masks, define the retained whole-word operation

$$a = x \wedge A \wedge V, \quad n = a \wedge N, \quad h = \text{popcount}(n \wedge B),$$

$$\text{ASA}(x; V, A, N, B) = \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases}$$

The two ordered sets and literal J–K update are

$$\begin{aligned} x_t &= E_X(q_t, d_t), & y_t^A &= \text{ASA}(x_t; V_A, A_A, N_A, B_A), \\ y_t &= \text{ASA}(y_t^A; V_B, A_B, N_B, B_B), \\ J_t &= E_J(q_t, y_t, d_t), & K_t &= E_K(q_t, y_t, d_t), \\ q_{t+1} &= [(J_t \wedge \neg_w q_t) \vee (\neg_w K_t \wedge q_t)] \wedge V_q. \end{aligned}$$

Here $\neg_w q = (2^w - 1) \oplus q$ and each word expression has $\text{Bits}(w)$ type. Both intermediate (a, n, h, y) tuples are observables. Absorption zeroes the whole output of its stage; it neither clears selected bits individually nor silently substitutes a hold transition. The mathematical width w is explicit; the retained executable orbital kernel still uses its specified 32-bit word profile.

Proposition 18.4 (Causal uniqueness and conditional finite periodicity). *For fixed initial states and an input stream, a typed finite instantaneous DAG whose required operations are defined determines at most one next state at each step. If all operations are total on the reachable states, it determines a unique state stream. A fixed autonomous transition on m active Boolean state bits is eventually periodic, with preperiod plus period at most 2^m .*

Proof. Topological induction determines every demanded instantaneous node on the admitted branches from old-state and input values; simultaneous commitment then determines the next tuple. Induction on steps gives the stream. For the finite autonomous case, among $2^m + 1$ states two coincide; determinism makes their subsequent tails coincide. Changing the input, strategy, timestamp, unbounded integer state or physical continuous state does not meet that finite-autonomous hypothesis. $\square$

18.8 Vectors, calculus and orbital expressions

Vector and matrix operators are typed compositions of scalar arithmetic: $u \cdot v = \sum_i u_i v_i$, $\|u\| = \sqrt{u \cdot u}$, and $(AB)_{ij} = \sum_k A_{ik} B_{kj}$. Cross product is restricted to three components. A normalized vector requires nonzero norm; a matrix inverse requires a square matrix and nonzero determinant. Polynomial, Chebyshev, Legendre, finite Fourier and rational expressions expand to declared finite templates. The catalog includes finite sum/product, differentiation, gradient and Jacobian binding, definite integration and an initial-value solution declaration.

$\text{DIFF}(f, x)$ denotes a derivative only where the stated limit exists. $\text{INTEGRAL}(f, a, b)$ denotes an oriented Riemann integral when the finite interval integrand is integrable; an improper integral has a separate limit declaration. $\text{ODE}(F, t_0, s_0, t, I)$ denotes the value at $t \in I$ of the unique continuous state satisfying

$$s(u) = s_0 + \int_{t_0}^u F(v, s(v)) dv \quad (u \in I),$$

with a Riemann-integrable right-hand side along that state. On continuous pieces the solution is differentiable; a coefficient jump need not admit a derivative at the junction. The named continuous FLOW uses this integral-equation meaning. State may be a tuple of position, velocity and other components with different dimensions, including explicitly framed vectors. Each derivative component retains its state's nominal frame and has its state dimension divided by time. Rotating-frame terms must occur in F ; the type annotation does not invent them. A local existence theorem or local regularity condition alone is not a certificate of a global interval. If existence, uniqueness or continuation is unresolved, the node remains a declaration with obligations, rather than an evaluated exact orbit.

A finite RK4 template can be represented exactly as expressions, for example

$$\begin{aligned} k_1 &= F(t, s), & k_2 &= F(t + h/2, s + hk_1/2), \\ k_3 &= F(t + h/2, s + hk_2/2), & k_4 &= F(t + h, s + hk_3), \\ \text{RK4}_h(t, s) &= s + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

Exact evaluation of this finite expression does not make it equal to the continuous ODE solution. Discretization, uncertain initial conditions, unmodelled forces and discontinuous external events remain separate mathematical or physical questions. All force and station expressions must retain their units, frames, absolute time and explicit singular domains. A finite symbolic term is not an error bound on its physical denotation.

18.9 Proof rules and strategy self-improvement

Let Γ contain typed assumptions and proven domain predicates. A permitted rewrite carries the judgement

$$\Gamma \vdash e \simeq e' : \tau, \quad \text{Dom}_\Gamma(e) = \text{Dom}_\Gamma(e'), \quad \llbracket e \rrbracket = \llbracket e' \rrbracket \text{ on that domain.}$$

Domain equality is essential: simplifying x/x to 1 without $x \neq 0$ changes the defined inputs. The proof vocabulary contains assumption, reflexivity, symmetry, transitivity, typed congruence, capture-avoiding substitution, definition expansion and explicit ordered-field/domain lemmas. Branch rules require their predicates; differentiation and integration rewrites additionally require their analytic hypotheses. A rule name or a source digest alone is not a proof of its side conditions.

Structural common-subexpression sharing, exact constant folding and eliminating a pure unused node preserve the denotation of the demanded exports when the typing, scope and domain obligations are satisfied. Reassociation is justified in exact integer/rational arithmetic but is not silently applied to R10 binary64 execution. Eliminating an observed ASA trace, changing a sample read, or changing query cadence is not observationally neutral merely because a final word matches.

For stateful replacement use a relation $\mathcal{R}$ between old and new states. Its proof must relate the initial states and, for every admitted input, preserve definedness, all declared observations and the relation after one synchronous step. Induction then gives equal observable streams. Absolute timestamps and selected sample identities belong to the observations whenever a query schedule is exported; a change in those requires a separately named strategy contract.

A strategy version is an immutable tuple

$$S_v = (\text{rule catalog, priority policy, cost model, proof system ID, resource policy}).$$

It may propose a new expression graph or a new strategy S_{v+1} . Acceptance requires a finite proof object under the already accepted proof system, a preserved input/output contract, and the stated termination/resource argument. The proposal cannot justify itself by replacing its own checker

or dropping an unmet premise. A proof-system change creates a new explicit trust boundary. Strategies and proposal histories may be unbounded, so finite Boolean periodicity does not apply to the combined evolving system.

Proposition 18.5 (Conditional preservation under strategy replacement). *Suppose the accepted proof rules are sound for the declared denotation, each accepted replacement carries a valid domain-preserving proof or stream simulation relation, and proof checking admits only valid derivations. Then any finite chain of accepted replacements preserves its declared observations.*

Proof. A sound proof preserves one replacement's domain and observations. Compose these equalities, or the stepwise stream relations, along the finite chain. Induction gives preservation after every accepted replacement. This establishes a conditional theorem; it supplies neither a proof-checker implementation nor an unconditional claim that proposed derivations are valid. $\square$

Logical strategy improvement can now be stated precisely. Sharing an identical pure subgraph reduces duplicated graph storage; memoizing it avoids a repeated evaluation only when the complete binding, state step and dependency identity are the same. A strictly decreasing natural-valued rewrite measure proves termination of that rewrite phase; a finite proposal budget also terminates a search without proving it finds a better expression. A predicted reduction in operator count is a claim under the declared cost model, not a measured speedup or a guarantee of lower big-integer time, memory use, physical error or global optimality. These are reasoning-based improvements in the retained XOP-R1 strategy contract. Full XOP execution and machine proof remain future obligations; R15 implements and checks the explicitly named subsets.

19 The exact kernel, physical fields and acquired observations

19.1 A shared state with explicit causal boundaries

The literal operators and one-bit word grammar are fixed in Sections 18 and A. Their application to physics uses the delayed state

$$s_n = (q_n, W_n, \xi_n, \ell_n, c_n; \mathcal{E}_n). \quad (79)$$

Here q_n is the ASA/NA-JK word, W_n an optional WANT word, ξ_n the selected finite physical-model state, ℓ_n consumed-record memory, c_n controller/calibration memory, and $\mathcal{E}_n$ their exact expression roots and immutable dependencies. Unused components are omitted explicitly. An actual plant state $x_{\text{phys}}(t)$ is outside this digital record. Its unknown parameters and disturbances do not become known literals by naming ξ .

Let R_n be a completed immutable measurement record available at control time t_n . Its acquisition interval, source timestamp, availability time, channel validity and calibration binding remain distinct. The following is the selected sample-and-hold chronology, with $h_n = t_{n+1} - t_n > 0$:

$$\begin{aligned} (\widehat{m}_n, e_n) &= C(R_n, c_n), & d_n &= E_D(\widehat{m}_n, e_n, t_n, c_n), \\ (x_n, y_n, J_n, K_n, q_{n+1}^{\text{base}}) &= \mathcal{W}(q_n, d_n), \\ q_{n+1} &= \begin{cases} [(J_n^* \wedge \neg_w q_n) \vee (\neg_w K_n^* \wedge q_n)] \wedge P, & \text{event adapter enabled,} \\ q_{n+1}^{\text{base}}, & \text{event adapter disabled,} \end{cases} \\ (W_{n+1}, \ell_{n+1}, c_{n+1}, u_n) &= \mathcal{A}(W_n, \ell_n, c_n, R_n, q_n, q_{n+1}). \end{aligned} \quad (80)$$

Here P is the present mask. The enabled J_n^*, K_n^* are defined by (83); the command adapter reads this same final q_{n+1} . The model step and its predicted observation are then

$$\begin{aligned} \xi_{n+1} &= \Psi_{\Pi, h_n}(\xi_n, u_n, v_{[t_n, t_{n+1}]}; \theta), \\ \widehat{M}_{n+1} &= O_{\Pi}(\xi_{[t_n, t_{n+1}]}, \theta). \end{aligned} \quad (81)$$

$\mathcal{W}$ is exactly the declared E_X , two ordered independent ASA/NA mask sets, E_J, E_K and synchronous JK composition. All its J/K expressions read the same old q_n ; y_n is the second whole-word output. $C, E_D, \mathcal{A}$ are pure typed templates with complete bodies, including their projection and domain rules. A profile without a WANT transition defines $W_{n+1} = W_n$. Enabling a historical WANT transition requires its complete versioned body; the name W64 alone selects none.

Ψ denotes either the separately specified continuous solution over the interval or a selected finite update. It cannot silently alternate between them. $\mathcal{O}$ needs the selected trajectory or an explicitly defined acquisition reconstruction; two finite endpoint states alone do not define its integral. v comprises declared external loads/inputs; a continuous input needs its finite function definition or an explicit acquisition/model binding. $\hat{M}$ is a model prediction. The actual next sample is separately acquired from x_{phys} and its instrument; it is never substituted by this prediction without an explicitly selected synthetic-input profile. Sensor latency enters the next availability time. A measurement completed after t_n cannot affect the command already applied at t_n . Changing h_n changes the recurrence.

All logical next roots are evaluated from one snapshot and commit together. The physical command and interval schedule form a separate timed interface; an unimplemented mathematical transition is not an actuator. The plant's control authority, command limits and timeout/held-command behavior must be specified by the chosen interface when that interface is used.

19.2 Events, non-values and exact information retention

For the retained Operator I observation, the Boolean event is

$$e_n = X_n \wedge Y_n \wedge C_{t,n} \wedge V_n \wedge \neg S_n \wedge \neg F_n. \quad (82)$$

These are acquired/calibrated threshold, coincidence, validity, saturation and fault facts; they are not inferred from the word masks. The optional event-lane adapter binds one ordered site/session stream, a reserved present-bit mask M_e , and the last consumed identity ℓ_n . For a valid ordered record,

$$\begin{aligned} p_n &= e_n \wedge \text{new}(R_n, \ell_n), & E_n &= \begin{cases} M_e, & p_n, \\ 0, & \neg p_n, \end{cases} \\ J_n^* &= (J_n \wedge \neg_w M_e) \vee E_n, & K_n^* &= (K_n \wedge \neg_w M_e) \vee E_n. \end{aligned} \quad (83)$$

The enabled reserved lane holds for $p_n = 0$ and toggles for $p_n = 1$; other lanes have the base one-step JK result. Later base expressions can read the changed lane, so this is not a global noninterference theorem. Disabling the adapter restores the base J/K expressions. Enabling starts with a recorded baseline without a pulse and consumes only later records. Tracker and word commit atomically. Repeated identical records are a no-new-pulse case; the same identity with changed bytes, calibration or time binding is invalid.

Missing, stale or calibration-unresolved demanded input is an explicit non-value, not a valid false event. Likewise an unknown real comparison is not zero. A selected application's defined failure result is different: the inherited CGK word-first failure can be represented by the single tagged value

$$(q^+, \xi, t, \text{chart history}, \text{failed} = 1), \quad (84)$$

which updates the word while keeping those other fields. A lazy failure branch does not demand its undefined inverse. This remains one atomic transition; it does not authorize a partial commit of a non-value operator. Physical time and external plant evolution continue if a digital transition does not commit.

An exact checkpoint retains sufficient exact operands, roots, source samples and dependencies for the declared state. A rounded rendering is an output projection. Feeding it back instead of

the retained state creates a different recurrence. Source rounding, clipping, modulo and ADC quantization remain explicit functions when selected; the exact word envelope preserves their input and output representations without reversing lost source information.

19.3 W64 is a finite chart/control record

The retained WANT.W64 fields are fixed below. They are distinct from the JK word, spatial UGTS keys and the Operator I measurement packet.

Bits	Field	Interpretation
63	χ	channel/control bit
62:52	ρ	signed 11-bit log-radius code
51:41	ϑ	unsigned phase modulo 2048
40:30	z	signed 11-bit axial code
29:20	Φ	unsigned 10-bit register
19:15; 14:10	depth; phase	two independent 5-bit registers
9:3; 2; 1:0	LUT; κ ; cycle	7-bit index; seam bit; 2-bit selector

Extraction is $(W \gg s) \wedge (2^w - 1)$; signed interpretation is u for $u < 2^{w-1}$ and $u - 2^w$ otherwise. With positive scales, an actual chart is

$$\begin{aligned} r &= r_W 2^{\Delta_\rho \text{signed}_{11}(\rho)}, & \theta &= 2\pi \vartheta / 2048, & z_{\text{phys}} &= \Delta_z \text{signed}_{11}(z), \\ x &= o + r \cos \theta e_1 + r \sin \theta e_2 + z_{\text{phys}} e_3. \end{aligned} \quad (85)$$

The origin o , orthonormal frame (e_1, e_2, e_3) , scales and their calibration are declared inputs. This code has no exact radius-zero point. Decoding the integer tuple is lossless; recovering an arbitrary pre-quantization location from that tuple is not defined. A changed origin or frame needs its full transform. The corresponding log-radius chart uses $\log(r/r_U) = \log(r_W/r_U) + \log(2)\Delta_\rho \text{signed}_{11}(\rho)$. Neither this geometry nor an OTAN2 spelling supplies an optical transfer.

The preserved integer pinion uses $s = (2H_4)v$ and

$$\widehat{A}(v) = \left(\mathcal{R}\left(\frac{106039s_0}{131072}\right), \mathcal{R}\left(\frac{40503s_1}{131072}\right), \mathcal{R}(s_2/2), \mathcal{R}(-s_3/2) \right), \quad \mathcal{R}(x) = \text{sgn}(x) \lfloor |x| + 1/2 \rfloor. \quad (86)$$

Its input is the selected normalized dimensionless coordinate tuple; any subsequent clip, seam or register projection belongs to its full transition template. This exact integer expression is distinct from the algebraic A_ϕ below. Retaining a pre-projection expression does not entitle a control decision to bypass the selected projection.

19.4 Optical fields close through a measured acquisition

The coupled optical chain is

$$U_{\text{det},\lambda} = H_{z,\lambda} \left[A_{\text{ap}} t_\lambda(T, a, \zeta) e^{i\varphi_{\text{lens},\lambda}} U_{\text{in},\lambda} \right]. \quad (87)$$

The electromagnetic chapter supplies Maxwell, material storage/loss and admitted propagation reductions. Here A_{ap} is the physical aperture, a, ζ material states, and φ_{lens} a declared phase field. In a scalar fixed-plane monochromatic profile, normalize U so $|U|^2$ is power per area. In its spectral version use power per area per wavelength; with that corresponding convention define

$$P_\lambda(t) = \int_{\mathcal{D}} |U_{\text{det},\lambda}(x, t)|^2 dA, \quad \mu_k = \int_{t^-}^{t^+} \int_{\Lambda_k} \eta_k(\lambda) \frac{\lambda P_\lambda(t)}{hc} d\lambda dt. \quad (88)$$

In the wavelength integral P_λ is spectral power density; for a discrete monochromatic mode use its power and a finite wavelength sum instead. The spectral power-sum profile requires mutually incoherent components, separately resolved orthogonal frequency channels, or acquisition averaging that removes inter-frequency beat terms. For arbitrary coherent multi-frequency illumination over a finite window, the detector's full field-response equation must retain the relevant cross terms instead. The efficiency $0 \leq \eta_k \leq 1$ assumes at most one primary detection per incident photon and the selected band. μ_k is an ideal expected primary count, not the observed count. Dark counts, gain, saturation, dead time and count statistics belong to the detector model when relevant. A thermal or analogue instrument instead uses its calibrated measurand and acquisition weighting. The actual code is imported with its original bits and exact code-unit map; the measurement chapter supplies its discrepancy, noise and uncertainty terms.

Use one input/field/material energy ledger: polarization storage, transmitted or reflected flux, absorbed energy and heat are the transfers already defined by the selected physical equations. Sensor gain or reporter output does not create a second incident budget. For an ideal binary aperture, $A_{\text{ap}}^2 = A_{\text{ap}}$ and $\|A_{\text{ap}}U\|^2 \leq \|U\|^2$. A general material transmission t is not idempotent; the ASA rule that one hit zeros an entire word is not supplied by this pointwise aperture equation.

19.5 An exact passive target for the algebraic pinion

For dimensionless coordinates, or separately declared power-normalized optical amplitudes, define

$$H_4 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad \phi = \frac{1 + \sqrt{5}}{2}, \quad A_\phi = \text{diag}(\phi, \phi^{-1}, 1, -1)H_4. \quad (89)$$

$H_4 H_4^T = I$, so the singular values of A_ϕ are $\phi, \phi^{-1}, 1, 1$. Thus A_ϕ is not a passive direct amplitude map for all inputs: $H_4^T e_1$ is a unit input amplified by $\phi > 1$. Instead

$$B = A_\phi / \phi = \text{diag}(1, \phi^{-2}, \phi^{-1}, -\phi^{-1})H_4, \quad B^T B \leq I. \quad (90)$$

For each diagonal entry d_j , pair the mode with an auxiliary mode using

$$O_j = \begin{pmatrix} d_j & \sqrt{1 - d_j^2} \\ -\sqrt{1 - d_j^2} & d_j \end{pmatrix}, \quad O_j^T O_j = I. \quad (91)$$

After H_4 , zero classical auxiliary input amplitude gives B on the selected outputs and carries the remaining power to side ports. This proves an ideal enlarged norm-preserving target with exact algebraic coefficients. A negative d_j requires relative phase; side-port energy is retained. A quantum use also needs its auxiliary states and noise model.

The identity $A_\phi v = \phi(Bv)$ permits an exact symbolic scale factor. It does not restore lost output power or remove measured amplitude uncertainty. An active direct map needs supplied gain and its noise model. A digital photonic encoding of the bits has a different relation between optical power and represented numeric magnitude; it is governed by the next contract.

19.6 Physical realization and minimal specializations

Let $\text{Dec}(p) = s$ relate an admitted physical carrier state to a finite digital state. A physical step $\mathcal{H}_\delta$ under admitted disturbance δ realizes the defined exact step F_Π only if

$$\text{Dec}(\mathcal{H}_\delta(p, \text{Enc}(u))) = F_\Pi(s, u) \quad \text{for every reached admitted } p \text{ with } \text{Dec}(p) = s. \quad (92)$$

The output must again be an admitted representative, or undergo a declared restoration. With the same acquired snapshots and initial decoded state, induction proves agreement for every defined finite prefix within the declared resources. Encoding, carry/borrow, memory, synchronization and atomic commit are obligations of a selected realization; the one-bit transport does not choose a physical carrier or discharge them. For an intensity bit with low level at most a_0 , high level at least a_1 and additive error bounded by ϵ , a threshold satisfying $a_0 + \epsilon < \theta < a_1 - \epsilon$ proves correct decoding. This conditional margin preserves a logical bit, not an unknown analogue measurand to infinite precision.

Only the dependency closure required by an application is imported:

- **Digital.** Omit the plant and detector; retain exact operands, word/expression state, sample ports and declared outputs. A physical hardware claim additionally needs (92).
- **Analogue/mechanical/thermal.** Use the selected finite or continuum physical equations, measured coefficients and boundary ports. A finite basis is an explicit approximation to the continuum, without prescribing a raster. The exact mechanical midpoint identity applies only to its own fixed linear profile; it does not assert conservation for every sampled coupled model.
- **Photonic.** Bind (87) to the material and propagation equations and (88) to the instrument. Choose an analogue transfer or a digital realization explicitly.
- **Orbital/spatial.** Choose $\dot{r} = v$, $\dot{v} = a_{\Pi}(t, r, v)$ with its gravity/forcing, time and frame model. A fixed station observes $z = L(M(t)r - r_g)$, $D = \|z\|$, azimuth $\text{atan2}(z_E, z_N)$ and elevation $\text{atan2}(z_U, \sqrt{z_E^2 + z_N^2})$ on their nonsingular domains. This is simultaneous geometry. A received-signal observation needs its emission time, propagation and clock model. In the inherited orbit seed, JK chooses query cadence and does not enter a_{Π} . Store full time as well as phase: $k - k_0 = Pw + r$, $0 \leq r < P$; (w, r) reconstructs k .
- **Optional attitude.** A unit quaternion Q has kinematic equation $\dot{Q} = \frac{1}{2}Q \otimes [0, \omega]$. A selected sampled correction defines $U = Q + h(\dot{Q}_\omega - \beta e)$ and $Q^+ = U/\sqrt{U^T U}$ for $U^T U > 0$. The residual/gradient e , calibrated rate, gain, frame, bias and missing-sample branches need their complete profile. This proves unit norm after a defined normalization; it does not supply heading, position or filter convergence.

These bindings retain the R6/R10 word lineage, R11 exact grammar, R12 word, optical and event derivations, and R13 physical/measurement contracts. Historical application recipes and results remain versioned dependencies when selected; none is silently promoted into evidence for a new physical device.

20 A minimal physical-law layer for application kernels

A physical law supplies relationships, constrains coupling and supports implementation choices only when its hypotheses are part of the selected model. R14 integrates this law-profile contract, introduced in R13, with the exact representation and state feedback restated in this document.

The unified profile is XOP-UNIFIED-FIELD-R1; its physical-law layer retains the XOP-PHYSICS-R1 definitions. Its mathematical bodies use the existing XOP-R1 operators; it does not allocate new opcodes or claim a new executable backend. A physical specialization imports only the laws and closures needed by its declared interface. A purely digital permutation does not need a continuum electromagnetic model merely because real hardware will eventually execute it.

20.1 Five distinct parts of a physical specialization

Part	What it supplies	Example in the existing model
Balance law	Conserved quantity, flux or production relation	Charge continuity; energy and momentum balance
Constitutive model	Material or component response under stated conditions	Polarization dynamics; thermoelastic free energy
Reduction	A finite or simplified model with its scope	Fixed modal basis; scalar optical field; lumped temperature
Discrete evolution	A selected finite state transition	Existing backward Euler; new energy-balanced midpoint
Observation	How physical state produces recorded evidence	Acquisition window, calibration, quantization and uncertainty

Conservation equations alone do not specify a material. A constitutive law alone does not provide boundary data. Exact arithmetic alone does not select a conservative time update. A recorded count alone does not fix a physical measurand. Keeping these parts explicit allows specialization without hidden changes of meaning.

20.2 Normative profile contract

A selected physical profile declares the finite record

$$\Pi = (\mathcal{T}, \mathcal{X}, \mathcal{U}, \mathcal{Y}, \Theta, \mathcal{B}, \mathcal{C}, \mathcal{A}, \mathcal{N}, \mathcal{O}, \mathcal{E}). \quad (93)$$

$\mathcal{T}$ contains units, frames and time conventions. $\mathcal{X}$ declares state and initial conditions; $\mathcal{U}, \mathcal{Y}$ declare inputs and outputs. Θ contains parameters with source identities and joint uncertainty. $\mathcal{B}$ lists applicable balance laws. $\mathcal{C}$ supplies constitutive closures. $\mathcal{A}$ lists approximation and admissibility hypotheses. $\mathcal{N}$ identifies a finite representation and the chosen update. $\mathcal{O}$ defines acquisition and calibrated observations. $\mathcal{E}$ records the claimed output bounds and evidence status.

Each law entry has a qualified name/version, argument types, equation body, domain, boundary terms, source and a status. Status distinguishes a definition, a derived identity, an adopted physical law, a constitutive assumption and a measured parameter. The same source tag cannot turn one status into another. The editable normative field contract is `formal/PHYSICS_PROFILE.md`. This is a record definition, not a claim that a new parser implements it.

For a finite state model, the mathematical content can be written

$$\dot{x} = f_{\Pi}(t, x, u; \Theta), \quad y = g_{\Pi}(t, x, u; \Theta), \quad B_j(t, x, \dot{x}, u; \Theta) = 0, \quad (94)$$

with declared inequalities and initial/boundary conditions. The B_j are relations to derive or impose consistently, not extra independent equations to append blindly to a fully determined system. Redundant balances are identified; inconsistent initial data or incompatible closures do not define an admitted solution. A finite ODE must meet the existing existence and uniqueness domain. Algebraic constraints need elimination or a separately defined constrained formulation; they are not silently handled by ODE.

20.3 Required connections to the existing seed

Existing quantity	Connection in the unified profile
$t_\lambda, H_{z,\lambda}$	Chosen electromagnetic transfer, material response, boundary data and power normalization
$P_{\text{abs}}, P_{\text{heat}}$	A shared energy ledger distinguishes absorption, stored excitation and dissipated heat
M, C, K and strain	Fixed geometry/basis or declared motion, force/work convention and constitutive response
T, ζ, a	Thermal/material energy stores and rates with the relevant temperature and coupling domains
ASA/NA, JK and actuator command	The word selects an action; a separate actuator equation supplies force, voltage, light or heat
WI measurement and calibration	Acquisition history, physical units, clock/frame binding, uncertainty and measurand equation
Exact operator graph	Captured physical inputs and hypotheses, complete finite operands, explicit unresolved outcomes

For example, a selector $b \in \{0, 1\}$ may choose $K(b) = K_0 + b(K_1 - K_0)$. It does not by itself specify the work required to switch stiffness. For an unchanged coordinate q , that storage change is $q^T(K_1 - K_0)q/2$ and must be attributed to the switching interaction. Signed damping or stiffness remains mathematically expressible; a passive energy theorem applies only when its sign hypotheses hold. An active element needs its actual supply and constitutive model.

20.4 Representation preserves laws only through an explicit map

Let $x = \mathcal{R}(z)$ reconstruct a finite physical model from encoded state z , and let F_h be a chosen digital transition. A continuous-law identity does not automatically give

$$\mathcal{B}_{j,h}(\mathcal{R}(z), \mathcal{R}(F_h(z, u)), u) = 0. \quad (95)$$

Here $\mathcal{B}_{j,h}$ is a separately defined discrete balance, rather than the continuous residual B_j with its arguments silently changed. The discrete relation has to be derived for the selected step. The law may instead provide an error bound or a nonzero residual ledger. Likewise, a finite basis is a declared approximation to a continuum field: exact evaluation of its coefficients and trajectory does not erase omitted modes. The electromagnetic and mechanical chapters state the necessary finite-representation and integral domains.

One-bit word packing preserves the encoded operators, parameters and inputs. It does not turn a finite word mask into an electromagnetic absorber, identify a code bit with a photon, or establish that a material follows a named law. Those are explicit realization mappings. The photonic refinement contract in Section 19 governs a proposed hardware executor.

21 An exact finite step with a derived mechanical energy balance

This addition supplies a specific useful kernel specialization. It removes algorithmic energy loss from the fixed linear model's discrete balance while retaining physical damping. It does not replace the existing CGK or orbital profiles silently and does not assert exact continuous-time trajectories.

21.1 Selected model and units

Choose a finite number of length-valued coordinates q in a fixed declared modal basis, velocity v , and a fixed reference configuration. Let

$$\dot{q} = v, \quad M\dot{v} = \bar{f} - Kq - Cv, \quad M = M^T > 0, \quad K = K^T \geq 0, \quad C = C^T \geq 0. \quad (96)$$

Here the displayed $\bar{f}$ denotes the force selected for the interval's discrete supply; a held-force physical model may use the same constant value. The quantities M, C, K have units kg, kg/s and kg/s² when all q coordinates have length units. Generalized coordinates of mixed dimensions require the inherited tuple/block rules, not a false uniformly dimensioned matrix. The storage is

$$E(q, v) = \frac{1}{2}v^T Mv + \frac{1}{2}q^T Kq, \quad \dot{E} = v^T \bar{f} - v^T Cv. \quad (97)$$

The continuous identity follows by differentiating the two quadratic terms and substituting the model. Its damping term becomes heat or another named output store in a coupled physical model, as derived in the energy chapter.

The parent already defines backward Euler for a more general signed CGK profile. Under fixed passive coefficients its storage change includes

$$-h(v^+)^T Cv^+ - \frac{1}{2}(v^+ - v)^T M(v^+ - v) - \frac{1}{2}(q^+ - q)^T K(q^+ - q) \quad (98)$$

in the unforced case. The last two terms are numerical dissipation from the selected update. Exact arithmetic preserves those terms exactly; it does not remove them. They cannot be reassigned to measured physical heat without a separate modeling decision. R13 therefore names an alternative finite profile.

21.2 PHYS-MIDPOINT-LINEAR-R1

For $h \geq 0$, write $\bar{q} = (q^+ + q)/2$ and $\bar{v} = (v^+ + v)/2$. Define

$$q^+ - q = h\bar{v}, \quad M(v^+ - v) = h(\bar{f} - K\bar{q} - C\bar{v}). \quad (99)$$

These undivided equations also define the $h = 0$ identity step. Eliminate q^+ to obtain the explicit finite solve

$$A_h = M + \frac{h}{2}C + \frac{h^2}{4}K, \quad (100)$$

$$b_h = (M - \frac{h}{2}C - \frac{h^2}{4}K)v - hKq + h\bar{f}, \quad (101)$$

$$v^+ = A_h^{-1}b_h, \quad q^+ = q + \frac{h}{2}(v^+ + v). \quad (102)$$

Proposition 21.1 (Existence and exact field closure). *For the stated symmetric sign conditions and $h \geq 0$, A_h is positive definite. Consequently the step is uniquely defined. If all supplied entries are rational, both output vectors have exact rational entries.*

Proof. For $w \neq 0$, $w^T A_h w = w^T M w + (h/2)w^T C w + (h^2/4)w^T K w > 0$. Thus A_h is invertible. A finite nonsingular linear solve over the rationals uses rational field operations and has rational output. No real root or transcendental decision is required for this selected rational profile. $\square$

Proposition 21.2 (Discrete power balance). *For the same fixed model,*

$$E(q^+, v^+) - E(q, v) = h\bar{v}^T \bar{f} - h\bar{v}^T C\bar{v}. \quad (103)$$

In particular, unforced undamped storage is preserved exactly, while the unforced damped step cannot increase it.

Proof. Symmetry gives $\Delta E = \bar{v}^T M(v^+ - v) + \bar{q}^T K(q^+ - q)$. Substitute (99); the stiffness terms cancel, leaving (103). Nonnegativity follows from $C \geq 0$. $\square$

The supplied work $h\bar{v}^T \bar{f}$ and dissipated energy $h\bar{v}^T C \bar{v}$ are the discrete profile's quantities. They are not automatically the exact work and heat integrals of the continuous trajectory. The theorem is a storage identity, not a proof of arbitrary-step trajectory accuracy or a guarantee about every other invariant.

21.3 Exact seed body and causal integration

The template arguments are

$$(q, v, M, C, K, h, \bar{f}) \mapsto (q^+, v^+, W_{\text{in}}, Q_d), \quad W_{\text{in}} = h\bar{v}^T \bar{f}, \quad Q_d = h\bar{v}^T C \bar{v}. \quad (104)$$

Use existing scalar arithmetic and matrix/vector construction/contraction, then `SOLVE_LINEAR` (opcode 87), to realize (102). The field operations, constants $1/2, 1/4$, and input lineage remain exact expressions. The output ledger is part of the template contract. No proof engine or compiled implementation is supplied here.

All operands read the same old snapshot. A completed event may select the interval's parameters according to the existing causal word policy. If that selection changes stiffness or other storage parameters, first account for the switch at the old state, then apply the fixed-parameter step. The force input and its actuator energy source stay explicit. Both output state roots commit together, and any rounded display remains an export.

This narrow profile makes no claim about time-varying mass, moving bases, nonlinear elasticity, contact, nonlinear reporter kinetics, orbital RK4 or arbitrary signed damping. Those applications retain their own definitions. Nonzero algorithmic residuals in another profile are reported as such; the word “exact” cannot turn them into a conservation proof.

21.4 Law residual as an output certificate

For the selected step, define the exact expression

$$r_E = E(q^+, v^+) - E(q, v) - W_{\text{in}} + Q_d. \quad (105)$$

The preceding algebra proves $r_E = 0$ on the template domain. A future compiler can use that identity as a rewrite or verification obligation without numerically comparing two nearly equal energies. If an implementation returns rounded state, its measured residual concerns that projection and cannot replace the expression-level proof. A residual of zero also does not establish that the physical parameters or observed device trajectory are correct.

22 Momentum, energy and admissible physical coupling

The exact seed needs a rule for composing physical models without creating energy at their interfaces. This chapter adopts established momentum and thermodynamic balances, then restates the finite energy-port contract introduced in R13 for the mechanical and thermal states. These are restrictions on a selected physical model. They do not assign physical meaning to a bit pattern by themselves.

22.1 Momentum uses a realized frame and a declared body

For a closed material body in classical continuum mechanics, let $\rho > 0$ be mass density, v velocity, b body force per unit mass, and σ Cauchy stress. In an inertial Cartesian frame the local balance laws are

$$\dot{\rho} + \rho \nabla \cdot v = 0, \quad \rho \dot{v} = \nabla \cdot \sigma + \rho b, \quad \sigma = \sigma^T. \quad (106)$$

Dots here are material derivatives. Stress symmetry assumes the ordinary continuum without independent body couples or couple stresses. A moving control volume instead requires the corresponding mass and momentum fluxes. These balances and their energy/entropy counterparts are stated in [EN1: Banerjee, Basic Thermoelasticity, sections 1.1–1.3](#).

For a rigid body of fixed mass m , centre of mass r_C , body-to-inertial rotation $R_B \in SO(3)$ and constant body-frame inertia $I_B > 0$, the corresponding finite specialization is

$$m \ddot{r}_C = F_{\text{ext}}, \quad I_B \dot{\omega}_B + \omega_B \times (I_B \omega_B) = \tau_B, \quad (107)$$

$$\dot{R}_B = R_B [\omega_B]_{\times}, \quad [\omega_B]_{\times} a = \omega_B \times a. \quad (108)$$

The torque is about the centre of mass; body components and inertial components must not be mixed. The cross term follows from differentiating angular momentum in rotating axes. See [EN2: MIT Dynamics, lecture 28, Euler's equations](#). For this specialization direct contraction gives

$$\frac{d}{dt} \left(\frac{1}{2} m |\dot{r}_C|^2 + \frac{1}{2} \omega_B^T I_B \omega_B \right) = F_{\text{ext}} \cdot \dot{r}_C + \tau_B \cdot \omega_B, \quad (109)$$

because $\omega_B \cdot [\omega_B \times (I_B \omega_B)] = 0$. An inertia estimate, applied torque and orientation measurement are different inputs; a Madgwick attitude estimate alone supplies no torque law.

The XOP frame annotations preserve coordinate identity. Their physical realization also requires origins, alignment and time convention. Transforming to accelerating or rotating coordinates changes the acceleration equation; renaming an inertial force law as a station-frame law is insufficient. Likewise a superposed rigid rotation must not generate elastic strain energy. For a finite-deformation hyperelastic model with deformation gradient $F = \nabla_X x$, $\det F > 0$, the selected objective free energy obeys

$$\Psi(QF, T) = \Psi(F, T), \quad Q \in SO(3), \quad (QF)^T (QF) = F^T F. \quad (110)$$

Material directions, if present, remain explicit reference data; objectivity does not imply isotropy. This condition and the two-point transformation of F are discussed in [EN3: Bouchbinder, Non-Equilibrium Continuum Physics, section 4](#). The existing $\epsilon = \text{sym } \nabla u$ model remains a small-gradient approximation; equation (110) does not extend it to arbitrary finite rotations. A finite-rotation specialization must supply its own objective constitutive equation and compatible finite state.

22.2 A finite port contract with an explicit power proof

Use the finite state x , differentiable storage $H(x)$, and matrices $\mathcal{J}(x) = -\mathcal{J}(x)^T$, $\mathcal{R}(x) = \mathcal{R}(x)^T \geq 0$ and $\mathcal{G}(x)$. Define

$$\dot{x} = (\mathcal{J} - \mathcal{R}) \nabla H + \mathcal{G} u, \quad y = \mathcal{G}^T \nabla H. \quad (111)$$

This is the standard finite port-Hamiltonian form; $u^T y$ is power supplied at the declared boundary. See [EN4: van der Schaft, Port-Hamiltonian nonlinear systems, equations 5–6](#). The symbol $\mathcal{J}$ is unrelated to the JK input word J . Each contraction must be dimensionally compatible. If components of x have different units, their derivative and matrix entries have corresponding component units; they are stored as typed tuples and explicit contractions, not forced into a homogeneous vector type.

Proposition 22.1 (Finite power balance and composition). *For a differentiable solution of (111),*

$$\dot{H} = u^T y - \mathcal{D}, \quad \mathcal{D} = (\nabla H)^T \mathcal{R} \nabla H \geq 0. \quad (112)$$

For two such systems whose ports admit a dimensionally compatible map L , connect $u_1 = L y_2$, $u_2 = -L^T y_1$. Their internal power cancels. Their combined storage is $H_1 + H_2$ and its rate is minus their dissipation plus any remaining external port power.

Proof. Apply the chain rule and use $z^T \mathcal{J} z = 0$ for a skew matrix. The two connected powers sum to $y_2^T L^T y_1 - y_1^T L y_2 = 0$. The entries of L carry any required unit and frame conversion; a force is not silently identified with a velocity. $\square$

If H has a lower bound, integration gives the usual passivity inequality $H(t) - H(0) \leq \int_0^t u^T y \, ds$. This limits extraction relative to storage and supply; it does not by itself prove asymptotic stability, accurate tracking, or a sufficient actuator. Algebraic port connections must also yield a well-defined finite evolution.

Physical port pairs include force/velocity, torque/angular velocity and voltage/current. In a thermal port the energy pairing is absolute temperature times entropy flow, $T \dot{S}_{\text{transfer}} = \dot{Q}$ at that port. Pairing temperature directly with heat power and calling the product power would have the wrong units. Entropy production inside the body is distinct from entropy transported through a port.

22.3 The existing modal mechanics fits this contract

R12 already declares a finite displacement expansion $u_m(X, t) = \sum_i q_i(t) \phi_i(X)$, mass and stiffness matrices from its weak form, and support conditions. For a fixed basis, fixed material parameters, isothermal or explicitly uncoupled mechanics, take

$$M \ddot{q} + D \dot{q} + K q = B u, \quad M = M^T > 0, \quad K = K^T \geq 0, \quad D = D^T \geq 0. \quad (113)$$

Let $p = M \dot{q}$ and

$$H_m(q, p) = \frac{1}{2} p^T M^{-1} p + \frac{1}{2} q^T K q, \quad \mathcal{J} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad \mathcal{R} = \begin{bmatrix} 0 & 0 \\ 0 & D \end{bmatrix}, \quad \mathcal{G} = \begin{bmatrix} 0 \\ B \end{bmatrix}. \quad (114)$$

Then (111) gives exactly (113), with

$$y = B^T \dot{q}, \quad \dot{H}_m = u^T B^T \dot{q} - \dot{q}^T D \dot{q}. \quad (115)$$

The PSD stiffness admits free modes; asymptotic displacement stability does not follow when rigid or undamped modes remain. If M depends on configuration, the derivative of its kinetic energy is part of the dynamics; merely replacing constant M by $M(q)$ in (113) generally misses terms.

Finite modal reduction preserves this displayed energy identity when its coefficient assumptions hold. It does not prove the basis resolves every physical mode. Spatial integrals still require the charts, measures and integrability obligations already specified by R12. A continuum PDE printed above is an origin for the model; it is not an implemented infinite-dimensional XOP operator.

An actual ASA/NA-JK specialization provides an explicit decode from its completed old state to a command, then a transduction law to the physical effort $u(t)$. A voltage command cannot be substituted as a force without that law. The resulting work is

$$W_n = \int_{t_n}^{t_{n+1}} u(t)^T y(t) \, dt. \quad (116)$$

The command may be held over the interval; the velocity generally is not. A bit selecting a positive effort can supply energy through the actuator's power source. Neither bit packing nor a delayed update makes that port passive.

22.4 Convert dissipated work to heat exactly once

For the uncoupled mechanical storage above let $P_d = \dot{q}^T D \dot{q} \geq 0$. Suppose all this dissipated mechanical power thermalizes in the modeled assembly, and fixed or state-declared fractions $w_i \geq 0$, $\sum_i w_i = 1$, distribute it among thermal nodes. Write

$$\dot{H}_m = P_{\text{mech,in}} - P_d, \quad (117)$$

$$C_i \dot{T}_i = P_{i,\text{other}} + w_i P_d + \sum_{j \neq i} G_{ij}(T_j - T_i) + G_{i0}(T_a - T_i), \quad (118)$$

where $C_i > 0$, $G_{ij} = G_{ji} \geq 0$ and $G_{i0} \geq 0$. The stores $U_i = C_i(T_i - T_{i,\text{ref}})$ are an uncoupled constant-capacity model. Their sum with H_m obeys the directly derived identity

$$\frac{d}{dt}(H_m + \sum_i U_i) = P_{\text{mech,in}} + \sum_i P_{i,\text{other}} + \sum_i G_{i0}(T_a - T_i). \quad (119)$$

Internal conduction cancels in pairs, as does mechanical-to-thermal power. Thus P_d is not added again as an external input. If only a fraction thermalizes locally, the remaining energy needs its destination, such as outgoing sound or a separate store. A resistor's VI and an optical absorber's thermalized energy are handled with the same nonduplicating boundary ledger. The distinction between work conversion and entropy creation is treated in [EN5: MIT, Work-to-heat Transduction](#).

22.5 Entropy supplies a second, independent condition

Conserving total energy does not make every proposed heat exchange physical. Restrict the node model to local equilibrium with $T_i > 0$, and define

$$S_i - S_{i,\text{ref}} = C_i \ln(T_i/T_{i,\text{ref}}), \quad T_{i,\text{ref}} > 0, \quad \dot{S}_i = C_i \dot{T}_i/T_i. \quad (120)$$

For one reciprocal conductive link the derived total entropy rate is

$$\dot{S}_{ij} = \frac{G_{ij}(T_j - T_i)}{T_i} + \frac{G_{ij}(T_i - T_j)}{T_j} = \frac{G_{ij}(T_i - T_j)^2}{T_i T_j} \geq 0. \quad (121)$$

A link to an ideal bath at $T_a > 0$ has the same expression after including the bath's entropy change. Local thermalization of $w_i P_d$ contributes $w_i P_d/T_i \geq 0$. Arbitrary external heat, light, matter and work ports must carry their own energy and entropy accounting; an incident optical field's entropy flux is not assigned the thermal formula P_{optical}/T_i merely because its energy is later absorbed.

In local continuum heat conduction, with heat flux $q_h = -\kappa \nabla T$ and positive-semidefinite symmetric conductivity κ , the analogous production density is

$$\xi_{\text{cond}} = q_h \cdot \nabla(1/T) = \frac{(\nabla T)^T \kappa \nabla T}{T^2} \geq 0. \quad (122)$$

Fourier's constitutive law is a local-equilibrium model, not a universal description of ballistic transport or ultrafast material response. Entropy flux and production are distinguished in [EN6: MIT, Entropy Production and Nonlinearity](#) and [EN7: MIT Advanced Thermodynamics, lecture 21, heat conduction](#).

22.6 Thermoelastic expansion requires reciprocal heat coupling

R12's thermal strain is useful, but a coupled model cannot independently choose an elastic energy, expansion force and unrelated heat capacity. The following small-gradient, local-equilibrium

specialization derives them from one Helmholtz free-energy density $\psi(\epsilon, T)$ per reference volume. All densities below use that same small-strain reference convention. Define

$$s = -\psi_T, \quad e = \psi + Ts = \psi - T\psi_T, \quad \sigma = \psi_\epsilon + \sigma_d, \quad c_\epsilon = -T\psi_{TT} > 0. \quad (123)$$

Here s is entropy density, e internal energy density, σ_d dissipative stress, and c_ϵ volumetric heat capacity at fixed strain. These thermodynamic conjugacies follow the free-energy formulation in [EN1, sections 1.3.3–1.3.4](#); the following compatibility calculation is the R13 specialization.

Combining the local balances $\dot{e} = \sigma : \dot{\epsilon} - \nabla \cdot q_h + r$ and $\dot{s} + \nabla \cdot (q_h/T) - r/T = \xi \geq 0$ gives

$$T\xi = \sigma : \dot{\epsilon} - \dot{\psi} - s\dot{T} - q_h \cdot \nabla T/T, \quad (124)$$

$$= \sigma_d : \dot{\epsilon} - q_h \cdot \nabla T/T \geq 0, \quad (125)$$

$$c_\epsilon \dot{T} = T\psi_{T\epsilon} : \dot{\epsilon} + \sigma_d : \dot{\epsilon} - \nabla \cdot q_h + r. \quad (126)$$

The last equation follows by differentiating $e = \psi - T\psi_T$ and canceling $\psi_\epsilon : \dot{\epsilon}$. Here r is heat supply after any declared local conversion, rather than a second copy of mechanical dissipation. With $\sigma_d = \mathbb{V} : \dot{\epsilon}$, symmetric $\mathbb{V} \geq 0$ on symmetric strains, and Fourier conductivity as above, both displayed dissipative contributions are nonnegative. The reversible term $T\psi_{T\epsilon} : \dot{\epsilon}$ need not be positive: it exchanges thermal and mechanical storage without being dissipation.

A concrete minimal profile uses constant elastic tensor $\mathbb{C}$ positive on admissible strains, constant symmetric thermal-stress tensor β , constant $c_v > 0$ and $T_0 > 0$:

$$\psi(\epsilon, T) = \frac{1}{2}\epsilon : \mathbb{C} : \epsilon - (T - T_0)\beta : \epsilon + c_v [T - T_0 - T \ln(T/T_0)], \quad (127)$$

$$\sigma = \mathbb{C} : \epsilon - (T - T_0)\beta + \sigma_d, \quad (128)$$

$$s = \beta : \epsilon + c_v \ln(T/T_0), \quad (129)$$

$$e = \frac{1}{2}\epsilon : \mathbb{C} : \epsilon + T_0\beta : \epsilon + c_v(T - T_0), \quad c_\epsilon = c_v, \quad (130)$$

$$c_v \dot{T} = -T\beta : \dot{\epsilon} + \sigma_d : \dot{\epsilon} - \nabla \cdot q_h + r. \quad (131)$$

Choosing $\beta = \mathbb{C} : (\alpha_T I)$ reproduces R12's constant-coefficient thermal-expansion stress. The linear term in e is required by the selected entropy convention and coupling; replacing e by an unrelated “elastic plus $c_v T$ ” formula breaks the same first-law calculation. The profile applies on a declared strain and positive-temperature range; it is not a fitted constitutive law for an unspecified specimen.

Additional reporter, damage or chemical state z changes this closure to $\psi(\epsilon, T, z)$ and adds $-\psi_z \cdot \dot{z}$ to $T\xi$. An interval-preserving reporter update from R12 proves its range, not this new thermodynamic inequality. Either specify the driving energy/rates that meet it, or keep the reporter as an empirical readout state with its energy exchange separately declared. No new chemical model is assumed here.

22.7 Switching and exact steps keep their own energy obligations

If a physical mode a changes the storage function, continuous parameter motion has the additional power term $H_a \dot{a}$ in the chain rule. At an idealized discrete switch, use one consistent reference for the stores and record

$$\Delta H_{\text{switch}} = H_{a^+}(x^+) - H_{a^-}(x^-) = W_{\text{switch,in}} - E_{\text{switch,out}}. \quad (132)$$

An ideal routing switch can have zero storage jump. A change of stiffness at fixed q, p and fixed M instead contributes $\frac{1}{2}q^T(K^+ - K^-)q$; a state reset or impulsive motion has its own mechanical balance. Labeling the selector as ASA, NA, JK or a thermal bit does not remove these terms. Passive-mode energy inequalities cannot simply be concatenated across unaccounted jumps.

Exact evaluation preserves the expression of a selected time-step map. It does not automatically preserve (112) between its endpoints. Section 21 supplies a finite midpoint specialization with its own derived step energy identity for fixed linear mechanical coefficients. The continuous coupled thermoelastic equations here retain their separate finite-reduction, existence and time-discretization obligations.

22.8 What is added to the kernel contract

No opcode is needed for these laws. The unified profile binds energy and entropy functions, their derivatives, domain conditions, port pairings and selected finite evolution as versioned templates using existing exact arithmetic, LN, finite integrals and finite ODE declarations. States of unlike units use tuples; matrix contractions expand componentwise. A valid specialization can now present its power identity, heat-transfer cancellation and nonnegative entropy production as explicit mathematical obligations. A false sign, unpowered actuator, duplicated heat term or incompatible thermal expansion can be identified before a kernel is run. This is the useful added foundation. Specimen parameters, model adequacy, runtime evaluation and physical performance remain separate evidence.

23 Electromagnetic closure of the optical interface

R12 defines $U_{\text{det},\lambda} = H_{z,\lambda}[A t_{\lambda} e^{i\phi_{\text{lens},\lambda}} U_{\text{in},\lambda}]$. This chapter supplies a physical route to its transfer, loss and material-state arguments. Maxwell's equations are the adopted classical field laws; a material law, geometry and boundary data select a particular device. The added templates fit the existing exact expression and finite-state interfaces.

23.1 Fields, charge and the data that close their evolution

Use SI quantities in one declared spatial frame and time coordinate. For a stationary macroscopic medium, with free charge ρ_f and free current $\mathbf{J}_f$, adopt

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}, & \nabla \times \mathbf{H} &= \mathbf{J}_f + \partial_t \mathbf{D}. \end{aligned} \quad (133)$$

The dimensions are $[\mathbf{E}] = \text{V/m}$, $[\mathbf{H}] = \text{A/m}$, $[\mathbf{D}] = \text{C/m}^2$, $[\mathbf{B}] = \text{T}$, $[\rho_f] = \text{C/m}^3$ and $[\mathbf{J}_f] = \text{A/m}^2$. The divergence of the last equation gives

$$\partial_t \rho_f + \nabla \cdot \mathbf{J}_f = 0. \quad (134)$$

Thus supplied charge and current histories cannot be chosen independently. The electric divergence constraint propagates when it holds initially and the supplied charge obeys continuity; the magnetic constraint propagates because the divergence of a curl vanishes. These statements require the corresponding classical derivatives or an explicitly declared weak solution. The laws and charge-conservation argument are documented in the [Caltech Feynman Lectures, II.18 \[EM1\]](#).

For a stationary interface, let $\mathbf{n}$ point from medium 1 to medium 2. With surface free charge ρ_s , surface free current $\mathbf{K}_s$, and no magnetic surface sources, the interface conditions are

$$\begin{aligned} \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) &= \rho_s, & \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) &= 0, \\ \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) &= 0, & \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) &= \mathbf{K}_s. \end{aligned} \quad (135)$$

They follow by shrinking Maxwell's integral pillboxes and loops across the interface. A moving boundary needs the corresponding moving-medium model; these stationary formulas cannot silently serve as that model. [MIT Electromagnetics and Applications, Section 2.6 \[EM2\]](#).

A device profile binds its material regions, interface geometry, constitutive relations, sources, compatible initial fields and initial material memory. It selects the needed independent boundary or incoming-wave data, including an outgoing-wave condition for an open problem. It does not prescribe every field component independently on every boundary. Existence and uniqueness belong to this completed problem; the four field equations alone do not resolve a resonant, contradictory or underspecified setup.

23.2 Power balance before optical loss is called heat

The vector identity for $\nabla \cdot (\mathbf{E} \times \mathbf{H})$ and equation (133) give the constitutive-independent identity

$$\nabla \cdot \mathbf{S} + \mathbf{E} \cdot \partial_t \mathbf{D} + \mathbf{H} \cdot \partial_t \mathbf{B} = -\mathbf{E} \cdot \mathbf{J}_f, \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (136)$$

For instantaneous, time-independent real symmetric positive-definite tensors ϵ, μ , with $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{B} = \mu \mathbf{H}$, set

$$u_{\text{em}} = \frac{1}{2} \mathbf{E} \cdot \epsilon \mathbf{E} + \frac{1}{2} \mathbf{H} \cdot \mu \mathbf{H},$$

$$\partial_t u_{\text{em}} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_f. \quad (137)$$

The quadratic energy follows by differentiating each form; time independence and symmetry are used at that exact step. This is Poynting's energy balance, with fields doing work on matter through $\mathbf{E} \cdot \mathbf{J}_f$. [Caltech Feynman Lectures, II.27 \[EM3\]](#).

Writing $\mathbf{J}_f = \sigma \mathbf{E} + \mathbf{J}_{\text{src}}$, where the symmetric part of σ is positive semidefinite, identifies a nonnegative Ohmic sink $q_{\text{Ohm}} = \mathbf{E} \cdot \sigma \mathbf{E}$. The impressed source term $-\mathbf{E} \cdot \mathbf{J}_{\text{src}}$ may supply field energy. In contrast, arbitrary $\mathbf{E} \cdot \mathbf{J}_f$ need not be positive or entirely heat: work can enter mechanical or other material stores. A dispersive dielectric also cannot use $\frac{1}{2} \mathbf{E} \cdot \mathbf{D}$ as its complete instantaneous energy without a justified material-state model.

23.3 A finite dispersive material law with an energy proof

An optional, explicit local linear closure uses finitely many polarization oscillators. In a fixed material region let $\epsilon_b, \mu > 0$ be scalar, time-independent background coefficients and set

$$\mathbf{D} = \epsilon_b \mathbf{E} + \sum_{j=1}^m \mathbf{P}_j, \quad \mathbf{B} = \mu \mathbf{H},$$

$$\dot{\mathbf{P}}_j = \mathbf{V}_j, \quad \dot{\mathbf{V}}_j = f_j \mathbf{E} - \gamma_j \mathbf{V}_j - \omega_j^2 \mathbf{P}_j, \quad (138)$$

where m is finite, $f_j > 0$, $\gamma_j \geq 0$ and $\omega_j > 0$. Here $[f_j] = [\epsilon_b]/s^2$, $[\gamma_j] = s^{-1}$ and $[\omega_j] = s^{-1}$. Each oscillator has supplied $\mathbf{P}_j(t_0), \mathbf{V}_j(t_0)$. This is a selected Lorentz material model, not a universal constitutive law. Its optical origin is described in [MIT Fundamentals of Photonics, Section 2.1.4 \[EM4\]](#).

Proposition 23.1 (Explicit storage and dissipation). *For the held coefficients in equation (138), define*

$$\begin{aligned} u_j &= \frac{|\mathbf{V}_j|^2 + \omega_j^2 |\mathbf{P}_j|^2}{2f_j}, \\ u_* &= \frac{1}{2}\epsilon_b |\mathbf{E}|^2 + \frac{1}{2}\mu |\mathbf{H}|^2 + \sum_j u_j, \\ q_{\text{diss}} &= \mathbf{E} \cdot \sigma \mathbf{E} + \sum_j \frac{\gamma_j}{f_j} |\mathbf{V}_j|^2 \geq 0. \end{aligned} \quad (139)$$

Then $\partial_t u_* + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_{\text{src}} - q_{\text{diss}}$.

Proof. Multiplying the oscillator equation by $\mathbf{V}_j/f_j$ gives $\dot{u}_j = \mathbf{E} \cdot \mathbf{V}_j - \gamma_j |\mathbf{V}_j|^2/f_j$. In equation (136), the polarization part of $\mathbf{E} \cdot \dot{\mathbf{D}}$ is $\sum_j \mathbf{E} \cdot \mathbf{V}_j$. Adding the oscillator balances cancels these exchanges exactly. Positivity follows from the coefficient and conductivity hypotheses. $\square$

This supplies a consistent source for R12's thermal ledger. Assign $P_{\text{heat,opt}} = \int_{\Omega} q_{\text{diss}} dV$ only when these dissipative channels thermalize in the modeled thermal store on the declared interval. If excitation, emission or another store participates, partition the sink and retain that store's evolution. Counting the same absorbed power independently as full heat and full chemical excitation would break the balance.

With the phasor convention $\mathbf{E}(t) = \text{Re}(\widehat{\mathbf{E}}e^{-i\omega t})$, $\omega > 0$, the scalar isotropic, steady harmonic closure is

$$\begin{aligned} \epsilon_{\text{eff}}(\omega) &= \epsilon_b + \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} + \frac{i\sigma}{\omega}, \\ \langle q_{\text{diss}} \rangle &= \frac{\omega}{2} \text{Im}(\epsilon_{\text{eff}}) |\widehat{\mathbf{E}}|^2. \end{aligned} \quad (140)$$

All denominators must be nonzero, and the selected steady response must exist; an undamped resonance does not satisfy these assumptions. Conductivity is included once, inside ϵ_{eff} in this frequency convention. The time-domain state in equation (138) remains the definition when a steady response is inappropriate.

Temperature, hysteresis and reporter state may parameterize a family of these material profiles. Holding them fixed during an optical subproblem is an explicit time-scale approximation. If coefficients change during propagation, their derivatives change stored energy: for example, $\partial_t(\epsilon_b |\mathbf{E}|^2/2)$ includes $\dot{\epsilon}_b |\mathbf{E}|^2/2$. The corresponding modulation work and material exchange must be included. Replacing a coefficient at each tick without that accounting is not the proved coupled energy law.

23.4 How Maxwell closes the existing propagation operator

In a homogeneous, source-free, lossless region with constant scalar $\epsilon, \mu > 0$, the divergence constraint gives $\nabla \cdot \mathbf{E} = 0$. Taking a curl and using $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ yields

$$\begin{aligned} \nabla^2 \mathbf{E} - \mu\epsilon \partial_t^2 \mathbf{E} &= 0, \\ (\nabla^2 + k^2) \widehat{\mathbf{E}} &= 0, \quad k = \omega\sqrt{\mu\epsilon}. \end{aligned} \quad (141)$$

For a linear dispersive homogeneous medium, use the corresponding frequency equation with its declared complex constitutive coefficients and radiation branch. A scalar component can stand for the optical field only within an appropriate polarization/geometry reduction. Spatially varying tensors, strong polarization conversion or near-field boundary interactions do not automatically reduce to the same scalar equation. [MIT Optics, Lecture 9 \[EM5\]](#).

For the scalar, forward, homogeneous lossless case write $U = e^{ikz}\psi$. Substitution gives

$$\partial_z^2 \psi + 2ik\partial_z \psi + \nabla_\perp^2 \psi = 0. \quad (142)$$

The paraxial approximation drops $\partial_z^2 \psi$ when it is negligible relative to the retained terms on the admitted field class. For $z > 0$, one resulting Fresnel transfer, with $\lambda_m = 2\pi/k$, is

$$(H_z U_0)(x, y) = \frac{e^{ikz}}{i\lambda_m z} \iint U_0(x', y') \exp\left(\frac{ik[(x - x')^2 + (y - y')^2]}{2z}\right) dx' dy'. \quad (143)$$

This states the normalization missing from a mere angular-shape transform. For R12's finite aperture, the integral is over its bounded support; its exact real-pair lift needs the section integrability conditions already specified there. A full-plane operator has its separately declared function-space domain. Fraunhofer additionally neglects the aperture-plane quadratic phase on the stated field/support scale. These approximations need application error allowances, not an assertion that exact encoding makes them exact Maxwell solutions. [MIT Optics, Lecture 16 \[EM6\]](#).

The same boundary-value problem fixes t_λ and reflected/outgoing channels from material parameters and geometry. A thin local multiplier is an adopted reduction; a general scattering device may require a matrix or integral transfer. The aperture A remains a geometric predicate. It does not specify the polarization currents or electromagnetic boundary response of the physical opaque material by itself.

23.5 Passivity at the same physical interface

For a fixed volume, let $\mathcal{U} = \int_\Omega u_* dV$. Separate its boundary flux into admitted incoming and outgoing power. With no impressed source, integration of the derived energy identity gives

$$E_{\text{out}}[t_0, t_1] \leq E_{\text{in}}[t_0, t_1] + \mathcal{U}(t_0) - \mathcal{U}(t_1). \quad (144)$$

Stored energy can temporarily power an output. In steady harmonic operation, the average storage change vanishes; for independent positive-flux modal coordinates this gives precisely R12's condition

$$S^\dagger W_{\text{out}} S \leq W_{\text{in}}. \quad (145)$$

Both metrics refer to the same physical normalization and selected ports. Unobserved reflected, radiated or coupled-away power is not automatically absorption. Only with all outgoing channels accounted for, no supplied gain and no net storage accumulation does $\mathcal{R} + \mathcal{T} + \mathcal{A} = 1$ identify the remaining fraction as material absorption. This derivation supplies the physical premise for the R12 scaled passive pinion construction; it does not change its algebra or demonstrate a manufactured device.

One optional mechanical coupling is already useful without a general electromagnetic stress model. For a stationary planar target in vacuum, normal incidence, forward transmission and specular backward reflection, the steady force along the incident beam is

$$F_{\parallel} = \frac{P_{\text{in}}}{c} (1 + \mathcal{R} - \mathcal{T}) = \frac{P_{\text{in}}}{c} (\mathcal{A} + 2\mathcal{R}). \quad (146)$$

It follows by subtracting outgoing from incoming momentum flux. It can enter the declared mechanical load. Oblique/diffuse scattering, moving media, directional re-emission or additional forces require their corresponding momentum balance; the scalar formula does not cover them. [Caltech Feynman Lectures, I.34.9 \[EM7\]](#).

23.6 Finite exact seed representation

Complex coefficients, amplitudes and matrices use R12's ordered real-pair templates. For a declared finite modal problem $Az = b$, with $A = X + iY$, $z = p + iq$ and $b = r + is$, the literal real system is

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} r \\ s \end{pmatrix}. \quad (147)$$

An inverse requires a proved nonzero determinant; an unresolved singularity or branch remains an explicit non-value outcome. The coefficients bind the chosen basis, geometry, boundary projection, material law and frequency. They may be retained exact expressions or supplied calibrated literals with uncertainty. For instance, select a bounded fixed cavity with a perfect electric conductor boundary and real admissible basis functions ψ_j whose tangential traces vanish. The expansion $\widehat{\mathbf{E}}_N = \sum_{j=1}^N z_j \psi_j$ and weak projection of the frequency curl-curl equation give

$$\begin{aligned} A_{ab} &= \int_{\Omega} [(\nabla \times \psi_a) \cdot \mu^{-1}(\nabla \times \psi_b) - \omega^2 \psi_a \cdot \epsilon_{\text{eff}} \psi_b] dV, \\ b_a &= i\omega \int_{\Omega} \psi_a \cdot \widehat{\mathbf{J}}_{\text{src}} dV. \end{aligned} \quad (148)$$

The boundary term vanishes under this particular trace condition. The finite space, source projection, divergence constraints and coefficient integral domains must be specified. Other boundaries require their own terms. These volume-integral definitions fit existing exact operators only when their actual integrands and bounded coordinate domains meet that operator contract; the weak notation alone is not an executable integral encoding.

The finite polarization coordinates use the first-order equations in (138). Together with a declared finite field reduction, they form an ordinary finite state evolution only after building algebraic constraints into the basis or explicitly eliminating them to obtain a nonsingular finite evolution. The fixed representation, selected elimination branch and regular domain remain explicit. XOP can denote the resulting ODE under its existence/domain contract or an explicitly selected finite update. Remaining algebraic constraints need their own defined constrained formulation. It does not acquire an arbitrary continuum PDE solver. Basis truncation, omitted modes and an approximate constitutive fit remain model errors even when every retained operation and packed bit is exact. The physical profile therefore exports the field transfer, dissipative/other energy channels and optional force from one common model, rather than fitting unrelated outputs that happen to share a seed identifier.

24 One coupled field state and one physical energy

The physical unification supplied here is an explicit composition of electromagnetic, polarization, mechanical and thermal state. Its coupling terms come from one energy function and its dissipative channels have named destinations. Maxwell's equations, material constitutive assumptions and thermodynamic balances keep their established meanings. No additional force law is inferred from a shared notation.

R13 already supplies separate field and mechanical stores, a Lorentz material law and a thermoelectric free energy. The missing connection addressed here is a common evolution when those stores depend on the same material or mechanical state. Independently changing optical coefficients, adding a force and depositing absorbed power as heat need not conserve their combined energy. The following construction makes those dependencies part of the formalization.

24.1 The common continuous model

Let z be a finite tuple of nonthermal state coordinates and let $s = (s_1, \dots, s_n)$ contain local-equilibrium entropy coordinates, in J/K. Choose a twice continuously differentiable total energy $\mathcal{H}(z, s)$, in joules, and define

$$e_z = \partial_z \mathcal{H}, \quad T_i = \partial_{s_i} \mathcal{H} > 0, \quad S_{\text{tot}} = \sum_i s_i. \quad (149)$$

These derivatives hold all other state coordinates fixed. Distinct physical units are retained componentwise; a matrix below represents the corresponding typed contractions. The temperature is the derivative of the *complete* energy, including its constitutive coupling, rather than an unrelated lookup. The use of thermodynamic potentials and their natural-variable derivatives is established in [UC1: Alberty et al., Use of Legendre Transforms in Chemical Thermodynamics](#).

Choose $\mathcal{J}^T = -\mathcal{J}$, dissipative maps $\mathcal{R}_\ell = \mathcal{R}_\ell^T \geq 0$, an input map $\mathcal{G}$ and nonnegative allocation fractions $\alpha_{i\ell}$ with $\sum_i \alpha_{i\ell} = 1$. Let $k_{ij} = k_{ji} \geq 0$ be thermal conductances in W/K, with $k_{ii} = 0$, and Q_i be externally supplied heat rates in watts. Define the versioned coupled model

$$\begin{aligned} \dot{z} &= \mathcal{J}e_z - \sum_\ell \mathcal{R}_\ell e_z + \mathcal{G}u, & y &= \mathcal{G}^T e_z, \\ D_\ell &= e_z^T \mathcal{R}_\ell e_z \geq 0, & \dot{s}_i &= \frac{\sum_\ell \alpha_{i\ell} D_\ell + \sum_j k_{ij}(T_j - T_i) + Q_i}{T_i}. \end{aligned} \quad (150)$$

The input/output pairing $u^T y$ is supplied nonthermal power. Each D_ℓ is power removed from the nonthermal coordinates and thermalized in the declared stores on the modeled time scale. A channel that stores excitation, emits radiation or exports power must retain that state or output instead of being wholly allocated as heat. The heat entropy term Q_i/T_i uses the receiving node's local-equilibrium temperature. An external reservoir at a different temperature needs its own entropy flux and interface production; coherent incident optical power is not assigned a heat entropy P/T_i .

This is a selected finite thermodynamic composition. Energy-preserving, entropy-producing coupling is part of established irreversible port-Hamiltonian modeling; see [UC2: Philipp et al., section 2.1](#). The explicit identities below follow directly from the equations here; no control, stability or convergence result from that source is imported.

Proposition 24.1 (Combined first and second laws). *For a differentiable solution in the stated positive-temperature domain,*

$$\dot{\mathcal{H}} = u^T y + \sum_i Q_i, \quad (151)$$

$$\dot{S}_{\text{tot}} - \sum_i \frac{Q_i}{T_i} = \sum_{i,\ell} \frac{\alpha_{i\ell} D_\ell}{T_i} + \sum_{i < j} \frac{k_{ij}(T_i - T_j)^2}{T_i T_j} \geq 0. \quad (152)$$

Proof. The chain rule gives $\dot{\mathcal{H}} = e_z^T \dot{z} + \sum_i T_i \dot{s}_i$. The skew contraction vanishes. Each $-D_\ell$ is canceled by $\sum_i \alpha_{i\ell} D_\ell$; the two heat rates on every reciprocal conduction link cancel. This proves the energy identity. Summing the entropy equations pairs a link's terms as $k_{ij}[(T_j - T_i)/T_i + (T_i - T_j)/T_j] = k_{ij}(T_i - T_j)^2/(T_i T_j)$. All remaining production terms are nonnegative. $\square$

Thus an isolated selected model conserves its total energy while dissipation increases entropy. Energy has not vanished when mechanical or field amplitude decays: it has entered the same model's thermal state. Positivity hypotheses are constitutive assumptions, not conclusions about every proposed device.

The maps may depend smoothly on state. Local Lipschitz regularity of the resulting right-hand side, regular finite coordinates, admissible initial data and a prescribed regular input give the usual local ODE obligation. These conditions do not prove global continuation, material stability, or that a trajectory remains in the positive-temperature domain. Lower-bounded energy, constitutive stability and any further operating-region invariance are checked for the selected application. Explicit time dependence adds $\partial_t \mathcal{H}$ to equation (151).

24.2 A concrete Maxwell–Lorentz, mechanical and thermal specialization

Select a finite electromagnetic representation on a fixed reference topology with power-dual coordinates. Let d be electric displacement-flux coordinates in coulombs, b magnetic flux coordinates in webers, and C a fixed signed curl/interconnection matrix. Their conjugates e, h have voltage and current units. Matrix entries and coordinate normalization are part of the selected field reduction; these are not point samples or pixel intensities. The skew pair $(C, -C^T)$ is the finite counterpart of curl integration by parts, with boundary power supplied through explicit ports. A compatible field reduction must derive this pair and its boundary terms. Such preservation of Maxwell power structure is developed in UC3: [Brugnoli, Rashad and Stramigioli, *Dual field structure-preserving discretization of port-Hamiltonian systems*](#).

Add mechanical coordinates q , momenta p , polarization-charge coordinates P_j of the same dimension as d , and their conjugate flux coordinates π_j . Here q is a local mechanical coordinate, distinct from the digital JK word q_n . Set $\delta = d - \sum_j P_j$ and choose

$$\begin{aligned} \mathcal{H} = & \frac{1}{2} p^T M^{-1} p + U(q, s) + \frac{1}{2} \delta^T C_e(q, s)^{-1} \delta + \frac{1}{2} b^T L_b(q, s)^{-1} b \\ & + \sum_{j=1}^m \left(\frac{f_j}{2} \pi_j^T \pi_j + \frac{\omega_j^2}{2 f_j} P_j^T P_j \right). \end{aligned} \quad (153)$$

Here $M = M^T > 0$ is fixed, $C_e = C_e^T > 0$ and $L_b = L_b^T > 0$ are selected electric and magnetic energy matrices, m is finite, and $f_j > 0$, $\omega_j > 0$ are fixed oscillator coefficients. Under the stated flux/charge normalization C_e, L_b, f_j have units F, H, F/s²; π_j has units V s. These finite coefficients are projected/material parameters and are not numerically identified with the continuum coefficients of equation (138) without the actual normalization map. Mixed mechanical dimensions retain the inherited typed-block convention.

The internal energy $U(q, s)$ includes elastic/thermal coupling once. In particular it may be the entropy-variable form of R13's thermoelastic free energy, rather than an independently added second elastic store. Define

$$\begin{aligned} v &= M^{-1} p, & e &= C_e^{-1} \delta, & h &= L_b^{-1} b, \\ V_j &= f_j \pi_j, & w_j &= -e + \frac{\omega_j^2}{f_j} P_j, & T_i &= \partial_{s_i} \mathcal{H}. \end{aligned} \quad (154)$$

For symmetric positive-semidefinite mechanical damping C_m and electric conductance G_e , and $\gamma_j \geq 0$, the nonthermal equations are

$$\begin{aligned} \dot{q} &= v, & \dot{p} &= -\partial_q \mathcal{H} - C_m v + F_{\text{ext}}, \\ \dot{\delta} &= C h - G_e e + i_{\text{ext}}, & \dot{b} &= -C^T e, \\ \dot{P}_j &= V_j, & \dot{\pi}_j &= -w_j - \frac{\gamma_j}{f_j} V_j. \end{aligned} \quad (155)$$

The displayed supply is $v^T F_{\text{ext}} + e^T i_{\text{ext}}$. Additional selected electromagnetic boundary ports are paired by the same power convention. The dissipation channels deposited through equation (150) are exactly

$$D_m = v^T C_m v, \quad D_e = e^T G_e e, \quad D_j = \frac{\gamma_j}{f_j} V_j^T V_j. \quad (156)$$

The curl powers cancel as $e^T \mathbf{C} h - h^T \mathbf{C}^T e = 0$; each mechanical or polarization canonical pair cancels in the same way. The remaining nonthermal power is the supply minus the displayed losses, so the preceding energy and entropy theorem applies without an additional unidentified transfer coefficient.

For the fixed f_j , differentiating $V_j = f_j \pi_j$ gives

$$\dot{V}_j = f_j e - \omega_j^2 P_j - \gamma_j V_j. \quad (157)$$

This is the finite-coordinate Lorentz oscillator. The same P_j enters δ in field energy, supplies the electromagnetic response and receives the work that its damping converts to heat. No separate full “absorption heat” is added on top of D_j . The constitutive origin and continuum energy proof remain those of equations (138)–(139).

24.3 Reciprocal optical force and thermal response

For fixed d, b, P_j, π_j, p, s , matrix inverse differentiation gives the mechanical load contributed by the two field stores:

$$(F_{\text{field}})_a = \frac{1}{2} e^T (\partial_{q_a} C_e) e + \frac{1}{2} h^T (\partial_{q_a} L_b) h. \quad (158)$$

Thus $-\partial_{q_a} \mathcal{H} = -\partial_{q_a} U + (F_{\text{field}})_a$ for the selected fixed M, f_j, ω_j . This derives the reciprocal force from the same geometry/material dependence that changes the field response. It does not insert a generic radiation-pressure formula into every medium. The scalar electroquasistatic restriction is the familiar $F = (e^2/2) \partial_{q_a} C_e$, with energy differentiation performed at fixed charge, as derived in [UC4: MIT Electromagnetic Fields, Forces, and Motion, lecture 12, pages 1–2](#). Holding voltage by a source changes the charge and supplies electrical work; one cannot differentiate a fixed-voltage expression as though that source were absent.

Likewise the common thermal derivative is

$$T_i = \partial_{s_i} U - \frac{1}{2} e^T (\partial_{s_i} C_e) e - \frac{1}{2} h^T (\partial_{s_i} L_b) h. \quad (159)$$

The positivity domain applies to this complete expression. Simply prescribing both arbitrary $C_e(q, s)$ and an unrelated $T_i(s_i)$ would overdetermine the thermodynamic constitutive model. As a simple uncoupled thermal store, $U_i(s_i) = c_i T_{i0} \exp[(s_i - s_{i0})/c_i]$ with $c_i, T_{i0} > 0$ gives $\partial_{s_i} U_i = T_{i0} \exp[(s_i - s_{i0})/c_i] > 0$; the exponent is dimensionless. Once field or elastic storage depends on entropy, its derivative must also be included. This identity is a constitutive illustration, not a numerical example or a universal material law.

Material data are often supplied versus temperature. For one thermal coordinate, a twice differentiable Helmholtz energy $\mathcal{F}(z, T)$ gives

$$s = -\partial_T \mathcal{F}(z, T), \quad c_z = -T \partial_T^2 \mathcal{F} > 0, \quad \mathcal{H}(z, s) = \mathcal{F}(z, T(z, s)) + T(z, s)s. \quad (160)$$

At fixed z , $\partial_T s = c_z/T > 0$ permits a local inverse on the declared branch. Differentiation cancels the ∂T terms and yields $\partial_s \mathcal{H} = T$ and $\partial_z \mathcal{H} = \partial_z \mathcal{F}|_T$. For several temperatures the corresponding $-\partial_T^2 \mathcal{F}$ must be nonsingular on the selected branch; positive definiteness supplies local thermodynamic stability of that temperature block. A constitutive inversion still needs its supplied exact finite definition and domain. This connects R13’s free-energy closure to the common entropy state without omitting its reciprocal thermoelastic term.

The force theorem is a theorem about the chosen reduced energy. To represent a moving physical device, q must describe a declared deformation or component coordinate with a valid energy reduction and boundary work. Fixed reference topology with parameterized energy matrices can describe selected small-motion or quasistatic couplings; it is not an arbitrary moving-medium Maxwell model. Unmodeled basis motion, motional electromotive terms, relativistic effects or unrepresented mechanical momentum are not supplied by writing a derivative.

24.4 Field constraints, material modes and exact seed binding

The Maxwell divergence conditions still constrain any proposed field representation. For an unreduced compatible flux description, let $N_e \mathbf{C} = 0$ and $N_b \mathbf{C}^T = 0$ express the boundary-of-boundary identities. If $N_e d = \rho$ initially and the free-charge coordinates obey

$$\dot{\rho} = N_e(-G_e e + i_{\text{ext}}), \quad N_b b = 0 \text{ initially}, \quad (161)$$

then differentiation preserves both constraints. Compatible boundary ports contribute their corresponding charge term. The signs use i_{ext} as an injected electric-flow input; an impressed current leaving the field has the opposite sign. Polarization is already part of d and is not an additional free-charge source.

For the exact finite ODE binding, the selected representation builds these constraints into a regular basis or supplies their explicit elimination and compatible independent inputs, as required by R13. The invariance calculation does not solve incompatible initial data, provide an unspecified boundary condition, or add a differential-algebraic solver to XOP. Modal truncation, constitutive fit and the mapping from finite fluxes to physical fields remain explicit approximation/evidence obligations.

ASA/NA and JK can select a constitutive mode χ , an actuator input or a boundary configuration. In a fixed mode the equations use $\mathcal{H}(z, s; \chi)$. At a mode change with unchanged physical coordinates,

$$W_{\text{switch}} = \mathcal{H}(z, s; \chi^+) - \mathcal{H}(z, s; \chi^-). \quad (162)$$

The value is a required signed supply or export, not proof that an actuator can deliver it. A reset of coordinates or entropy has its own complete before/after balance. A dynamic reporter or hysteretic state affecting energy must likewise have its conjugate exchange and evolution; a declared empirical readout may remain an observation without being counted as an energy store.

The seed specialization records $\mathcal{H}$, its explicit conjugate bodies, the selected matrices, thermal allocations, inputs, initial state and domains in one typed operator graph. Existing scalar arithmetic, finite sums, EXP, and nonsingular SOLVE_LINEAR express the displayed finite bodies when their operands meet the inherited contracts. Gradients in this chapter are mathematical definitions; the specialization supplies their derived finite bodies. Existing DIFF can denote a derivative on its declared domain; it does not by itself provide an automatic differentiation algorithm. The finite ODE notation retains the inherited existence and denotation contract. One-bit words preserve this authoritative graph and its operands.

R13's midpoint balance applies to its fixed linear mechanical specialization. It does not automatically extend to the nonlinear coupled energy (153). Any selected finite update for the combined model needs its own discrete work, heat and entropy identities or stated defects. This chapter adds the integrated physical equations and their continuous proofs; it does not silently claim a new time-step implementation.

25 A finite exact field kernel with work and heat balance

The common continuous energy law needs a separately derived finite update. This section supplies one for a fixed quadratic electromagnetic/mechanical reduction with one thermal store. It extends the linear mechanical midpoint profile to a coupled passive finite field. Temperature-dependent coefficients, nonlinear energy and moving bases retain the continuous model and need their own selected update; they are not covered by this fixed-coefficient proof.

25.1 FIELD-MIDPOINT-QUADRATIC-R1

Let x contain the retained field and mechanical coordinates after resolving algebraic constraints. Declare constant matrices

$$Q = Q^T > 0, \quad J^T = -J, \quad R = R^T \geq 0, \quad A = (J - R)Q, \quad (163)$$

and a constant input map B . The model is

$$\dot{x} = Ax + Bu, \quad E_f(x) = \frac{1}{2}x^T Qx, \quad y = B^T Qx. \quad (164)$$

Each coordinate keeps its own dimensions. In particular $[Q_{ij}] = J/([x_i][x_j])$, the input/output pairing $u^T y$ has power units, and A_{ij} has units $[x_i]/([x_j] \text{ s})$. Positive definiteness refers to the energy form on independent coordinates; it cannot be asserted before removing gauge or other null directions. Fixed constrained Maxwell–Lorentz modes and a fixed positive mechanical stiffness are examples when their energy has this form.

For $h \geq 0$, held discrete input $\bar{u}$ and $\bar{x} = (x^+ + x)/2$, define the undivided midpoint equation

$$x^+ - x = h(A\bar{x} + B\bar{u}). \quad (165)$$

Its complete finite body is

$$\begin{aligned} D_h &= I - \frac{h}{2}A, \\ b_h &= (I + \frac{h}{2}A)x + hB\bar{u}, \\ x^+ &= \text{SOLVE_LINEAR}(D_h, b_h), \quad \bar{x} = \frac{1}{2}(x^+ + x). \end{aligned} \quad (166)$$

For $h = 0$ this gives $x^+ = x$ without a division by time.

Proposition 25.1 (Unique finite solve). *Under the stated hypotheses, D_h is invertible for every $h \geq 0$. If the matrix entries, state, step and input are rational, the output is rational.*

Proof. If $D_h w = 0$, then $w = (h/2)(J - R)Qw$. Multiplication by $w^T Q$ gives

$$w^T Qw = -\frac{h}{2}(Qw)^T R(Qw) \leq 0.$$

Positive definiteness forces $w = 0$, so the square matrix is nonsingular. A finite nonsingular rational linear system has a rational solution. $\square$

25.2 The same update supplies its exact energy ledger

Define the interval work and deposited dissipative energy

$$W = h\bar{u}^T B^T Q\bar{x}, \quad D = h(Q\bar{x})^T R(Q\bar{x}) \geq 0. \quad (167)$$

Symmetry and the midpoint equation give

$$\begin{aligned} E_f(x^+) - E_f(x) &= \bar{x}^T Q(x^+ - x) \\ &= h(Q\bar{x})^T (J - R)Q\bar{x} + h\bar{x}^T QB\bar{u} = W - D. \end{aligned} \quad (168)$$

Thus a lossless unforced instance preserves the selected quadratic field energy, and a passive unforced instance cannot increase it. The skew term exchanges energy between retained coordinates without creating it.

Here R represents only losses thermalized in the selected store. Outgoing radiation or power delivered to an external load belongs in boundary ports or its own store; it is not automatically

deposited as local heat. Connect all of this profile's dissipation to one thermal store of fixed heat capacity $C_\theta > 0$, with $T > 0$, no external heat flow and

$$E_\theta = C_\theta T, \quad S_\theta = C_\theta \log(T/T_\star), \quad T_\star > 0, \quad T^+ = T + D/C_\theta. \quad (169)$$

The arbitrary energy reference is omitted. The temperature remains positive, and direct substitution proves

$$\begin{aligned} \Delta(E_f + E_\theta) &= W, \\ \Delta S_\theta &= C_\theta \log\left(1 + \frac{D}{C_\theta T}\right) \geq 0. \end{aligned} \quad (170)$$

This is a complete finite field/heat update. It deposits the energy D once; adding the same loss again as an independent heater would violate the identity. Heat leaving the selected thermal store requires an additional explicit exchange model. This one-store closure deliberately has no feedback from its temperature into Q, R during the interval.

25.3 Constraints and representation

For a constant linear constraint map L such that $LA = 0$ and $LB\bar{u} = 0$, multiplication of (165) yields $Lx^+ = Lx$. This is a conditional preservation result; an arbitrary modal discretization need not satisfy those identities. For charge-driven Maxwell fields, changing charge and currents need a compatible continuity state and constraint law, rather than a falsely constant constraint.

The template exports (x^+, T^+, W, D) and reads one old snapshot. Its body uses finite scalar operations, typed matrix contraction and the existing `SOLVE_LINEAR` operator. Rational C_θ, T also give a rational next temperature; the logarithm may remain exact syntax if entropy is exported. The packing transform then preserves these operators and operands literally. No PDE opcode, arbitrary-sign oracle or new executable backend is required by the mathematical definition.

The exact residual

$$r_{\text{total}} = E_f(x^+) + C_\theta T^+ - E_f(x) - C_\theta T - W$$

vanishes by the displayed proof. These are discrete work and heat quantities, not a claim that they equal the integrals along the continuous trajectory. Conservation does not establish trajectory accuracy at an arbitrary step size. Changing a coefficient first requires its parameter-work ledger; nonlinear coupling cannot be added to this step while retaining the fixed quadratic proof.

26 Gravity, clocks and station observations from one state

The satellite specialization shares the orbit, gravitational coefficients, station motion, frame realization and clock parameters with its observation map. This closes a specific gap in R11: its orbital station query is simultaneous geometry, whereas its live GPS query uses emission and reception events. The following equations define their common physical interface.

26.1 A source potential with explicit finite exterior models

For an isolated Newtonian mass density $\rho_m \geq 0$, fix the potential zero at infinity. In SI units the source law and its boundary realization are

$$\nabla^2 \Phi = 4\pi G \rho_m, \quad \Phi(x) = -G \int \frac{\rho_m(x')}{\|x - x'\|} d^3 x', \quad g = -\nabla \Phi. \quad (171)$$

Outside the source $\nabla^2 \Phi = 0$. Boundary conditions or measured exterior coefficients are required; the Poisson equation alone does not identify them. [\[GT1: MIT, potential theory\]](#)

Let $R(t) \in SO(3)$ map Earth-fixed components to a common geocentric, nonrotating spatial chart. For $p = R^T r$, $s = \|p\| > a_E > 0$ and $\mu_E > 0$, one closed exterior choice is

$$\Phi_E(t, r) = -\frac{\mu_E}{s} \left[1 - J_2 \left(\frac{a_E}{s} \right)^2 P_2(p_z/s) \right], \quad P_2(u) = \frac{3u^2 - 1}{2}. \quad (172)$$

$J_2 = 0$ selects the monopole. Alternatively select the finite R11 solid-harmonic body with $\Phi_E = -U_{R2}$, retaining its normalization, degree, coefficient epoch and exterior domain. These are alternative Earth fields; adding them would count the monopole twice. $\mu_E = GM_E$ is a gravitational parameter with a declared realization, not an exact SI defining constant. Arbitrary fitted coefficients need not correspond to a nonnegative physical density; consistency with the selected source and coefficient uncertainty is a separate obligation.

For a point body at geocentric $r_b(t) \neq 0$, the appropriate geocentric *tidal* potential and its force are

$$\Phi_b^{\text{tid}}(t, r) = -\frac{\mu_b}{\|r_b - r\|} + \frac{\mu_b}{\|r_b\|} + \frac{\mu_b r_b \cdot r}{\|r_b\|^3}, \quad (173)$$

$$-\nabla_r \Phi_b^{\text{tid}} = \mu_b \left(\frac{r_b - r}{\|r_b - r\|^3} - \frac{r_b}{\|r_b\|^3} \right), \quad r \neq r_b. \quad (174)$$

The constant and linear terms remove the external potential and acceleration at the geocenter. Set $\Phi_{\text{geo}} = \Phi_E + \sum_b \Phi_b^{\text{tid}}$ and

$$\dot{r} = v, \quad \dot{v} = -\nabla_r \Phi_{\text{geo}} + a_{\text{ng}} + a_{1\text{PN}}. \quad (175)$$

a_{ng} binds the selected radiation-pressure, thrust or other physical force body and its parameters. Set absent terms to zero explicitly. The existing R11 spherical-Earth a_{GR} may bind $a_{1\text{PN}}$; it is a specified post-Newtonian correction, not the whole relativistic force model. All retained force choices remain subject to their source/model error.

26.2 Frame acceleration and gravitational energy

For a general Newtonian inertial chart $x = o(t) + R(t)p$, define $\Omega = R^T \dot{R}$ and $\Omega z = \omega \times z$. Differentiating twice gives

$$\dot{x} = \dot{o} + R(\dot{p} + \omega \times p), \quad (176)$$

$$\ddot{p} = R^T(\ddot{x} - \ddot{o}) - 2\omega \times \dot{p} - \dot{\omega} \times p - \omega \times (\omega \times p). \quad (177)$$

Thus a rotating-frame propagation includes translation, Coriolis, Euler and centrifugal terms. In (175) the geocenter acceleration has already been removed by the tidal potential; it is not subtracted a second time. The identity follows from $R^T R = I$, which also makes Ω skew. Independently fitted matrix entries in R11 do not establish this identity; an R14 frame realization must prove it or retain a bounded frame discrepancy. [\[GT2: MIT, rotating axes\]](#)

For the Newtonian part and constant satellite mass m , differentiation yields

$$\frac{d}{dt} \left(\frac{m\|v\|^2}{2} + m\Phi_{\text{geo}}(t, r) \right) = m v \cdot a_{\text{ng}} + m \partial_t \Phi_{\text{geo}}. \quad (178)$$

This equation assumes $a_{1\text{PN}} = 0$; otherwise its work must be retained and a relativistic energy model selected for a relativistic conservation claim. Moving external sources exchange energy through $\partial_t \Phi_{\text{geo}}$. It is not heat or arithmetic drift. An Earth-fixed constant field can still be time dependent in the nonrotating chart. A finite integration operator needs its own energy identity; the continuous balance does not certify RK4.

26.3 Proper time and the same gravitational state

Take t as TCG and the spatial/force data as compatible geocentric coordinates. In the weak-field, slow-motion regime the leading proper-time rate of an ideal clock on trajectory $r_A(t)$ is

$$\frac{d\tau_A}{dt} = 1 + \frac{\Phi_{\text{geo}}(t, r_A) - \|v_A\|^2/2}{c^2} + \epsilon_{\text{clk},A}(t). \quad (179)$$

$\epsilon_{\text{clk},A}$ accounts for omitted higher-order/model terms; it is not an arbitrary control input. The negative-potential convention fixes the sign: a stationary clock deeper in a monopole field has a smaller rate. The tidal potential above, rather than the full absolute Sun/Moon potential, is appropriate to this geocentric convention. A barycentric clock model uses its own coordinate transformation and potential. [GT3: IERS, chapter 10, equations 10.6–10.9]

For a scaled coordinate $T = \alpha t + \beta$, $\alpha > 0$, the chain rule gives $d\tau_A/dT = (d\tau_A/dt)/\alpha$. TT uses $\alpha = 1 - L_G$, with exact $L_G = 6.969290134 \times 10^{-10}$; station, orbit and gravitational parameter values must be compatible with the chosen coordinate scale. GPST tags require their declared realization and mapping. A stored rational timestamp does not prove a physical clock exact or turn (179) into R11's rational time-conversion template.

The readout is modeled by $C_A(t) = t + \delta_A(t)$, where a declared oscillator and steering model connects C_A to τ_A . Clock initial offset, intrinsic frequency error and steering belong to the shared uncertainty state. A broadcast clock product may already contain physical corrections: apply its documented convention once, rather than adding (179) again. [GT4: ESA, relativistic clock correction]

26.4 Causal reception, range and pointing

For satellite emission at t_e and station reception at $t_r > t_e$, define

$$d = r_s(t_e) - r_g(t_r), \quad \rho = \|d\| > 0, \quad t_r - t_e = \frac{\rho}{c} + \Delta_{\text{grav}} + \Delta_{\text{med}}. \quad (180)$$

This is a two-event equation in one coordinate chart. For the leading static Earth monopole propagation correction, let $s_e = \|r_s(t_e)\|$, $s_r = \|r_g(t_r)\|$ and use

$$\Delta_{\text{grav}} = \frac{2\mu_E}{c^3} \log \frac{s_e + s_r + \rho}{s_e + s_r - \rho}. \quad (181)$$

It requires $s_e + s_r > \rho$ and a nonocculted exterior ray in the weak-field domain. Δ_{med} binds a frequency-dependent propagation model appropriate to the measured signal, or is zero in the declared vacuum model. Higher gravity terms, bending, atmospheric error and multipath remain in the observation discrepancy where omitted. [GT5: IERS, chapter 11, ranging equation 11.17]

For the flat vacuum choice $\Delta_{\text{grav}} = \Delta_{\text{med}} = 0$ and fixed t_r , put $F(u) = u + \|r_s(u) - r_g(t_r)\|/c - t_r$. On a collision-free interval with $\|\dot{r}_s\| < c$, $F'(u) = 1 + \hat{d} \cdot \dot{r}_s/c > 0$. A bracket $F(a) \leq 0 \leq F(b)$ therefore specifies a unique emission time. Additional delays require a corresponding monotonicity bound; a fixed number of correction iterations is not identical by definition to this root.

When positions are supplied in Earth-fixed components and both charts share the geocenter, the retarded geometric vector in reception axes is

$$d_E = R(t_r)^T R(t_e) r_{s,E}(t_e) - r_{g,E}(t_r), \quad z = L_g(t_r) d_E = (E, N, U)^T. \quad (182)$$

$L_g \in SO(3)$ is the declared local ENU rotation. Then $\text{az} = \text{atan2}(E, N)$ and $\text{el} = \text{atan2}(U, \sqrt{E^2 + N^2})$ on their nonsingular domains. This product includes Earth rotation during flight; adding a second Sagnac correction would duplicate it. These are retarded geometric angles. Physical

detector pointing additionally binds aberration, ray bending, refraction and mounting terms when required. [GT6: ESA, reception-frame coordinates]

For code readouts C_r, C_s , a corresponding pseudorange observation is

$$P = \rho + c(\Delta_{\text{grav}} + \Delta_{\text{med}}) + c[\delta_r(t_r) - \delta_s(t_e)] + d_p + \eta_p. \quad (183)$$

Here d_p is the remaining modeled discrepancy/bias and η_p the declared measurement error. Orbit coefficients, both clock states, station motion, Earth orientation and propagation parameters enter the joint uncertainty set of Section 28. Their correlations are preserved. Changing gravity, time or frame in one branch changes every dependent branch. Literal one-bit encoding preserves these equations and their operands; a physical tolerance still requires the bound in Equation (206) for the chosen output and interval.

27 An atomic-time observation, literal latch and domain bindings

An actual NIST internet-time observation is included as a fixed input to this formalization. Its literal words provide a reproducible event anchor for the physical models. They do not cause a running clock, calibrate an oscillator, or supply the initial state of every physical system at that event.

The SI atomic reference is the unperturbed ground-state hyperfine transition frequency of cesium-133:

$$\Delta\nu_{\text{Cs}} = 9192631770 \text{ Hz}, \quad 1 \text{ s} = 9192631770/(\Delta\nu_{\text{Cs}}). \quad (184)$$

The defining value is exact; a realization has measurement uncertainty. [CL4: BIPM, SI defining constants] UTC(NIST) uses an ensemble of cesium-beam and hydrogen-maser clocks calibrated against NIST's primary frequency standard. Its internet service distributes that atomic reference; this capture does not install an atomic oscillator in aTOMos. [CL5: NIST, UTC(NIST) realization]

27.1 The observed words and their exact rational coordinate

The recorded exchange with `time.nist.gov`, endpoint `132.163.96.4:123`, used a version-4 request and returned a version-3 response, mode 4, stratum 1, leap indicator 0 and reference ID NIST. The complete request, reply and local acquisition stamps are preserved in `source/clock_capture/nist_capture.json`. The server transmit field is

$$w_{\text{tx}} = 17173213285462662115 \quad \text{as an unsigned 64-bit word}, \quad e = 0. \quad (185)$$

The era e is bound by the recorded contemporary-date context; the packet does not carry it. The reply's SHA-256 is

$$26a429327733f35d8b67abfb9bbb33f54fae9fa590c4ba0e5a7947fe6946c3e4.$$

For the 32-bit seconds and fraction fields $s, f \in \{0, \dots, 2^{32} - 1\}$, the mathematical decoder is

$$w = 2^{32}s + f, \quad q_N = e \cdot 2^{32} + s + \frac{f}{2^{32}}, \quad q_U = q_N - 2208988800. \quad (186)$$

Here q_N uses the 1900 NTP epoch and q_U the corresponding 1970 civil coordinate. Every operation is integer/rational; no floating conversion is needed. The observed word gives the reduced rational

$$q_U = \frac{7685678632232377315}{4294967296}, \quad (187)$$

whose UTC display, truncated to microseconds, is `2026-09-15T08:38:22.859456Z`. The display is not the authoritative value. The NTP fixed-point format and era distinction follow [CL1: RFC 5905, section 6].

NTP's coordinate is affected by leap seconds; differences across leap events are not automatically elapsed SI durations. A complete mapping binds the era evidence, leap table, applicable interval and any ambiguity/smear policy. [CL2: RFC 8877, NTP timestamps] NIST documents its positive-leap convention as repetition of the final second's binary timestamp. [CL3: NIST Internet Time Service] This convention is part of the source identity; replacing the provider does not silently inherit it. The earlier Cloudflare capture is retained separately as acquisition provenance; the NIST record is the selected latch.

27.2 Complete nominal mapping at the captured epoch

The retained `source/clock_capture/leap-seconds.list` has last transition $q_N = 3692217600$ with TAI–UTC offset 37 seconds and declared expiry $q_N = 4023129600$. Its SHA-256 is

db5a895f16853b03bfc865e8d68f9fc8710ef1740e3400c701cd46a5bbbc3433.

For the captured non-leap instant in this bound interval, the nominal conversion body is explicitly

$$\begin{aligned} Q_{\text{TAI}} &= q_U + 37, \\ t_{\text{GPST}} &= q_U - 315964800 + (37 - 19), \\ Q_{\text{TT}} &= q_U + 37 + \frac{4023}{125}, \\ Q_0 &= 220924800 + \frac{4023}{125}, \quad t_{\text{TCG}} = \frac{Q_{\text{TT}} - Q_0}{1 - L_G}. \end{aligned} \quad (188)$$

GPST is elapsed seconds from 1980-01-06 00:00:00 GPST. Q_{TAI} and Q_{TT} use 1970-01-01 00:00:00 calendar-coordinate origins on their respective scales. t_{TCG} is elapsed TCG seconds from 1977-01-01 00:00:32.184, when TT and TCG are defined equal; Q_0 is that event's TT coordinate and $L_G = 6.969290134 \times 10^{-10}$ is exact. These are finite rational expressions for the nominal scales. The leap file is a dated dependency, not authority for an unannounced future leap or an ambiguous repeated second. UTC(NIST)–UTC and GNSS realization differences remain physical uncertainty. [CL6: IANA, retained leap table] [GT3: IERS, time coordinates]

27.3 What the clock observation establishes

The timestamp denotes the server's transmit event. Its observed local monotonic round-trip interval is the exact recorded count $127594000/10^9$ seconds under the local timer convention. Neither that count nor stratum, precision or dispersion metadata certifies a UTC error bound. The UDP reply was not cryptographically authenticated. The host clock was not changed and no oscillator calibration was performed.

Let q_* be the nominal converted source-event coordinate and write

$$t_{\text{tx}} = q_* + \epsilon_s, \quad t_{\text{rx}} = t_{\text{tx}} + d_{\leftarrow}, \quad d_{\leftarrow} \geq 0. \quad (189)$$

Source clock error ϵ_s , inbound delay $d_{\leftarrow}$, local readout error and conversion/realization error enter one joint uncertainty set. An interval for any of them needs supporting assumptions or calibration. One exchange does not determine asymmetric one-way delays or prove source authenticity. Literal retention preserves the observation actually received.

For a complete four-stamp exchange on compatible, unfolded coordinates, the conventional offset and round-trip expressions are

$$\hat{\theta} = \frac{(T_2 - T_1) + (T_3 - T_4)}{2}, \quad \hat{d} = (T_4 - T_1) - (T_3 - T_2). \quad (190)$$

These expressions can remain exact rationals. Interpreting $\hat{\theta}$ as the physical offset requires a clock/path model; unequal inbound and outbound delays contribute an offset error. [CL1: RFC 5905, section 8]

27.4 Immutable XOP snapshot and a one-time latch

Define a finite typed tuple C containing both raw packet words, source identity, era and its evidence, acquisition stamps, protocol fields, the rational coordinate and the complete time-mapping dependency identities. The packet words use `Bits(384)` for this 48-octet capture. Other fields use existing integer, rational, Boolean and fixed structural types; the binding fixes their widths and ordering. No new timestamp opcode is needed.

The completed capture may be a closed dependency value. A newly acquired capture instead enters through a sample port. Every consumer at a logical step reads the same immutable old-state/input snapshot. The physical mapping is an imported complete pure template, with its declared domain, rather than an instruction to consult the network again. The materialized `source/clock_capture/anchor_state.json` records the selected NIST observation and nominal rational mappings. The equations here specify its XOP binding; no running XOP clock executor is claimed.

Let a_n be an accepted-record pulse, v_n the admission predicate for its typed record, ℓ_n the old latched Boolean, and H_n the stored tuple. The explicit one-time latch is

$$\begin{aligned} e_n &= \text{IF}(\ell_n, \text{false}, \text{IF}(a_n, v_n, \text{false})), \\ \ell_{n+1} &= \ell_n \vee e_n, \quad H_{n+1} = \text{IF}(e_n, C_n, H_n). \end{aligned} \quad (191)$$

Initially $\ell_0 = \text{false}$ and H_0 is a fixed closed placeholder of the declared type, which has no observed-clock meaning while unlatched. For defined inputs the pulse equals $a_n \wedge v_n \wedge \neg \ell_n$. The displayed lazy form demands no new record, admission check or pulse once latched, and demands no record when an unlatched acceptance pulse is false. Both next roots commit together. Once ℓ_n is true, e_n is false at every later step and induction proves $H_{n+k} = H_n$ for all $k \geq 0$. Thus future internet replies cannot alter this seed's latched anchor. If an electrical rising edge supplies acceptance, a declared old Boolean b_n gives $a_n = c_n \wedge \neg b_n$ and $b_{n+1} = \text{IF}(\ell_n, b_n, c_n)$ in the same commit. Thus the optional edge memory also demands no new level once latched. A later anchor is a separately versioned capture, not an implicit rewrite.

The raw sample time and canonical-time value remain distinct: R11 tag 5 exposes the source rational and tag 6 applies the complete `TimeConvention` mapping to GPST-SI. An unresolved era or leap branch produces an explicit non-value, not a guessed instant. Closing the captured record as a seed dependency also closes its mapping table and source interval. No live lookup is hidden in a pure expression.

27.5 One anchor, explicit time maps for each physical domain

Let t_0 denote the nominal latched event on the selected canonical scale. Each domain uses a declared map $t_d = \chi_d(t)$ and its own state, frame, boundary conditions and observation history. For a differentiable increasing map, the equation transforms by the chain rule:

$$\frac{dx_d}{dt} = \frac{d\chi_d}{dt} F_d(\chi_d(t), x_d, u_d), \quad x_d(t_0) = x_{d,0}. \quad (192)$$

The supplied state $x_{d,0}$ must refer to that event. An old observation cannot become a current state merely by changing its timestamp.

Domain	Explicit binding to the latched event
Electromagnetism and photonics	$E(x, t), B(x, t), P_j(x, t)$ and their source/boundary histories use the mapped field time. A carrier phase $\varphi(t) = \varphi_0 + \int_{t_0}^t \omega(s) ds$ also needs its phase/frequency reference; NTP does not measure optical phase.
Mechanics and control	$q(t_0), p(t_0)$, attitude, loads and actuator events retain their initial-event identity. ASA/NA and JK transitions act through their explicit physical ports.
Thermal and material history	$T(t_0)$, internal variables and accumulated work/heat require the actual state/history binding. A clock latch does not reset energy or entropy.
Gravity and orbit	The GPST/TT/TCG mapping, compatible frame/force parameters and station/satellite events remain those of Section 26. A coordinate-time conversion does not measure proper clock time.
Sensors and calibration	Acquisition endpoints, integration windows, latency and calibration age use mapped recorded events, with the shared clock error in the observation uncertainty.
Digital state and replay	The integer transition index n is causal order. A time $t_n = t_0 + \sum_{j < n} h_j$ requires explicit durations h_j ; n alone is not seconds.

A common event does not require equal sample rates or spatial rasterization. Field expressions may retain their continuous variables and exact implicit denotation. A selected finite mode basis or finite step remains a separately specified approximation with its own discrepancy. Clock anchoring supplies the temporal relationship; it does not turn a finite model into every physical domain or remove source, state or model uncertainty.

28 Measurement, constants and output tolerances

The existing XOP dimension, frame, sample and dependency types already carry the information needed here. This chapter adds a physical interpretation and conditional output certificates, without adding a measurement oracle or a new packet format. An exactly stored indication, an exact nominal model and the physical quantity being inferred remain distinct. The purpose is operational: preserve what is known, keep shared uncertainties coupled, and determine what an output can claim within a selected application tolerance.

28.1 Exact defining constants and measured coefficients

Five SI defining constants connect the present time, optical, electrical and thermal profiles. Their numerical values are fixed by the SI; unit realizations and measurements made with them still have uncertainty. The following dimension vectors use the inherited order $(L, M, T, I, \Theta, N, J_{\text{lum}})$. [\[ME1: BIPM defining constants\]](#)

Constant	Exact SI quantity	XOP dimension vector
$\Delta\nu_{\text{Cs}}$	9 192 631 770 Hz	(0, 0, -1, 0, 0, 0, 0)
c	299 792 458 m s ⁻¹	(1, 0, -1, 0, 0, 0, 0)
h	6.626 070 15 × 10 ⁻³⁴ J s	(2, 1, -1, 0, 0, 0, 0)
e	1.602 176 634 × 10 ⁻¹⁹ C	(0, 0, 1, 1, 0, 0, 0)
k_{B}	1.380 649 × 10 ⁻²³ J K ⁻¹	(2, 1, -2, 0, -1, 0, 0)

The decimal coefficients are finite rational literals, reduced canonically by the XOP grammar. For example, the numerical coefficient of h is 662607015/10⁴² before reduction. A binary floating-point approximation is unnecessary. The elementary charge e is positive; the charge of an electron is $-e$. The namespace distinguishes it from the Boolean accepted-event bit.

The gravitational constant G and fine-structure constant α_{fs} are measured quantities, with dimensions $L^3 M^{-1} T^{-2}$ and 1, respectively. A specialization that needs them binds a dated value, uncertainty and provenance; this core does not hard-code a new estimate. [ME2: NIST CODATA table] In present SI, vacuum permeability and permittivity are also not independent exact decimal constants. For positive α_{fs} , retain their common dependency:

$$\mu_0 = \frac{2\alpha_{\text{fs}}h}{e^2c}, \quad \epsilon_0 = \frac{e^2}{2\alpha_{\text{fs}}hc}, \quad \mu_0\epsilon_0 = \frac{1}{c^2}. \quad (193)$$

Here $[\mu_0] = MLT^{-2}I^{-2}$ and $[\epsilon_0] = M^{-1}L^{-3}T^4I^2$. Multiplication cancels the shared α_{fs} exactly. Replacing the two occurrences by unrelated uncertain inputs would lose that constraint. The former exact assignment $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ is not the present SI definition. [ME3: SI Brochure, 9th edition, 2026, section 2.3.2]

28.2 An observation is a modeled acquisition

Let $x(t)$ be the selected physical state, θ its sensor/calibration parameters, and m the transduction law into an analogue indication. An averaging acquisition over $[t_-, t_+]$ has the model

$$M[x, \theta, w] = \int_{t_-}^{t_+} w(t) m(x(t), \theta, t) dt, \quad w(t) \geq 0, \quad \int_{t_-}^{t_+} w(t) dt = 1, \quad (194)$$

$$y = Q_{\Delta}(M[x, \theta, w] + d_M + \eta). \quad (195)$$

The weighting w has dimension T^{-1} ; y, M , the model discrepancy d_M and analogue noise η have the same indication dimension. For a counter or accumulated dose, use its declared accumulation law instead of a normalized average. An instantaneous sample is a separately defined profile, not an implicit substitution of the interval endpoint for the integral. Each integral requires finite bounds and its stated integrability conditions. The profile supplies finite function templates or explicit external bindings for the trajectory and weighting; it cannot encode an arbitrary unknown continuous signal as a fully known finite literal.

Q_{Δ} includes its step, offset, ties rule, range and saturation behavior. For an unsaturated uniform nearest quantizer with step $\Delta > 0$, put $v = M + d_M + \eta$. Then

$$y = M + d_M + \eta + q, \quad q = Q_{\Delta}(v) - v, \quad |q| \leq \Delta/2. \quad (196)$$

This is a deterministic bound, not an assumption that q is uniform, independent or newly random at every sample. Saturated codes require the actual preimage of the saturation law; the half-step bound no longer follows. The stored integer and its rational unit conversion can be exact while the corresponding physical indication occupies a range.

The measurement model also declares its valid specimen, operating conditions and reference conditions. A discrepancy bound is supplied evidence or a stated hypothesis, never a number

generated merely by writing d_M . Unrecognized effects are not covered automatically. This makes explicit the model scope emphasized by JCGM GUM-6, sections 6–7. [ME4: [Developing and using measurement models](#)]

28.3 Calibration and joint uncertainty

Let p collect the finite parameters of the chosen profile: initial state, calibration, clock and frame realization, input history parameters and relevant residuals. A joint admissible set $\mathcal{U}$ retains their actual shared dependencies. With observation map F built from (195), the data-compatible set is

$$\mathcal{U}_y = \{p \in \mathcal{U} : F(p) = y\}, \quad \mathcal{Z}_y = \{g(p) : p \in \mathcal{U}_y\}. \quad (197)$$

For multiple records, impose all their observation equations on the same shared calibration parameters. A repeated common offset does not become a new independent offset. If the set is proved empty, the declared model, admissible bounds and observations are inconsistent; empty-set containment must never be used to certify an output. If solving it is unresolved, retain the unresolved status. Neither conclusion identifies which physical assumption failed.

A simple calibration model $y^{\text{cal}} = A\theta + \varepsilon$ illustrates what equations alone can establish. With known design matrix $A \in \mathbb{R}^{m \times r}$, for consistent noiseless data the parameters are unique exactly when

$$\text{rank } A = r. \quad (198)$$

For a chosen positive-definite covariance Σ and full column rank, the generalized least-squares expression is

$$\hat{\theta} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} y^{\text{cal}}. \quad (199)$$

The inverse exists since $z^T A^T \Sigma^{-1} A z = (Az)^T \Sigma^{-1} (Az) > 0$ for $z \neq 0$. This proves the algebraic expression's domain, not a calibration accuracy. Uncertain reference settings make A uncertain and must enter the joint model. For a differentiable nonlinear calibration, a full-column-rank Jacobian permits a local regular inverse by selecting independent observation components. It does not give global uniqueness. Rank deficiency alone does not decide nonlinear identifiability.

An alternative statistical profile supplies a joint distribution or covariance with its basis. For a differentiable output map and its linearization $J = Dg(\hat{p})$,

$$\Sigma_z \simeq J \Sigma_p J^T, \quad u^2(z_i) \simeq \sum_{j,k} J_{ij}(\Sigma_p)_{jk} J_{ik}. \quad (200)$$

Cross terms retain correlations. This is exact for an affine map with the stated covariance; in general it is a first-order approximation and a standard uncertainty is not a worst-case bound. [ME5: [JCGM 100, sections 5.1–5.2](#)] Significant nonlinearity can also affect the output estimate itself; exact evaluation of $g(\hat{p})$ does not make it the distributional mean. [ME6: [JCGM 100, Amendment 1:2026](#)]

28.4 Acquisition time and realized coordinates

R12's WI record, immutable registry and session consumption rules remain unchanged. The calibration identifier resolves complete calibration content; its presence is not proof of freshness or traceability. Add the chosen measurement model, acquisition weighting, quantizer and joint-uncertainty definition to that existing dependency closure.

For recorded source times $\tau_{\pm}$, a declared affine clock model is

$$t_{\pm} = a\tau_{\pm} + b + \delta t_{\pm}, \quad a > 0. \quad (201)$$

The same a, b appear at both endpoints and in every record covered by that calibration. The physical acquisition interval, conversion validity interval and availability time remain distinct.

Subtracting endpoints cancels the shared offset b ; treating it as two independent offsets would miss this exact dependency. The uncertainty of a and endpoint residuals remains.

The analogous physical frame map is

$$x_B(t) = R_{BA}(t)x_A(t) + o_{BA}(t), \quad R_{BA}^T R_{BA} = I, \quad \det R_{BA} = 1. \quad (202)$$

The XOP frame identity specifies the convention; R_{BA} and o_{BA} specify its realization. A shared uncertain mounting is retained once in p , and an interval enclosure of its entries must retain, or conservatively enclose, the rotation constraints. A nominal matrix is not made physically aligned merely by encoding it exactly. Coordinate values from different acquisition times require the selected dynamical/time-transfer model before they can represent simultaneous geometry.

28.5 A bounded output certificate

The core needs a reusable rule for deciding a functional tolerance without inventing a universal numerical tolerance. Let K be a nonempty compact convex set in the finite parameter coordinates; require $\hat{p} \in K$ and a nonempty $\mathcal{U}_y \subseteq K$. The output g_i is continuously differentiable on an open neighborhood of K . Suppose the following finite bounds have been established for all $p \in K$:

$$|p_j - \hat{p}_j| \leq r_j, \quad \left| \frac{\partial g_i}{\partial p_j}(p) \right| \leq L_{ij}. \quad (203)$$

L_{ij} has output-unit per parameter-unit dimensions. No covariance or independence assumption is used. By integrating the derivative along the segment from $\hat{p}$ to p ,

$$g_i(p) - g_i(\hat{p}) = \int_0^1 \sum_j \frac{\partial g_i}{\partial p_j}(\hat{p} + s(p - \hat{p}))(p_j - \hat{p}_j) ds, \quad (204)$$

$$|g_i(p) - g_i(\hat{p})| \leq \sum_j L_{ij} r_j. \quad (205)$$

The compact domain stays clear of undefined divisions, branch changes or other points where the derivative hypotheses fail. A box can conservatively enclose a correlated set; using the original joint constraints can tighten the result. Piecewise or discrete outputs use direct image/branch enclosure instead of differentiating through a switch.

To apply the certificate to a physical result, assume its actual parameter vector lies in $\mathcal{U}_y$. Nonemptiness alone does not establish that physical inclusion. Suppose the intended physical output is $z_i^{\text{phys}} = g_i(p) + d_i^{\text{out}}$ with $|d_i^{\text{out}}| \leq b_i$, and the delivered numerical output obeys $|z_i^{\text{del}} - g_i(\hat{p})| \leq \rho_i$. Then

$$|z_i^{\text{phys}} - z_i^{\text{del}}| \leq B_i := \sum_j L_{ij} r_j + b_i + \rho_i. \quad (206)$$

This triangle-inequality certificate separates input/calibration uncertainty, output-model discrepancy and representation/evaluation error. Effects already included in p are not counted again in b_i . The XOP exact expression can be retained with no numerical rendering error in its denotation; a finite rendering needs its own ρ_i certificate. Existing binary64 execution is not assigned $\rho_i = 0$ by this new formalization.

For a requested absolute error tolerance $\tau_i \geq 0$, the condition $B_i \leq \tau_i$ is sufficient on the stated domain and acquisition interval. For an allowed physical operating band $[\ell_i, u_i]$, sufficient containment is

$$z_i^{\text{del}} - B_i \geq \ell_i, \quad z_i^{\text{del}} + B_i \leq u_i. \quad (207)$$

Failure to prove this containment does not prove violation: a conservative enclosure may overlap both regions. A wholly disjoint enclosing interval proves exclusion; overlap remains unresolved. This is the general form of R12's bounded bridge and timing decisions. It certifies the declared quantity and interval, not every later state or a different physical application.

28.6 Minimal specialization record

Each specialization binds only: the measurand and conditions; the complete measurement/calibration expressions and actual source records; the unit, frame and acquisition-time realizations; the joint uncertainty or statistical model; and the requested output tolerance with its scoped evidence. The certificate binds those same dependency versions. Changing a calibration, clock, model basis, duration or tolerance requires reevaluating the affected claim, not changing the one-bit packing format.

These are metadata and template/dependency obligations under XOP-R1. Set images, universal bounds and physical evidence are not secretly new executable opcodes. A backend may return UNKNOWN, MISSING_INPUT or RESOURCE_LIMIT when it cannot establish an obligation. The bridge, thermal and optical expressions remain useful before that backend exists.

Metrological traceability is a property of a measurement result supported by its calibration chain and uncertainty; source hashing and exact arithmetic preserve provenance but do not supply that chain. Fitness for a particular use additionally depends on the required uncertainty or tolerance. [\[ME7: NIST traceability policy\]](#) The additions here make that distinction actionable through (206), rather than adding unrelated domain rules.

29 Specialization and remaining engine scope

The engine now executes literal finite operator programs on the GPU and closes their emitted-word feedback loop. R18 adds complete ordered native EMIT journals, verified seeded artifact bytes and exact bounded phase arithmetic. The host still evaluates the original formal expression trees. R17 composes those complete transitions inside lane-local chunks and publishes the same final state and receipt with fewer launches. The retained reference path remains available where intermediate observation is needed. The exact event adapter also derives affine crossing times, applies simultaneous state/rate changes and replays their mathematical certificates. These additions support deterministic procedural machines, discrete control logic, exact piecewise-affine events, reproducible state transport and the retained batched spherical-query profiles.

The imported learner is studied and source-pinned, but its authored finite candidate families and publication machinery are not incorporated into the runtime. A finite exact-consistency search does not establish noisy calibration or statistical confidence. The release adds the operator and event mechanisms that are directly executable here.

The retained Maxwell, mechanical, thermal, gravity and measurement equations support physical specializations with supplied parameters, boundary conditions and observations. They do not determine an unknown satellite state, material response or optical transfer function. Application accuracy must be measured against that application's declared output and tolerance. R18 compares the preserved frozen orbital models with an independent Orekit numerical adapter and the same later reference samples. A separate SPICE experiment measures ephemeris reconstruction. Neither supplies observations, orbit determination, an operational clock, or a universal physical error bound.

The next engine work is broader native expression lowering, exact or certified non-affine root isolation, versioned dynamic geometry refit and cache rebinding, and measured operation coverage beyond the existing spherical radius/count profiles. The R16 event adapter is a CPU rational implementation with once-only relations. Its immutable epoch transition is not yet a GPU BVH refit. The TOMAGI word machine and the inherited $a + b\varphi$ evaluator retain distinct instruction and overflow contracts.

Editable source is docs/unified.tex; formal/ defines the contracts and review/ binds the evidence. The published PDF matches its reviewed bytes.

A Normative XOP-R1 grammar and operator registry

This chapter embeds the literal, type, wire and operator contract from `formal/OPERATOR_CATALOG.md`. It supplies the actual fields and numbered operators needed to read the proposed profile without another file. The mathematical denotations and conditional proofs are in Section 18; the application connection is in Section 19. This is a specification, not an implemented parser, exact evaluator or proof checker. Unassigned tags and opcodes are rejected; the new grammar does not reinterpret the preserved R10 format.

A.1 Sorts, outcomes and canonical byte primitives

The scalar embeddings are explicit and value preserving: $\text{Int} \hookrightarrow \text{Rat} \hookrightarrow \text{Alg} \hookrightarrow \text{RealTerm}$. Booleans and finite words are separate types. `RealTerm` carries finite syntax with a possibly partial real denotation; it is not a literal for an unspecified arbitrary real number. Structural types carry their shapes. A `Dim` scalar additionally carries seven rational SI exponents. A `Frame` annotation encodes nominal coordinate identity; prose alone cannot supply it. Angle conventions remain declared interface metadata.

Outcome	Meaning
VALUE	An exact literal with evidence of its construction.
TERM	Retained typed syntax, assumptions and outstanding obligations.
UNDEFINED	A violated domain condition has been established.
UNKNOWN	A required comparison, domain or existence claim is undecided.
MISSING_INPUT	A required dependency or sample is unavailable.
RESOURCE_LIMIT	The declared memory, time or proof-search limit is reached.

These outcomes are separate from scalar values. Unknown or exhausted resources mean neither false nor zero. A retained term is not a proved real value. `COMPARE` is a decision-service interface returning justified LT, EQ, GT, or UNKNOWN; it is not an arithmetic oracle.

Concatenation in the grammar has no implicit alignment. The byte primitives are:

Primitive	Encoding
U(n)	Minimal unsigned base-128 integer: seven payload bits per octet, least significant digit first; high bit one on every nonterminal octet and zero on the last. Zero is one zero octet. A multi-octet encoding cannot end in a zero payload. Counts have no format-level 64-bit ceiling.
O(b)	One octet, $0 \leq b \leq 255$.
B(bytes)	U(length) followed by exactly those octets.
Seq_T(items)	U(count) followed by the items in order, each using schema <i>T</i> .
Z(z)	O(sign) U(limb_count), then that many eight-octet, little-endian limbs; canonical sign and leading-limb rules below.
Q(q)	Z(numerator) Z(denominator), with positive denominator and reduced fraction.
Name	A B byte string containing only printable ASCII octets <code>0x21..0x7e</code> ; nonempty and case sensitive. Free-form provenance is opaque B data and has no name-binding semantics.

Primitive	Encoding (continued)
ProfileId	Name(namespace) U(major_version) U(minor_version) Name(label). Complete fields form the nominal type/binding identity; they are not executable prose.

An implementation may announce finite count, width, depth and memory limits. A larger valid mathematical record is then unsupported, not truncated or partly accepted. Check lengths before allocation and retain exact count arithmetic.

A.2 Literal payloads and type tags

For integer limbs a_i in base $\beta = 2^{64}$, $z = (-1)^s \sum_{i=0}^{k-1} a_i \beta^i$, with $0 \leq a_i < \beta$. Zero is $s = 0, k = 0$; otherwise $a_{k-1} \neq 0$. A rational p/q requires $q > 0$ and $\gcd(|p|, q) = 1$; rational zero is $0/1$.

Literal type	Exact payload and admission rules
Bool	O(value), with value zero or one.
Int	Z(value), with the canonical integer form above.
Rat	Q(value), with the canonical rational form above.
Alg	Seq_Z(coefficients) U(root_index). Coefficients run from constant to highest degree. The polynomial has degree at least one, is primitive and irreducible over $\mathbb{Q}$, and has positive leading coefficient. The zero-based index selects a distinct real root in increasing order.
Bits(w)	Exactly $\lceil w/8 \rceil$ octets; bit zero is the low bit of the first octet. Unused high bits in the final octet are zero. Width is in the type, not repeated in this payload.
Dim(T,d)	Exactly the scalar payload of T ; the seven exponents already occur in its type. No extra numeric field or raw RealTerm literal is authorized by the wrapper.
Tuple, vector, matrix	Ordered Seq_Literal components with type/shape agreement; each component retains its own type.
Frame	The underlying Vec/Mat literal payload, with the complete frame annotation retained in Type; encoding and sharing cannot erase it.

An algebraic isolating interval is a witness, not an alternative value spelling. Exact factorization, minimal-polynomial selection, root isolation and selected root identification remain implementation obligations. A resultant alone does not select the intended root. Raw finite IEEE bits can be converted to their exact dyadic value; this does not recover a pre-rounding physical quantity. Signed-zero payloads remain distinct as bits but both denote rational zero. NaN and infinity have no real conversion.

Each type is U(tag) followed by the fields in this table:

Tag	Type	Remaining fields
0	Bool	None.
1	Int	None.
2	Rat	None.
3	Alg	None.

Tag	Type	Remaining fields (continued)
4	RealTerm	None.
5	Bits	$U(w)$, $w > 0$.
6	Tuple	$\text{Seq_Type}(\text{component_types})$.
7	Vec	$U(n)$ $\text{Type}(T)$, $n > 0$.
8	Mat	$U(\text{rows})$ $U(\text{cols})$ $\text{Type}(T)$; both sizes positive.
9	Fn	$\text{Seq_Type}(\text{parameter_types})$ $\text{Type}(\text{result})$.
10	Dim	$\text{Type}(\text{scalar})$, followed by exactly seven Q exponents.
11	Stream	$\text{Type}(\text{element})$.
12	Frame	$\text{Type}(\text{base})$ $\text{ProfileId}(\text{source})$, plus $\text{ProfileId}(\text{target})$ when base is Mat. Base is an unannotated Vec or Mat.

Dim accepts only an unwrapped numeric scalar; nested dimension wrappers are rejected. A function's domain obligations remain part of its interface/body. Function, stream and ungrounded real-term literals are rejected.

A framed vector's identity denotes its coordinate basis; a framed matrix's source/target denote the input/output bases of its linear map. Matching nominal identity does not prove physical alignment, origin or calibration. Axis/origin realization and uncertainty are explicit template/dependency inputs. Qualified namespaces must agree across the dependency closure.

Bare vectors/matrices are frame-free objects. Once an interface declares a frame, its annotation is mandatory. COERCE cannot add, change or remove it. Literal input may declare a frame; a computed frame change requires an explicitly typed TEMPLATE, its full body and coordinate/domain obligations. Extracting coordinates and relabelling the reassembled result is not an implicit conversion. This also governs removal of an annotation for use in geometry.

Frame-aware addition/subtraction and dot/cross require identical vector frame identities. Norm returns a scalar. A matrix $A \rightarrow B$ maps a vector in A to B through an explicit contraction template; composing $A \rightarrow B$ then $B \rightarrow C$ produces $A \rightarrow C$ in the corresponding product order. Transpose and inverse swap source/target annotations under their algebraic domains; interpreting transpose as inverse also requires orthonormality. Determinant refers to the declared coordinate matrix, without a basis-independence assertion. Projection retains its annotated source in node identity even when its result is scalar. Sharing/memoization include complete annotated types and immutable bindings.

A.3 Scopes, references and complete record order

The following fields are concatenated in the displayed order:

```

Scope = Seq_Node(nodes) Seq_Ref(exports)
Node = U(opcode) Type(result) Seq_Ref(operands) B(opcode_payload)
Literal(T) = Type(T) B(payload)

ASCII("XOPSEED1") U(1) U(0)
Seq_Dependency(dependencies)
Seq_Template(templates)
Seq_State(states)
Seq_Sample(samples)
Scope(main)
Seq_Export(exports)
Strategy(strategy)
B(provenance)

```

The magic is exactly eight ASCII octets; the two naturals specify major/minor profile (1,0). No trailing octets are allowed. Every opcode payload must decode exactly under its selected schema, with no extra fields. Empty payload means a zero-length B; the stated result type must satisfy the operator rule.

A reference is U(tag) followed by:

Tag	Reference kind	Fields and scope rule
0	Local node	U(index). A node operand refers only to an earlier node in its own scope.
1	Bound variable	U(de_bruijn_index). Zero is the nearest lexical binder.
2	Old state	U(slot); main dynamic scope only.
3	Current sample	U(slot); main dynamic scope only.
4	Dependency export	U(dependency) U(export); complete bytes and export interface must resolve.
5	Current sample raw timestamp	U(slot); main dynamic scope only. Returns the binding's exact Q(absolute_time) as Rat in the slot's source TimeConvention coordinate.
6	Current sample GPST timestamp	U(slot); main dynamic scope only. Explicitly applies the declared time mapping and returns Dim(Rat, (0, 0, 1, 0, 0, 0, 0)) in GPST SI seconds.

A scope export or state root may use any already declared node in its complete scope, but this does not authorize forward operands inside a node. Nested scopes cannot reach an outer local node through an accidental numeric index: captures enter as explicit binders. Binder order follows the payload; the last parameter has de Bruijn index zero. Substitution is capture avoiding and shifts indices beneath binders. Templates receive state/sample values as arguments.

Tags 3, 5 and 6 read the same immutable sample snapshot. Tag 5 exposes recorded source time, without asserting that a custom coordinate is SI seconds. Tag 6 preserves the registered GPST-SI rational with its time units, or invokes the complete custom mapping. Missing input and unmet/unresolved conversion domains remain explicit non-value outcomes; no offset is guessed. The SI exponent order is length, mass, time, electric current, thermodynamic temperature, amount of substance, luminous intensity.

An SI-second integrator's time arguments are declared expressions, for example

$$t = \text{SAMPLE_GPST}(j) - t_{\text{epoch}}, \quad h = \text{SAMPLE_GPST}(j) - t_{\text{previous}},$$

with matching time dimensions and GPST epoch conventions on every term. Here SAMPLE_GPST names reference tag 6, not an additional opcode. Other integration coordinates require explicit typed mappings and domain/unit hypotheses. Chronology and $h > 0$ remain recurrence obligations. Initial roots and pure template bodies exclude tags 5 and 6 as they exclude tag 3; a template receives needed timestamps as arguments. Raw ticks, phase wraps and civil-clock strings cannot silently supply these integration arguments.

```

Dependency = Name(label) O(kind) B(canonical_seed_bytes)
Template = Name(label) Seq_Type(parameters) Type(result) Scope(body)
State = Name(label) Type(T) Ref(init) Ref(next)
Sample = Name(label) Type(T) TimeConvention
         O(mode) B(binding) B(provenance)
Row = Z(logical_step) Q(absolute_time) Literal(T)
Export = Name(label) U(main_export_index) O(observation)
TimeConvention = ProfileId(identity) O(kind) B(definition)

```

Dependencies. Kind zero imports a seed value export; kind one imports a template-library seed. Content is a complete XOPSEED1 record, labels are unique, and imports are finite and acyclic. Imported values must be closed and stateless in their demanded graph. A snapshot may be closed literal/expression data; a live stream must enter through an explicit sample. Importing a seed neither executes its state machine nor silently selects a clock tick.

Templates. A body has exactly one export of its declared result type. It contains no main-state/sample references and calls only earlier local or explicitly imported templates. The encoded body fixes its meaning; a label alone does not provide an equation or authorize recursion.

State. Initial and next roots refer to the main scope. Initial roots are closed with respect to state and samples and must have proved values of the declared type. Next roots read the same old-state/input snapshot. The state table is the sole causal DELAY; all successful next values commit together. No instantaneous cycle or partial commit is defined. A required input/domain failure remains an explicit outcome.

Samples. Mode zero binding is Seq_Row(rows) with nonnegative, strictly increasing logical-step indices. Mode one binding is Name(external_port); every demanded step supplies a typed literal and absolute timestamp under the named convention. Missing rows or ports yield MISSING_INPUT. No implicit interpolation, nearest-row choice, leap-second conversion or future extrapolation is supplied. Repeated reads in one step bind the same sample. The timestamp and sample identity are retained.

Time convention. Kind zero has empty definition and exactly the registered identity ("XOP-TIME", 1, 0, "GPST-SI"). Its rational coordinate is elapsed SI seconds on continuous GPS time from 1980-01-06 00:00:00 GPST; no UTC conversion occurs. Kind one is custom, with definition U(dependency) U(template) ProfileId(target). The dependency is a complete template-library seed and the target is the registered GPST-SI identity. The selected pure template maps unambiguous rational source time to rational target time. Its mapping, interval, epoch and any leap/disambiguation tables must be in its body or complete dependencies. Ambiguous civil readings need additional binding. Repeated custom identities in the dependency closure must resolve to identical canonical definition/template bytes. Ordering and elapsed time claims require the mapping's monotonicity/domain proof, not version matching alone.

Exports. Labels are unique and observation is zero or one. Stateful equivalence preserves observation-one exports, sample identities and declared output timestamps. Even unobserved exports remain part of the ordinary value interface unless an explicit interface change is adopted.

Canonical encoding means one spelling for the same ordered scoped record, not for every mathematically equivalent expression or sharing pattern. Dependency hashes are lookup aids; complete canonical bytes establish required identity. A digest-only open manifest is not this self-contained encoding.

A.4 The one-bit plane envelope

Let C be a canonical semantic stream of positive length L . Pad it with zero octets to $512\lceil L/512 \rceil$ octets. Each block holds 64 little-endian uint64 patterns w_r . Its plane words are

$$p_c = \sum_{r=0}^{63} ((w_r \gg c) \& 1) 2^r, \quad 0 \leq c < 64.$$

```
ASCII("XOPPLAN1") U(L)
plane words: eight octets each, little endian,
              block-major then plane-major order
```

Exactly $512\lceil L/512 \rceil$ payload octets are required. Apply the same transpose to invert; reject nonzero

decoded padding, take exactly L octets, then parse XOPSEED1. Nonminimal lengths, redundant integer forms, invalid references, nonzero padding and trailing bytes have no canonical decoding. The transpose proof is in Section 18. Compression is a separate transport that must reconstruct this envelope exactly. A bit permutation does not reduce its uncompressed information content or erase high limbs.

A.5 Structural, numeric and Boolean opcodes

In the tables, S is a common underlying scalar sort reached by explicit coercions. Dimension transformations below refine the scalar signatures; matching S does not require multiplication operands to have equal units. T is a complete type. Unless stated otherwise, payload is empty. Ordinary numeric operands are strict in demanded subexpressions; IF is lazy and a lambda body is demanded only through application. A node table is a dependency order, not a command to evaluate unused branches.

ID / operator	Operands $\rightarrow$ result	Domain and payload
0 LITERAL	None $\rightarrow$ literal type	Payload is that type's canonical literal payload.
1 COERCE	Scalar $\rightarrow$ higher scalar	Only the stated injective scalar embeddings.
2 TUPLE	Ordered values $\rightarrow$ tuple	Result component types match.
3 PROJECT	Tuple/Vec/Mat $\rightarrow$ component	Payload $U(\text{index})$, in bounds. Matrix index is row-major $r \cdot \text{cols} + c$ and returns one scalar.
4 VECTOR	n scalars $\rightarrow \text{Vec}(n, S)$	Positive size; common scalar sort and dimension.
5 MATRIX	rc scalars $\rightarrow \text{Mat}(r, c, S)$	Row-major order; result fixes shape.
6 LAMBDA	Captures $\rightarrow$ Fn	Payload $\text{Seq_Type}(\text{parameters}) \text{ Scope}(\text{body})$; one body export; capture binders precede parameter binders.
7 APPLY	Function, arguments $\rightarrow$ result	Exact arity/types; substitute into the body and its domain.
8 TEMPLATE	Arguments $\rightarrow$ template result	Payload $0(\text{location}) \ U(\text{index})$. Location 0 selects local template. Location 1 makes index a dependency and adds $U(\text{imported_template_index})$.
16 ADD	$S, S \rightarrow S$	Equal dimensions.
17 SUB	$S, S \rightarrow S$	Equal dimensions.
18 NEG	$S \rightarrow S$	Numeric scalar.
19 MUL	$S, S \rightarrow S$	Add dimension vectors.
20 DIV	$S, S \rightarrow S$	Denominator nonzero; Int result forbidden, requiring explicit Rat promotion; subtract dimensions.
21 IDIV	Int, Int $\rightarrow$ Int	Nonzero divisor; Euclidean quotient.
22 MOD	Int, Int $\rightarrow$ Int	Nonzero divisor d ; $0 \leq r < d $.
23 POW_INT	$S, \text{Int} \rightarrow S$	Nonnegative exponent, or nonzero base and rational/higher result; $x^0 = 1$ by finite-product convention.
24 ABS	$S \rightarrow S$	Ordered real scalar.
25 SIGN	$S \rightarrow \text{Int}$	-1, 0, 1 according to exact sign; decision can remain unresolved.
26 MIN	$S, S \rightarrow S$	Matching dimensions; order evidence required for selection.
27 MAX	$S, S \rightarrow S$	Matching dimensions; order evidence required for selection.
28 FLOOR	Dimensionless scalar $\rightarrow$ Int	Unique n with $n \leq x < n + 1$; decision can remain unresolved.
29 CEIL	Dimensionless scalar $\rightarrow$ Int	$-\lfloor -x \rfloor$.
30 ROOT	Int n , scalar $x \rightarrow$ Alg/RealTerm	$n \geq 1$. Even n : $x \geq 0$, nonnegative root. Odd n : unique real root. Divide dimensions by n .
31 SQRT	Scalar $\rightarrow$ Alg/RealTerm	$x \geq 0$, nonnegative root; explicit scalar promotion and halved dimensions.
32 EQ	Compatible equal types $\rightarrow$ Bool	Exact value equality; no terminating general RealTerm decision is promised.

ID / operator	Operands → result	Domain and payload (continued)
33 LT	$S, S \rightarrow \text{Bool}$	Ordered scalar; matching dimensions.
34 LE	$S, S \rightarrow \text{Bool}$	Ordered scalar; matching dimensions.
35 AND_BOOL	$\text{Bool}, \text{Bool} \rightarrow \text{Bool}$	Logical conjunction, not a word operation.
36 OR_BOOL	$\text{Bool}, \text{Bool} \rightarrow \text{Bool}$	Logical disjunction.
37 NOT_BOOL	$\text{Bool} \rightarrow \text{Bool}$	Logical negation.
38 IF	$\text{Bool}, T, T \rightarrow T$	Lazy: only the selected branch must be defined. An unknown predicate retains the conditional term.
40 FULL_ADDER	$\text{Bool } a, b, c \rightarrow \text{Bool pair}$	Sum bit and carry bit, equations below.
41 FULL_SUBTRACTOR	$\text{Bool } a, b, k \rightarrow \text{Bool pair}$	Difference bit and borrow bit, equations below.
48 BIT_AND	$\text{Bits}(w) \text{ pair} \rightarrow \text{Bits}(w)$	Equal explicit widths.
49 BIT_OR	$\text{Bits}(w) \text{ pair} \rightarrow \text{Bits}(w)$	Equal explicit widths.
50 BIT_XOR	$\text{Bits}(w) \text{ pair} \rightarrow \text{Bits}(w)$	Equal explicit widths.
51 BIT_NOT	$\text{Bits}(w) \rightarrow \text{Bits}(w)$	Complement inside the declared width only.
52 BIT_SHIFT	$\text{Bits}(w), \text{Int } k \rightarrow \text{Bits}(w)$	$k \geq 0$; payload 0(direction): 0 left, 1 logical right. Shifted-out bits are deliberately discarded.
53 POPCOUNT	$\text{Bits}(w) \rightarrow \text{Int}$	Sum of the w Boolean bits.
54 BITS_TO_UINT	$\text{Bits}(w) \rightarrow \text{Int}$	$\sum_i b_i 2^i$.
55 UINT_TO_BITS	$\text{Int} \rightarrow \text{Bits}(w)$	$0 \leq x < 2^w$; no implicit modular truncation.
56 INT_SHIFT	$\text{Int}, \text{Int } k \rightarrow \text{Int}$	$k \geq 0$; payload direction 0 multiplies by 2^k , 1 takes floor division by 2^k .
57 IEEE64_TO_RAT	$\text{Bits}(64) \rightarrow \text{Rat}$	Finite IEEE binary64 pattern only; exact dyadic conversion below.
58 DECIMAL_TO_RAT	$\text{Int mantissa}, \text{Int scale} \rightarrow \text{Rat}$	Exact mantissa $\cdot 10^{-\text{scale}}$, reduced.

For IEEE conversion let s be the sign bit, e the unsigned 11-bit exponent and f the unsigned 52-bit fraction. The exact value is

$$\begin{cases} (-1)^s f 2^{-1074}, & e = 0, \\ (-1)^s (2^{52} + f) 2^{e-1075}, & 1 \leq e \leq 2046. \end{cases}$$

Exponent 2047 is outside the real conversion domain. Keep the raw operand if signed-zero or other bit provenance must remain observable. Rational operations reduce by gcd and normalize sign; algebraic operations preserve selected-root identity. The source equation-language token 1 means an all-ones word only under its separately declared language adapter.

For Boolean carry c and borrow k , the one-bit arithmetic operators are

$$\begin{aligned} s &= a \oplus b \oplus c, & c' &= (a \wedge b) \vee (c \wedge (a \oplus b)), \\ d &= a \oplus b \oplus k, & k' &= (\neg a \wedge b) \vee (k \wedge \neg(a \oplus b)), \\ a + b + c &= s + 2c', & a - b - k &= d - 2k'. \end{aligned}$$

Complements here act on one Boolean bit. Chain carry/borrow toward more significant bits, starting at zero. Multiply by placing $a_i \wedge b_j$ at position $i + j$ and adding every shifted row with retained carries. To compare padded magnitudes from high to low bits, start $(E, G) = (1, 0)$ and update simultaneously $G' = G \vee (E \wedge a \wedge \neg b)$ and $E' = E \wedge \neg(a \oplus b)$. Long division uses this exact comparison and subtraction. These are finite definitions for every finite input; no executed bit test is asserted.

A.6 Elementary, tensor, calculus and kernel opcodes

Elementary functions have the real denotations and branch conventions in Section 18. They require explicit promotion to `RealTerm`; angle results are dimensionless with declared angle convention.

ID / operator	Operands $\rightarrow$ result	Domain and payload
64 SIN	$\text{RealTerm} \rightarrow \text{RealTerm}$	Entire sine series; dimensionless argument.
65 COS	$\text{RealTerm} \rightarrow \text{RealTerm}$	Entire cosine series; dimensionless argument.
66 TAN	$x \rightarrow \sin x / \cos x$	$\cos x \neq 0$.
67 EXP	$\text{RealTerm} \rightarrow \text{RealTerm}$	Entire exponential series.
68 LOG	$x \rightarrow \int_1^x du/u$	$x > 0$.
69 ATAN	$x \rightarrow \int_0^x du/(1+u^2)$	Every real x .
70 ATAN2	$y, x \rightarrow \text{angle}$	$(x, y) \neq (0, 0)$; range $(-\pi, \pi]$ with the declared quadrant cases.
71 ASIN	$x \rightarrow \text{angle}$	$-1 \leq x \leq 1$; inverse sine on $[-\pi/2, \pi/2]$.
72 ACOS	$x \rightarrow \text{angle}$	$-1 \leq x \leq 1$; inverse cosine on $[0, \pi]$.
73 PI	$\text{None} \rightarrow \text{RealTerm}$	$4 \text{atan}(1)$.
74 POW_REAL	$x, y \rightarrow \exp(y \log x)$	$x > 0$; both arguments dimensionless. Distinct from integer powers and root branches.
80 DOT	$\text{Vec}(n, S) \text{ pair} \rightarrow S$	Finite sum of pairwise products.
81 CROSS	$\text{Vec}(3, S) \text{ pair} \rightarrow \text{Vec}(3, S)$	Right-handed determinant in the declared frame.
82 NORM	$\text{Vec}(n, S) \rightarrow \text{Alg}/\text{RealTerm}$	Nonnegative square root of dot product; retains the vector dimension.
83 TRANSPPOSE	$\text{Mat}(r, c, S) \rightarrow \text{Mat}(c, r, S)$	Swap indices.
84 MATMUL	$\text{Mat}(r, k, S), \text{Mat}(k, c, S) \rightarrow \text{Mat}(r, c, S)$	Finite contraction over k .
85 DET	Square matrix $\rightarrow S$	Finite signed permutation sum.
86 INVERSE	$\text{Mat}(n, n, S) \rightarrow \text{Mat}(n, n, S)$	Nonzero determinant; scalar sort closed under division.
87 SOLVE_LINEAR	$A, b \rightarrow x, Ax = b$	Square A , nonzero determinant, compatible vector and division-capable scalar sort.
96 FIN_SUM	$\text{Int } a, b, \text{captures} \rightarrow S$	Inclusive bounds. Payload $\text{Type}(S) \text{ Scope}(\text{body})$; index binder last. Empty interval gives additive identity.
97 FIN_PRODUCT	$\text{Int } a, b, \text{captures} \rightarrow S$	Same payload/binding; empty interval gives multiplicative identity. Terms dimensionless in this base rule.
98 DIFF	$\text{Point}, \text{captures} \rightarrow \text{RealTerm}$	Payload $\text{Type}(\text{variable}) \text{ Scope}(\text{body})$; variable binder last. Derivative must exist at the point.
99 INTEGRAL	$a, b, \text{captures} \rightarrow \text{RealTerm}$	Same payload/binder; oriented finite Riemann integral. Integrability is required.
100 LIMIT	$\text{Point}, \text{captures} \rightarrow \text{RealTerm}$	Payload $\text{Type}(\text{variable}) \text{ } 0(\text{side}) \text{ Scope}(\text{body})$. Side 0 two-sided, 1 from below, 2 from above. Unique finite limit required.
101 ODE	$t_0, s_0, t, \text{captures} \rightarrow \text{state}$	Payload $Q(\text{interval_left}) \text{ } Q(\text{interval_right}) \text{ Scope}(F)$. Closed interval includes t_0, t ; time binder then state last. Unique continuous integral-equation solution required.
102 SOLVE_ROOT	$\text{Captures} \rightarrow \text{RealTerm}$	Payload $Q(\text{left}) \text{ } Q(\text{right}) \text{ Scope}(f)$; binder x last. Exactly one real root in the open rational interval; existence and uniqueness required.
112 ASA_NA	$x, V, A, N, B \rightarrow (a, n, h, y)$	Five $\text{Bits}(w)$ operands; result has word components a, n, y and $\text{Int } h$. Whole-word absorption below.
113 JK	$q, J, K, V \rightarrow \text{Bits}(w)$	Four same-width words, reading the same old q ; complement restricted to width.

Calculus binders and solution domain. Captures bind in operand order, followed by the displayed integration/differentiation variables. Each body has one export. Body and variable dimensions determine derivative/integral units. ODE supports a scalar, uniform vector, framed vector, or finite nested tuple of those dimensioned components; every scalar result leaf has sort RealTerm . Initial algebraic values are coerced componentwise while preserving all frame annotations. F has the same structure and identical nominal frame on each framed vector, with each scalar leaf dimension equal to its state dimension minus the time dimension. Rotating-frame derivatives still require the appropriate terms in F ; the type supplies none. Interval rationals use

the same declared time unit as t_0 and t .

Specifically, ODE denotes the unique continuous s on the declared closed interval I such that, componentwise,

$$s(u) = s_0 + \int_{t_0}^u F(v, s(v)) dv \quad (u \in I),$$

with a Riemann-integrable integrand along s . It is differentiable where the composite right-hand side is continuous; a derivative at a coefficient jump is not required. The derived name FLOW uses this same declaration. State-dependent discontinuities still require existence/uniqueness under the event policy. A local theorem does not certify the whole interval. General root and ODE declarations are not guaranteed solution algorithms. Unbounded sums, improper integrals, complex roots and distributional derivatives are outside these opcodes.

Dimensions. If operands have dimension vectors d_A, d_B , then MUL, DOT, CROSS and MATMUL return $d_A + d_B$; DIV returns $d_A - d_B$. NORM retains dimension. For a uniformly dimensioned $n \times n$ matrix, determinant has dimension nd_A , inverse $-d_A$, and SOLVE_LINEAR has dimension $d_b - d_A$. Addition, subtraction and order require equal dimensions. Heterogeneous Jacobians use dimensioned tuples/blocks and explicit contraction templates, not one falsely uniform matrix type. Dimensioned powers/roots must establish the annotated output dimensions from their degree at admission; a variable degree cannot contradict a fixed annotation. Dimensionless inputs avoid that degree/unit issue.

The retained kernel operators have literal equations

$$\begin{aligned} a &= x \wedge A \wedge V, & n &= a \wedge N, & h &= \text{popcount}(n \wedge B), \\ y &= \begin{cases} 0, & h > 0, \\ n, & h = 0, \end{cases} & \text{ASA_NA}(x, V, A, N, B) &= (a, n, h, y), \\ \text{JK}(q, J, K, V) &= ((J \wedge \neg_w q) \vee (\neg_w K \wedge q)) \wedge V. \end{aligned}$$

Here $\neg_w q = (2^w - 1) \oplus q$. The second mask set receives the first stage's y ; both traces remain exported when required. J, K use the same old state and final y . Absorption zeroes the whole stage output and does not silently substitute a hold rule. The state table supplies the delay.

A.7 Named templates and the proof-record interface

Named templates require explicit typed parameters and bodies. Labels alone provide no operator implementation. WRAP is $x - P \lfloor x/P \rfloor$ for $P > 0$, paired with integer winding; PHI is $(1 + \sqrt{5})/2$; FIBONACCI has initial $(0, 1)$ and delayed pair update $(a, b) \mapsto (b, a+b)$. NORMALIZE requires nonzero norm. GRADIENT/JACOBIAN assemble coordinate DIFF terms. Polynomial/Horner and Chebyshev templates preserve their finite coefficients and degrees, with $T_0 = 1, T_1 = x, T_{n+1} = 2xT_n - T_{n-1}$. Legendre, gravity, frame, third-body, pressure/shadow and station templates retain their source equations and singular/branch domains. RK4 is the finite rule in Section 18, not the continuous ODE declaration. OTAN2, both UGTS key layouts and the source geometry remain separate templates with their literal source identities. OTAN2 is not ATAN2. Modular keys do not replace absolute coordinates, time or sample bytes.

The strategy and proof records have this exact outer order:

```
Strategy = U(version) Name(proof_system) Seq_Rule(rules)
          B(priority_policy) B(cost_model) B(resource_policy)
          Seq_Proposal(history)
Rule = Name(label) B(typed_statement) B(proof_object)
Proposal = U(parent_version) U(new_version)
          B(old_canonical_graph) B(new_canonical_graph)
          B(statement) B(proof_object) B(cost_argument)
```

The named proof-system/policy profile resolves the B contents; they are not executable arbitrary source. The baseline with empty rules, policy and history admits identity-only execution strategy. XOP-R1 defines the interface, not a completed proof language or checker. Uninterpreted policy bytes confer no authority to admit a nonidentity rewrite.

For $\Gamma \vdash e \simeq e' : T$, domains under Γ and denotations on those domains must be identical. The proof interface has these rule families:

Rule family	Required premise
ASSUMPTION	The exact proposition occurs in the typed environment.
REFL, SYM, TRANS	Typed equality and compatible domain judgements.
CONGRUENCE	Equal corresponding operands, same operator/bindings and preserved definedness.
SUBSTITUTION, BETA	Capture-avoiding typed substitution; function/body domain preserved.
EXPAND	Exact catalog/template definition, including its branches and domains.
INTEGER_RING, RATIONAL_FIELD	Exact laws with every nonzero-denominator obligation retained.
ORDER, ROOT_BRANCH	Required sign, range and unique-root facts.
ANALYTIC_LEMMA	Explicit derivative/integral/limit hypotheses and proof in the accepted calculus.
STATE_SIMULATION	Initial relation, step-domain equivalence and equal declared observations for every admitted input.

A theorem identifier must resolve to an accepted theorem with discharged substitutions and premises. A named or absent checker cannot admit nonidentity proofs. Strategy versions are immutable; the previously accepted checker reviews a proposal. Replacing that checker creates a new trust boundary and cannot be certified merely by the proposal's own changed rules. Hashes identify bytes, not soundness or termination.

Memoization binds complete operator/dependency identity, arguments, sample snapshot and state step. Resource outcomes and traces also count when promised by the interface. A decreasing natural-valued measure proves termination of that rewrite phase; a finite search budget proves only bounded search. Neither establishes global optimality, speed, bounded limb growth or physical accuracy. The causal and conditional replacement proofs remain those of Section 18; no new algorithm validation is asserted here.